

✓ **Solution**

Figure 2.25 illustrates the function $f(x) = \sqrt{x-3}$ and aids in our understanding of these limits.

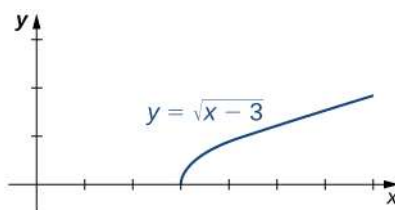


Figure 2.25 The graph shows the function $f(x) = \sqrt{x-3}$.

- a. The function $f(x) = \sqrt{x-3}$ is defined over the interval $[3, +\infty)$. Since this function is not defined to the left of 3, we cannot apply the limit laws to compute $\lim_{x \rightarrow 3^-} \sqrt{x-3}$. In fact, since $f(x) = \sqrt{x-3}$ is undefined to the left of 3, $\lim_{x \rightarrow 3^-} \sqrt{x-3}$ does not exist.
- b. Since $f(x) = \sqrt{x-3}$ is defined to the right of 3, the limit laws do apply to $\lim_{x \rightarrow 3^+} \sqrt{x-3}$. By applying these limit laws we obtain $\lim_{x \rightarrow 3^+} \sqrt{x-3} = 0$.

In [Example 2.22](#) we look at one-sided limits of a piecewise-defined function and use these limits to draw a conclusion about a two-sided limit of the same function.

EXAMPLE 2.22

Evaluating a Two-Sided Limit Using the Limit Laws

For $f(x) = \begin{cases} 4x-3 & \text{if } x < 2 \\ (x-3)^2 & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits:

- a. $\lim_{x \rightarrow 2^-} f(x)$
 b. $\lim_{x \rightarrow 2^+} f(x)$
 c. $\lim_{x \rightarrow 2} f(x)$

✓ **Solution**

Figure 2.26 illustrates the function $f(x)$ and aids in our understanding of these limits.

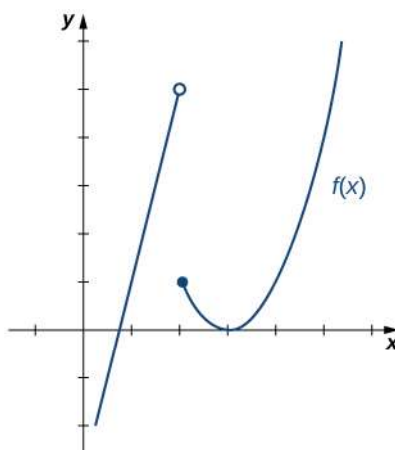


Figure 2.26 This graph shows a function $f(x)$.

- a. Since $f(x) = 4x - 3$ for all x in $(-\infty, 2)$, replace $f(x)$ in the limit with $4x - 3$ and apply the limit laws:

Step 5. Then, we cancel the common factors of $(x - 1)$:

$$= \lim_{x \rightarrow 1} \frac{-1}{2(x+1)}.$$

Step 6. Last, we evaluate using the limit laws:

$$\lim_{x \rightarrow 1} \frac{-1}{2(x+1)} = -\frac{1}{4}.$$

✓ 2.15 Evaluate $\lim_{x \rightarrow -3} \frac{\frac{1}{x+2} + 1}{x+3}$.

[Example 2.20](#) does not fall neatly into any of the patterns established in the previous examples. However, with a little creativity, we can still use these same techniques.

EXAMPLE 2.20

Evaluating a Limit When the Limit Laws Do Not Apply

Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} + \frac{5}{x(x-5)} \right)$.

✓ Solution

Both $1/x$ and $5/(x-5)$ fail to have a limit at zero. Since neither of the two functions has a limit at zero, we cannot apply the sum law for limits; we must use a different strategy. In this case, we find the limit by performing addition and then applying one of our previous strategies. Observe that

$$\begin{aligned} \frac{1}{x} + \frac{5}{x(x-5)} &= \frac{x-5+5}{x(x-5)} \\ &= \frac{x}{x(x-5)}. \end{aligned}$$

Thus,

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{x} + \frac{5}{x(x-5)} \right) &= \lim_{x \rightarrow 0} \frac{x}{x(x-5)} \\ &= \lim_{x \rightarrow 0} \frac{1}{x-5} \\ &= -\frac{1}{5}. \end{aligned}$$

✓ 2.16 Evaluate $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{4}{x^2-2x-3} \right)$.

Let's now revisit one-sided limits. Simple modifications in the limit laws allow us to apply them to one-sided limits. For example, to apply the limit laws to a limit of the form $\lim_{x \rightarrow a^-} h(x)$, we require the function $h(x)$ to be defined over an open interval of the form (b, a) ; for a limit of the form $\lim_{x \rightarrow a^+} h(x)$, we require the function $h(x)$ to be defined over an open interval of the form (a, c) . [Example 2.21](#) illustrates this point.

EXAMPLE 2.21

Evaluating a One-Sided Limit Using the Limit Laws

Evaluate each of the following limits, if possible.

- $\lim_{x \rightarrow 3^-} \sqrt{x-3}$
- $\lim_{x \rightarrow 3^+} \sqrt{x-3}$

EXAMPLE 2.18**Evaluating a Limit by Multiplying by a Conjugate**

Evaluate $\lim_{x \rightarrow -1} \frac{\sqrt{x+2}-1}{x+1}$.

 Solution

Step 1. $\frac{\sqrt{x+2}-1}{x+1}$ has the form $0/0$ at -1 . Let's begin by multiplying by $\sqrt{x+2}+1$, the conjugate of $\sqrt{x+2}-1$, on the numerator and denominator:

$$\lim_{x \rightarrow -1} \frac{\sqrt{x+2}-1}{x+1} = \lim_{x \rightarrow -1} \frac{\sqrt{x+2}-1}{x+1} \cdot \frac{\sqrt{x+2}+1}{\sqrt{x+2}+1}.$$

Step 2. We then multiply out the numerator. We don't multiply out the denominator because we are hoping that the $(x+1)$ in the denominator cancels out in the end:

$$= \lim_{x \rightarrow -1} \frac{x+1}{(x+1)(\sqrt{x+2}+1)}.$$

Step 3. Then we cancel:

$$= \lim_{x \rightarrow -1} \frac{1}{\sqrt{x+2}+1}.$$

Step 4. Last, we apply the limit laws:

$$\lim_{x \rightarrow -1} \frac{1}{\sqrt{x+2}+1} = \frac{1}{2}.$$

 2.14 Evaluate $\lim_{x \rightarrow 5} \frac{\sqrt{x-1}-2}{x-5}$.

EXAMPLE 2.19**Evaluating a Limit by Simplifying a Complex Fraction**

Evaluate $\lim_{x \rightarrow 1} \frac{\frac{1}{x+1} - \frac{1}{2}}{x-1}$.

 Solution

Step 1. $\frac{\frac{1}{x+1} - \frac{1}{2}}{x-1}$ has the form $0/0$ at 1 . We simplify the algebraic fraction by multiplying by $2(x+1)/2(x+1)$:

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x+1} - \frac{1}{2}}{x-1} = \lim_{x \rightarrow 1} \frac{\frac{1}{x+1} - \frac{1}{2}}{x-1} \cdot \frac{2(x+1)}{2(x+1)}.$$

Step 2. Next, we multiply through the numerators. Do not multiply the denominators because we want to be able to cancel the factor $(x-1)$:

$$= \lim_{x \rightarrow 1} \frac{2 - (x+1)}{2(x-1)(x+1)}.$$

Step 3. Then, we simplify the numerator:

$$= \lim_{x \rightarrow 1} \frac{-x+1}{2(x-1)(x+1)}.$$

Step 4. Now we factor out -1 from the numerator:

$$= \lim_{x \rightarrow 1} \frac{-(x-1)}{2(x-1)(x+1)}.$$

$$\begin{aligned}
 \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} &= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} \\
 &= \lim_{x \rightarrow 1} (x + 1) \\
 &= 2.
 \end{aligned}$$

The limit has the form $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, where $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$. (In this case, we say that $f(x)/g(x)$ has the indeterminate form $0/0$.) The following Problem-Solving Strategy provides a general outline for evaluating limits of this type.

Problem-Solving Strategy

Calculating a Limit When $f(x)/g(x)$ has the Indeterminate Form $0/0$

1. First, we need to make sure that our function has the appropriate form and cannot be evaluated immediately using the limit laws.
2. We then need to find a function that is equal to $h(x) = f(x)/g(x)$ for all $x \neq a$ over some interval containing a . To do this, we may need to try one or more of the following steps:
 - a. If $f(x)$ and $g(x)$ are polynomials, we should factor each function and cancel out any common factors.
 - b. If the numerator or denominator contains a difference involving a square root, we should try multiplying the numerator and denominator by the conjugate of the expression involving the square root.
 - c. If $f(x)/g(x)$ is a complex fraction, we begin by simplifying it.
3. Last, we apply the limit laws.

The next examples demonstrate the use of this Problem-Solving Strategy. [Example 2.17](#) illustrates the factor-and-cancel technique; [Example 2.18](#) shows multiplying by a conjugate. In [Example 2.19](#), we look at simplifying a complex fraction.

EXAMPLE 2.17

Evaluating a Limit by Factoring and Canceling

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 3x}{2x^2 - 5x - 3}$.

Solution

Step 1. The function $f(x) = \frac{x^2 - 3x}{2x^2 - 5x - 3}$ is undefined for $x = 3$. In fact, if we substitute 3 into the function we get $0/0$, which is indeterminate. Factoring and canceling is a good strategy:

$$\lim_{x \rightarrow 3} \frac{x^2 - 3x}{2x^2 - 5x - 3} = \lim_{x \rightarrow 3} \frac{x(x - 3)}{(x - 3)(2x + 1)}$$

Step 2. For all $x \neq 3$, $\frac{x^2 - 3x}{2x^2 - 5x - 3} = \frac{x}{2x + 1}$. Therefore,

$$\lim_{x \rightarrow 3} \frac{x(x - 3)}{(x - 3)(2x + 1)} = \lim_{x \rightarrow 3} \frac{x}{2x + 1}.$$

Step 3. Evaluate using the limit laws:

$$\lim_{x \rightarrow 3} \frac{x}{2x + 1} = \frac{3}{7}.$$

 2.13 Evaluate $\lim_{x \rightarrow -3} \frac{x^2 + 4x + 3}{x^2 - 9}$.

Example 2.16 applies this result.

EXAMPLE 2.16

Evaluating a Limit of a Rational Function

Evaluate the $\lim_{x \rightarrow 3} \frac{2x^2 - 3x + 1}{5x + 4}$.

✓ Solution

Since 3 is in the domain of the rational function $f(x) = \frac{2x^2 - 3x + 1}{5x + 4}$, we can calculate the limit by substituting 3 for x into the function. Thus,

$$\lim_{x \rightarrow 3} \frac{2x^2 - 3x + 1}{5x + 4} = \frac{10}{19}.$$

✓ 2.12 Evaluate $\lim_{x \rightarrow -2} (3x^3 - 2x + 7)$.

Additional Limit Evaluation Techniques

As we have seen, we may evaluate easily the limits of polynomials and limits of some (but not all) rational functions by direct substitution. However, as we saw in the introductory section on limits, it is certainly possible for $\lim_{x \rightarrow a} f(x)$ to exist when $f(a)$ is undefined. The following observation allows us to evaluate many limits of this type:

If for all $x \neq a$, $f(x) = g(x)$ over some open interval containing a , then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$.

To understand this idea better, consider the limit $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

The function

$$\begin{aligned} f(x) &= \frac{x^2 - 1}{x - 1} \\ &= \frac{(x - 1)(x + 1)}{x - 1} \end{aligned}$$

and the function $g(x) = x + 1$ are identical for all values of $x \neq 1$. The graphs of these two functions are shown in Figure 2.24.

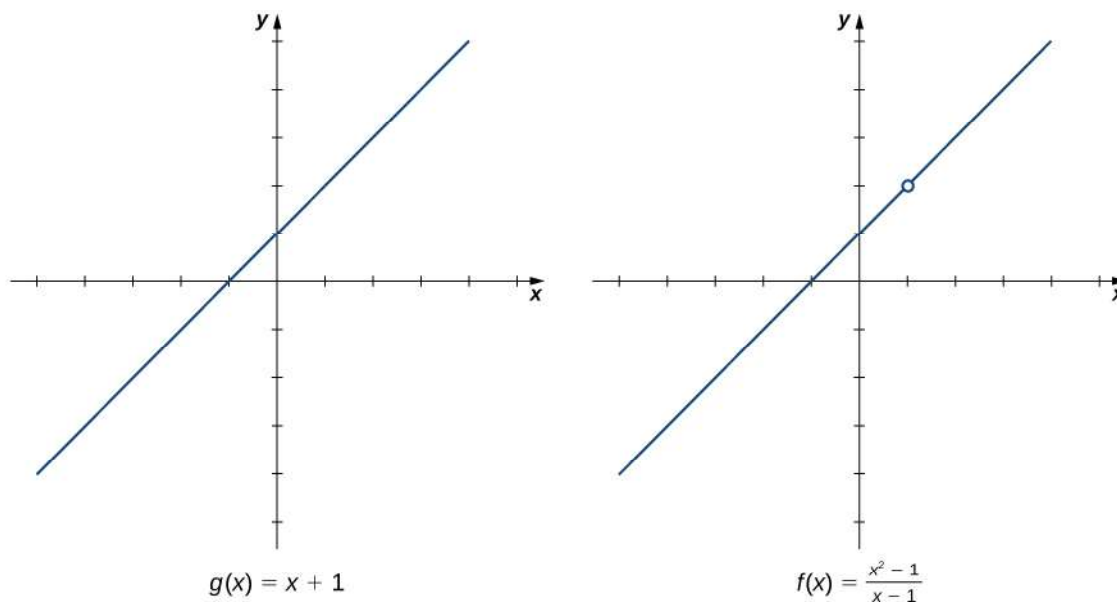


Figure 2.24 The graphs of $f(x)$ and $g(x)$ are identical for all $x \neq 1$. Their limits at 1 are equal.

We see that

EXAMPLE 2.15**Using Limit Laws Repeatedly**

Use the limit laws to evaluate $\lim_{x \rightarrow 2} \frac{2x^2 - 3x + 1}{x^3 + 4}$.

✓ Solution

To find this limit, we need to apply the limit laws several times. Again, we need to keep in mind that as we rewrite the limit in terms of other limits, each new limit must exist for the limit law to be applied.

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{2x^2 - 3x + 1}{x^3 + 4} &= \frac{\lim_{x \rightarrow 2} (2x^2 - 3x + 1)}{\lim_{x \rightarrow 2} (x^3 + 4)} && \text{Apply the quotient law, making sure that } (2)^3 + 4 \neq 0 \\ &= \frac{2 \cdot \lim_{x \rightarrow 2} x^2 - 3 \cdot \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1}{\lim_{x \rightarrow 2} x^3 + \lim_{x \rightarrow 2} 4} && \text{Apply the sum law and constant multiple law.} \\ &= \frac{2 \cdot \left(\lim_{x \rightarrow 2} x \right)^2 - 3 \cdot \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1}{\left(\lim_{x \rightarrow 2} x \right)^3 + \lim_{x \rightarrow 2} 4} && \text{Apply the power law.} \\ &= \frac{2(4) - 3(2) + 1}{(2)^3 + 4} = \frac{1}{4}. && \text{Apply the basic limit laws and simplify.} \end{aligned}$$

✓ 2.11 Use the limit laws to evaluate $\lim_{x \rightarrow 6} (2x - 1) \sqrt{x + 4}$. In each step, indicate the limit law applied.

Limits of Polynomial and Rational Functions

By now you have probably noticed that, in each of the previous examples, it has been the case that $\lim_{x \rightarrow a} f(x) = f(a)$.

This is not always true, but it does hold for all polynomials for any choice of a and for all rational functions at all values of a for which the rational function is defined.

Theorem 2.6**Limits of Polynomial and Rational Functions**

Let $p(x)$ and $q(x)$ be polynomial functions. Let a be a real number. Then,

$$\begin{aligned} \lim_{x \rightarrow a} p(x) &= p(a) \\ \lim_{x \rightarrow a} \frac{p(x)}{q(x)} &= \frac{p(a)}{q(a)} \text{ when } q(a) \neq 0. \end{aligned}$$

To see that this theorem holds, consider the polynomial $p(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$. By applying the sum, constant multiple, and power laws, we end up with

$$\begin{aligned} \lim_{x \rightarrow a} p(x) &= \lim_{x \rightarrow a} (c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0) \\ &= c_n \left(\lim_{x \rightarrow a} x \right)^n + c_{n-1} \left(\lim_{x \rightarrow a} x \right)^{n-1} + \cdots + c_1 \left(\lim_{x \rightarrow a} x \right) + \lim_{x \rightarrow a} c_0 \\ &= c_n a^n + c_{n-1} a^{n-1} + \cdots + c_1 a + c_0 \\ &= p(a). \end{aligned}$$

It now follows from the quotient law that if $p(x)$ and $q(x)$ are polynomials for which $q(a) \neq 0$, then

$$\lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}.$$

EXAMPLE 2.13**Evaluating a Basic Limit**

Evaluate each of the following limits using [Basic Limit Results](#).

- a. $\lim_{x \rightarrow 2} x$
 b. $\lim_{x \rightarrow 2} 5$

✓ Solution

- a. The limit of x as x approaches a is a : $\lim_{x \rightarrow 2} x = 2$.
 b. The limit of a constant is that constant: $\lim_{x \rightarrow 2} 5 = 5$.

We now take a look at the **limit laws**, the individual properties of limits. The proofs that these laws hold are omitted here.

Theorem 2.5**Limit Laws**

Let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a . Assume that L and M are real numbers such that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Let c be a constant. Then, each of the following statements holds:

Sum law for limits: $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L + M$

Difference law for limits: $\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L - M$

Constant multiple law for limits: $\lim_{x \rightarrow a} cf(x) = c \cdot \lim_{x \rightarrow a} f(x) = cL$

Product law for limits: $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L \cdot M$

Quotient law for limits: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}$ for $M \neq 0$

Power law for limits: $\lim_{x \rightarrow a} (f(x))^n = \left(\lim_{x \rightarrow a} f(x) \right)^n = L^n$ for every positive integer n .

Root law for limits: $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L}$ for all L if n is odd and for $L \geq 0$ if n is even and $f(x) \geq 0$.

We now practice applying these limit laws to evaluate a limit.

EXAMPLE 2.14**Evaluating a Limit Using Limit Laws**

Use the limit laws to evaluate $\lim_{x \rightarrow -3} (4x + 2)$.

✓ Solution

Let's apply the limit laws one step at a time to be sure we understand how they work. We need to keep in mind the requirement that, at each application of a limit law, the new limits must exist for the limit law to be applied.

$$\begin{aligned} \lim_{x \rightarrow -3} (4x + 2) &= \lim_{x \rightarrow -3} 4x + \lim_{x \rightarrow -3} 2 && \text{Apply the sum law.} \\ &= 4 \cdot \lim_{x \rightarrow -3} x + \lim_{x \rightarrow -3} 2 && \text{Apply the constant multiple law.} \\ &= 4 \cdot (-3) + 2 = -10. && \text{Apply the basic limit results and simplify.} \end{aligned}$$

82. A track coach uses a camera with a fast shutter to estimate the position of a runner with respect to time. A table of the values of position of the athlete versus time is given here, where x is the position in meters of the runner and t is time in seconds. What is $\lim_{t \rightarrow 2} x(t)$? What does it mean physically?

t (sec)	x (m)
1.75	4.5
1.95	6.1
1.99	6.42
2.01	6.58
2.05	6.9
2.25	8.5

2.3 The Limit Laws

Learning Objectives

- 2.3.1 Recognize the basic limit laws.
- 2.3.2 Use the limit laws to evaluate the limit of a function.
- 2.3.3 Evaluate the limit of a function by factoring.
- 2.3.4 Use the limit laws to evaluate the limit of a polynomial or rational function.
- 2.3.5 Evaluate the limit of a function by factoring or by using conjugates.
- 2.3.6 Evaluate the limit of a function by using the squeeze theorem.

In the previous section, we evaluated limits by looking at graphs or by constructing a table of values. In this section, we establish laws for calculating limits and learn how to apply these laws. In the Student Project at the end of this section, you have the opportunity to apply these limit laws to derive the formula for the area of a circle by adapting a method devised by the Greek mathematician Archimedes. We begin by restating two useful limit results from the previous section. These two results, together with the limit laws, serve as a foundation for calculating many limits.

Evaluating Limits with the Limit Laws

The first two limit laws were stated in [Two Important Limits](#) and we repeat them here. These basic results, together with the other limit laws, allow us to evaluate limits of many algebraic functions.

Theorem 2.4

Basic Limit Results

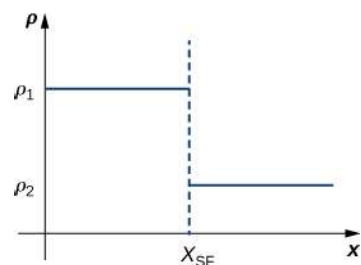
For any real number a and any constant c ,

i. $\lim_{x \rightarrow a} x = a$ (2.14)

ii. $\lim_{x \rightarrow a} c = c$ (2.15)

80. As $x \rightarrow -\infty$, $f(x) \rightarrow 0$, $\lim_{x \rightarrow -1^-} f(x) = \infty$, $\lim_{x \rightarrow -1^+} f(x) = -\infty$,
 $f(0) = -1$, $\lim_{x \rightarrow 1^-} f(x) = -\infty$, $\lim_{x \rightarrow 1^+} f(x) = \infty$, As $x \rightarrow \infty$, $f(x) \rightarrow 0$

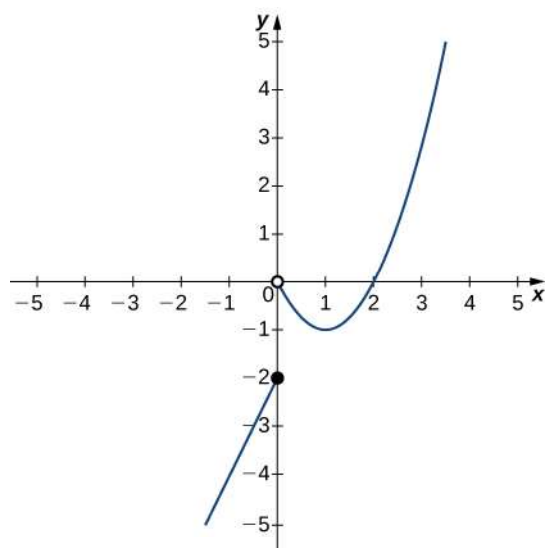
81. Shock waves arise in many physical applications, ranging from supernovas to detonation waves. A graph of the density of a shock wave with respect to distance, x , is shown here. We are mainly interested in the location of the front of the shock, labeled x_{SF} in the diagram.



- Evaluate $\lim_{x \rightarrow x_{SF}^+} \rho(x)$.
- Evaluate $\lim_{x \rightarrow x_{SF}^-} \rho(x)$.
- Evaluate $\lim_{x \rightarrow x_{SF}} \rho(x)$.

Explain the physical meanings behind your answers.

In the following exercises, use the graph of the function $y = f(x)$ shown here to find the values, if possible. Estimate when necessary.



71. $\lim_{x \rightarrow 0^-} f(x)$

72. $\lim_{x \rightarrow 0^+} f(x)$

73. $\lim_{x \rightarrow 0} f(x)$

74. $\lim_{x \rightarrow 1} f(x)$

75. $\lim_{x \rightarrow 2} f(x)$

In the following exercises, sketch the graph of a function with the given properties.

76. $\lim_{x \rightarrow 2} f(x) = 1$, $\lim_{x \rightarrow 4^-} f(x) = 3$, $\lim_{x \rightarrow 4^+} f(x) = 6$, $f(4)$ is not defined.

77. As $x \rightarrow -\infty$, $f(x) \rightarrow 0$, $\lim_{x \rightarrow -1^-} f(x) = -\infty$,

$\lim_{x \rightarrow -1^+} f(x) = \infty$, $\lim_{x \rightarrow 0} f(x) = f(0)$, $f(0) = 1$, As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

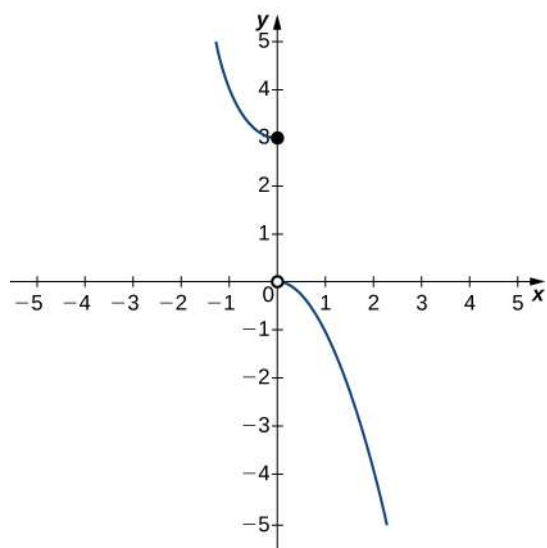
78. As $x \rightarrow -\infty$, $f(x) \rightarrow 2$, $\lim_{x \rightarrow 3^-} f(x) = -\infty$,

$\lim_{x \rightarrow 3^+} f(x) = \infty$, As $x \rightarrow \infty$, $f(x) \rightarrow 2$, $f(0) = \frac{-1}{3}$

79. As $x \rightarrow -\infty$, $f(x) \rightarrow 2$, $\lim_{x \rightarrow -2} f(x) = -\infty$,

As $x \rightarrow \infty$, $f(x) \rightarrow 2$, $f(0) = 0$

In the following exercises, use the graph of the function $y = g(x)$ shown here to find the values, if possible. Estimate when necessary.

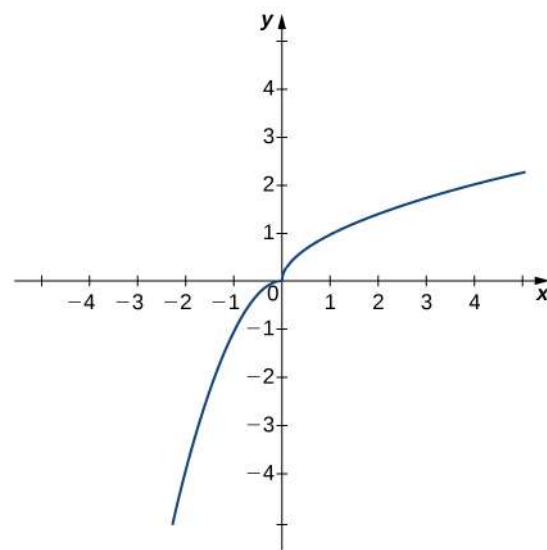


65. $\lim_{x \rightarrow 0^-} g(x)$

66. $\lim_{x \rightarrow 0^+} g(x)$

67. $\lim_{x \rightarrow 0} g(x)$

In the following exercises, use the graph of the function $y = h(x)$ shown here to find the values, if possible. Estimate when necessary.

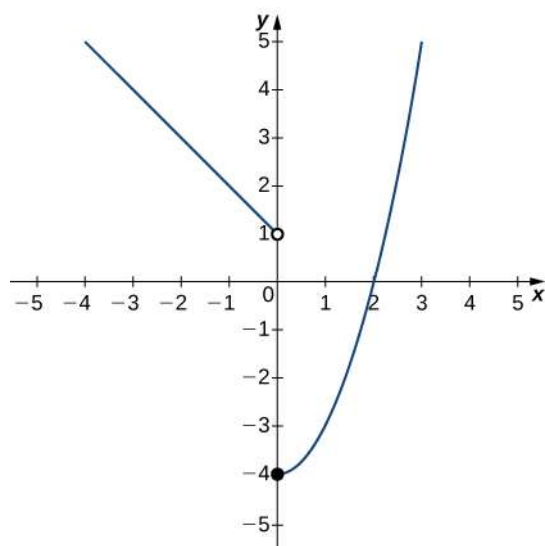


68. $\lim_{x \rightarrow 0^-} h(x)$

69. $\lim_{x \rightarrow 0^+} h(x)$

70. $\lim_{x \rightarrow 0} h(x)$

In the following exercises, use the graph of the function $y = f(x)$ shown here to find the values, if possible. Estimate when necessary.



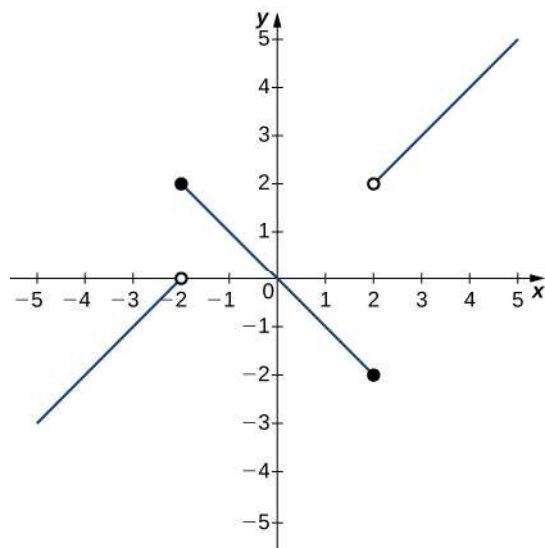
55. $\lim_{x \rightarrow 0^-} f(x)$

56. $\lim_{x \rightarrow 0^+} f(x)$

57. $\lim_{x \rightarrow 0} f(x)$

58. $\lim_{x \rightarrow 2} f(x)$

In the following exercises, use the graph of the function $y = f(x)$ shown here to find the values, if possible. Estimate when necessary.



59. $\lim_{x \rightarrow -2^-} f(x)$

60. $\lim_{x \rightarrow -2^+} f(x)$

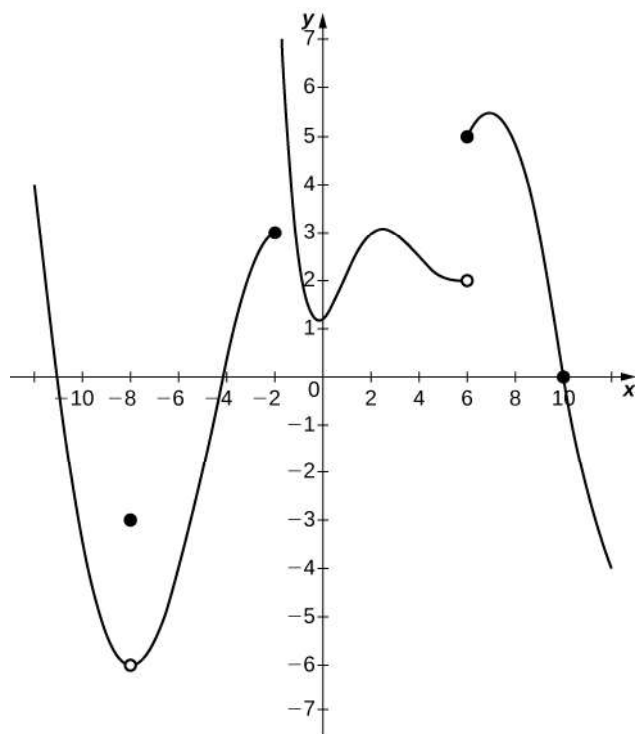
61. $\lim_{x \rightarrow -2} f(x)$

62. $\lim_{x \rightarrow 2^-} f(x)$

63. $\lim_{x \rightarrow 2^+} f(x)$

64. $\lim_{x \rightarrow 2} f(x)$

In the following exercises, consider the graph of the function $y = f(x)$ shown here. Which of the statements about $y = f(x)$ are true and which are false? Explain why a statement is false.



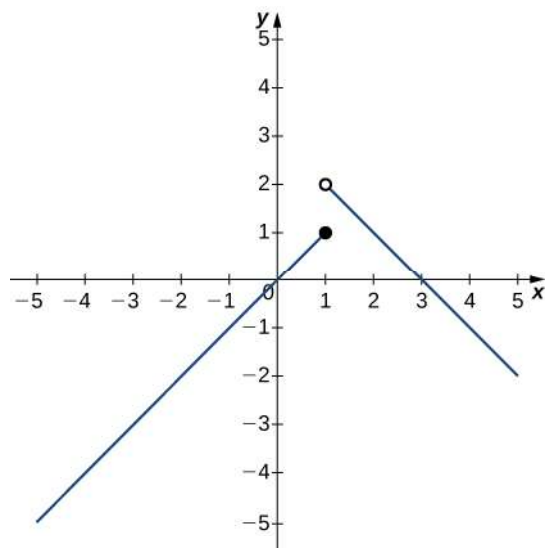
46. $\lim_{x \rightarrow 10} f(x) = 0$

47. $\lim_{x \rightarrow -2^+} f(x) = 3$

48. $\lim_{x \rightarrow -8} f(x) = f(-8)$

49. $\lim_{x \rightarrow 6} f(x) = 5$

In the following exercises, use the following graph of the function $y = f(x)$ to find the values, if possible. Estimate when necessary.



50. $\lim_{x \rightarrow 1^-} f(x)$

51. $\lim_{x \rightarrow 1^+} f(x)$

52. $\lim_{x \rightarrow 1} f(x)$

53. $\lim_{x \rightarrow 2} f(x)$

54. $f(1)$

42. $\lim_{t \rightarrow 0^+} \frac{\cos t}{t}$

t	$\frac{\cos t}{t}$
0.1	a.
0.01	b.
0.001	c.
0.0001	d.

43. $\lim_{x \rightarrow 2} \frac{1 - \frac{2}{x}}{x^2 - 4}$

x	$\frac{1 - \frac{2}{x}}{x^2 - 4}$	x	$\frac{1 - \frac{2}{x}}{x^2 - 4}$
1.9	a.	2.1	e.
1.99	b.	2.01	f.
1.999	c.	2.001	g.
1.9999	d.	2.0001	h.

[T] In the following exercises, set up a table of values and round to eight significant digits. Based on the table of values, make a guess about what the limit is. Then, use a calculator to graph the function and determine the limit. Was the conjecture correct? If not, why does the method of tables fail?

44. $\lim_{\theta \rightarrow 0} \sin\left(\frac{\pi}{\theta}\right)$

θ	$\sin\left(\frac{\pi}{\theta}\right)$	θ	$\sin\left(\frac{\pi}{\theta}\right)$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

45. $\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} \cos\left(\frac{\pi}{\alpha}\right)$

α	$\frac{1}{\alpha} \cos\left(\frac{\pi}{\alpha}\right)$
0.1	a.
0.01	b.
0.001	c.
0.0001	d.

In the following exercises, use the given values to set up a table to evaluate the limits. Round your solutions to eight decimal places.

35. [T] $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}; \pm 0.1, \pm 0.01, \pm 0.001, \pm 0.0001$

x	$\frac{\sin 2x}{x}$	x	$\frac{\sin 2x}{x}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

36. [T] $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} \pm 0.1, \pm 0.01, \pm 0.001, \pm 0.0001$

x	$\frac{\sin 3x}{x}$	x	$\frac{\sin 3x}{x}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

37. Use the preceding two exercises to conjecture (guess) the value of the following limit: $\lim_{x \rightarrow 0} \frac{\sin ax}{x}$ for a , a positive real value.

[T] In the following exercises, set up a table of values to find the indicated limit. Round to eight digits.

38. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 + x - 6}$

x	$\frac{x^2 - 4}{x^2 + x - 6}$	x	$\frac{x^2 - 4}{x^2 + x - 6}$
1.9	a.	2.1	e.
1.99	b.	2.01	f.
1.999	c.	2.001	g.
1.9999	d.	2.0001	h.

39. $\lim_{x \rightarrow 1} (1 - 2x)$

x	$1 - 2x$	x	$1 - 2x$
0.9	a.	1.1	e.
0.99	b.	1.01	f.
0.999	c.	1.001	g.
0.9999	d.	1.0001	h.

40. $\lim_{x \rightarrow 0} \frac{5}{1 - e^{1/x}}$

x	$\frac{5}{1 - e^{1/x}}$	x	$\frac{5}{1 - e^{1/x}}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

41. $\lim_{z \rightarrow 0} \frac{z - 1}{z^2(z + 3)}$

z	$\frac{z - 1}{z^2(z + 3)}$	z	$\frac{z - 1}{z^2(z + 3)}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

Thus, according to [Table 2.8](#), if an object with mass 100 kg is traveling at $0.9999c$, its mass becomes 7071 kg. Since no object can have an infinite mass, we conclude that no object can travel at or more than the speed of light.



SECTION 2.2 EXERCISES

For the following exercises, consider the function $f(x) = \frac{x^2 - 1}{|x - 1|}$.

30. [T] Complete the following table for the function. Round your solutions to four decimal places.

x	$f(x)$	x	$f(x)$
0.9	a.	1.1	e.
0.99	b.	1.01	f.
0.999	c.	1.001	g.
0.9999	d.	1.0001	h.

31. What do your results in the preceding exercise indicate about the two-sided limit $\lim_{x \rightarrow 1} f(x)$? Explain your response.

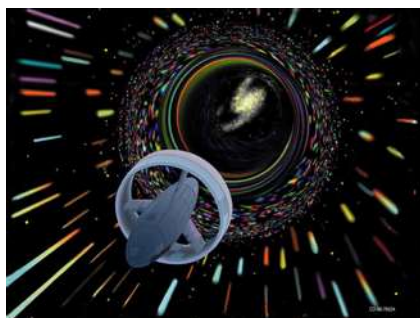
For the following exercises, consider the function $f(x) = (1 + x)^{1/x}$.

32. [T] Make a table showing the values of f for $x = -0.01, -0.001, -0.0001, -0.00001$ and for $x = 0.01, 0.001, 0.0001, 0.00001$. Round your solutions to five decimal places.

x	$f(x)$	x	$f(x)$
-0.01	a.	0.01	e.
-0.001	b.	0.001	f.
-0.0001	c.	0.0001	g.
-0.00001	d.	0.00001	h.

33. What does the table of values in the preceding exercise indicate about the function $f(x) = (1 + x)^{1/x}$?

34. To which mathematical constant does the limit in the preceding exercise appear to be getting closer?

EXAMPLE 2.12**Chapter Opener: Einstein's Equation****Figure 2.22** (credit: NASA)

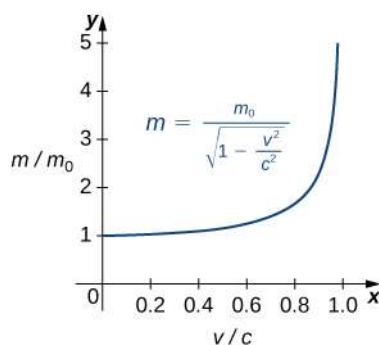
In the chapter opener we mentioned briefly how Albert Einstein showed that a limit exists to how fast any object can travel. Given Einstein's equation for the mass of a moving object, what is the value of this bound?

✓ Solution

Our starting point is Einstein's equation for the mass of a moving object,

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}},$$

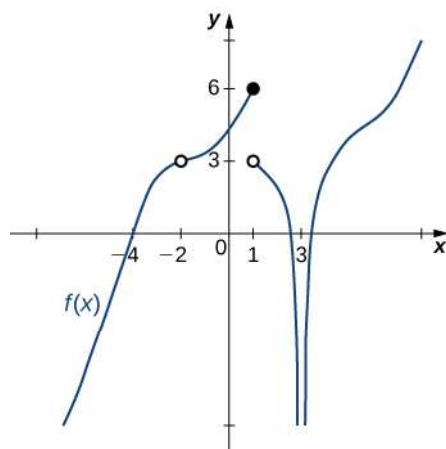
where m_0 is the object's mass at rest, v is its speed, and c is the speed of light. To see how the mass changes at high speeds, we can graph the ratio of masses m/m_0 as a function of the ratio of speeds, v/c (Figure 2.23).

**Figure 2.23** This graph shows the ratio of masses as a function of the ratio of speeds in Einstein's equation for the mass of a moving object.

We can see that as the ratio of speeds approaches 1—that is, as the speed of the object approaches the speed of light—the ratio of masses increases without bound. In other words, the function has a vertical asymptote at $v/c = 1$. We can try a few values of this ratio to test this idea.

$\frac{v}{c}$	$\sqrt{1 - \frac{v^2}{c^2}}$	$\frac{m}{m_0}$
0.99	0.1411	7.089
0.999	0.0447	22.37
0.9999	0.0141	70.71

Table 2.8 Ratio of Masses and Speeds for a Moving Object

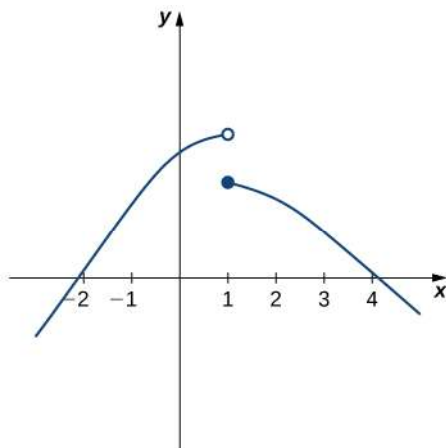
Figure 2.21 The graph shows $f(x)$.

✓ **Solution**

Using [Infinite Limits from Positive Integers](#) and the graph for reference, we arrive at the following values:

- $\lim_{x \rightarrow -4^-} f(x) = 0$; $\lim_{x \rightarrow -4^+} f(x) = 0$; $\lim_{x \rightarrow -4} f(x) = 0$; $f(-4) = 0$
- $\lim_{x \rightarrow -2^-} f(x) = 3$; $\lim_{x \rightarrow -2^+} f(x) = 3$; $\lim_{x \rightarrow -2} f(x) = 3$; $f(-2)$ is undefined
- $\lim_{x \rightarrow 1^-} f(x) = 6$; $\lim_{x \rightarrow 1^+} f(x) = 3$; $\lim_{x \rightarrow 1} f(x)$ DNE; $f(1) = 6$
- $\lim_{x \rightarrow 3^-} f(x) = -\infty$; $\lim_{x \rightarrow 3^+} f(x) = -\infty$; $\lim_{x \rightarrow 3} f(x) = -\infty$; $f(3)$ is undefined

- ✓ 2.10 Evaluate $\lim_{x \rightarrow 1} f(x)$ for $f(x)$ shown here:



$$\lim_{x \rightarrow a^-} f(x) = +\infty \text{ or } -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = +\infty \text{ or } -\infty$$

or

$$\lim_{x \rightarrow a} f(x) = +\infty \text{ or } -\infty$$

EXAMPLE 2.10

Finding a Vertical Asymptote

Evaluate each of the following limits using [Infinite Limits from Positive Integers](#). Identify any vertical asymptotes of the function $f(x) = 1/(x+3)^4$.

- $\lim_{x \rightarrow -3^-} \frac{1}{(x+3)^4}$
- $\lim_{x \rightarrow -3^+} \frac{1}{(x+3)^4}$
- $\lim_{x \rightarrow -3} \frac{1}{(x+3)^4}$

Solution

We can use [Infinite Limits from Positive Integers](#) directly.

- $\lim_{x \rightarrow -3^-} \frac{1}{(x+3)^4} = +\infty$
- $\lim_{x \rightarrow -3^+} \frac{1}{(x+3)^4} = +\infty$
- $\lim_{x \rightarrow -3} \frac{1}{(x+3)^4} = +\infty$

The function $f(x) = 1/(x+3)^4$ has a vertical asymptote of $x = -3$.

-  2.9 Evaluate each of the following limits. Identify any vertical asymptotes of the function $f(x) = \frac{1}{(x-2)^3}$.

- $\lim_{x \rightarrow 2^-} \frac{1}{(x-2)^3}$
- $\lim_{x \rightarrow 2^+} \frac{1}{(x-2)^3}$
- $\lim_{x \rightarrow 2} \frac{1}{(x-2)^3}$

In the next example we put our knowledge of various types of limits to use to analyze the behavior of a function at several different points.

EXAMPLE 2.11

Behavior of a Function at Different Points

Use the graph of $f(x)$ in [Figure 2.21](#) to determine each of the following values:

- $\lim_{x \rightarrow -4^-} f(x)$; $\lim_{x \rightarrow -4^+} f(x)$; $\lim_{x \rightarrow -4} f(x)$; $f(-4)$
- $\lim_{x \rightarrow -2^-} f(x)$; $\lim_{x \rightarrow -2^+} f(x)$; $\lim_{x \rightarrow -2} f(x)$; $f(-2)$
- $\lim_{x \rightarrow 1^-} f(x)$; $\lim_{x \rightarrow 1^+} f(x)$; $\lim_{x \rightarrow 1} f(x)$; $f(1)$
- $\lim_{x \rightarrow 3^-} f(x)$; $\lim_{x \rightarrow 3^+} f(x)$; $\lim_{x \rightarrow 3} f(x)$; $f(3)$

- a. $\lim_{x \rightarrow 0^-} \frac{1}{x^2}$
 b. $\lim_{x \rightarrow 0^+} \frac{1}{x^2}$
 c. $\lim_{x \rightarrow 0} \frac{1}{x^2}$

It is useful to point out that functions of the form $f(x) = 1/(x - a)^n$, where n is a positive integer, have infinite limits as x approaches a from either the left or right (Figure 2.20). These limits are summarized in [Infinite Limits from Positive Integers](#).

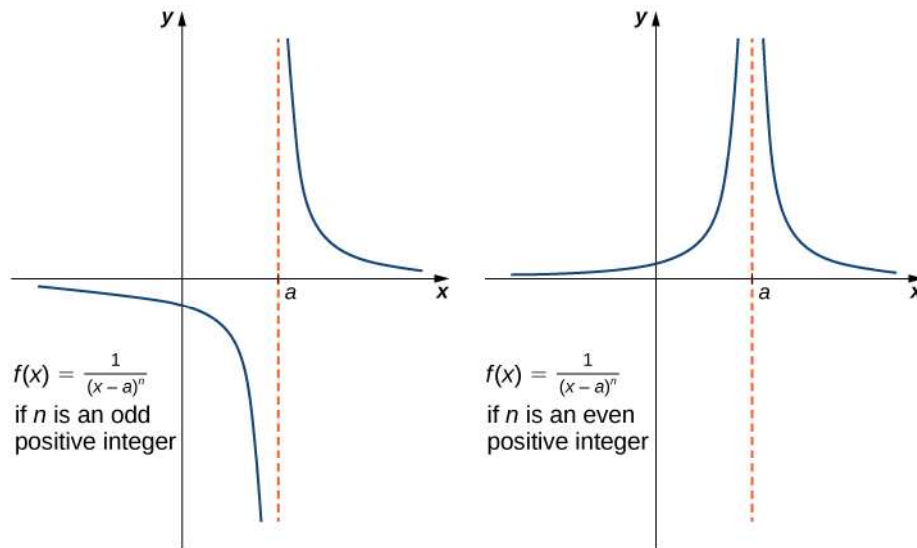


Figure 2.20 The function $f(x) = 1/(x - a)^n$ has infinite limits at a .

Theorem 2.3

Infinite Limits from Positive Integers

If n is a positive even integer, then

$$\lim_{x \rightarrow a} \frac{1}{(x - a)^n} = +\infty.$$

If n is a positive odd integer, then

$$\lim_{x \rightarrow a^+} \frac{1}{(x - a)^n} = +\infty$$

and

$$\lim_{x \rightarrow a^-} \frac{1}{(x - a)^n} = -\infty.$$

We should also point out that in the graphs of $f(x) = 1/(x - a)^n$, points on the graph having x -coordinates very near to a are very close to the vertical line $x = a$. That is, as x approaches a , the points on the graph of $f(x)$ are closer to the line $x = a$. The line $x = a$ is called a **vertical asymptote** of the graph. We formally define a vertical asymptote as follows:

Definition

Let $f(x)$ be a function. If any of the following conditions hold, then the line $x = a$ is a **vertical asymptote** of $f(x)$.

✓ **Solution**

Begin by constructing a table of functional values.

x	$\frac{1}{x}$	x	$\frac{1}{x}$
-0.1	-10	0.1	10
-0.01	-100	0.01	100
-0.001	-1000	0.001	1000
-0.0001	-10,000	0.0001	10,000
-0.00001	-100,000	0.00001	100,000
-0.000001	-1,000,000	0.000001	1,000,000

Table 2.7 Table of Functional Values for $f(x) = \frac{1}{x}$

- a. The values of $1/x$ decrease without bound as x approaches 0 from the left. We conclude that

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

- b. The values of $1/x$ increase without bound as x approaches 0 from the right. We conclude that

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty.$$

- c. Since $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$ and $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$ have different values, we conclude that

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ DNE.}$$

The graph of $f(x) = 1/x$ in [Figure 2.19](#) confirms these conclusions.

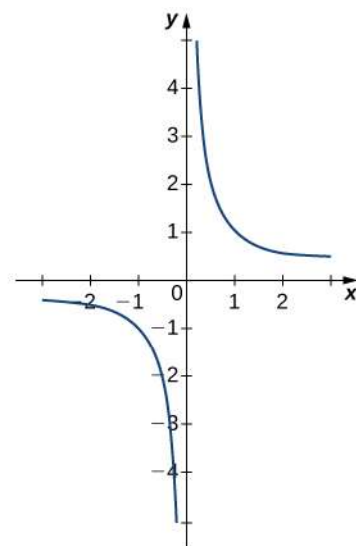


Figure 2.19 The graph of $f(x) = 1/x$ confirms that the limit as x approaches 0 does not exist.

- ✓ 2.8 Evaluate each of the following limits, if possible. Use a table of functional values and graph $f(x) = 1/x^2$ to confirm your conclusion.

$$\lim_{x \rightarrow 2} h(x) = +\infty.$$

More generally, we define **infinite limits** as follows:

Definition

We define three types of **infinite limits**.

Infinite limits from the left: Let $f(x)$ be a function defined at all values in an open interval of the form (b, a) .

- i. If the values of $f(x)$ increase without bound as the values of x (where $x < a$) approach the number a , then we say that the limit as x approaches a from the left is positive infinity and we write

$$\lim_{x \rightarrow a^-} f(x) = +\infty. \quad (2.8)$$

- ii. If the values of $f(x)$ decrease without bound as the values of x (where $x < a$) approach the number a , then we say that the limit as x approaches a from the left is negative infinity and we write

$$\lim_{x \rightarrow a^-} f(x) = -\infty. \quad (2.9)$$

Infinite limits from the right: Let $f(x)$ be a function defined at all values in an open interval of the form (a, c) .

- i. If the values of $f(x)$ increase without bound as the values of x (where $x > a$) approach the number a , then we say that the limit as x approaches a from the right is positive infinity and we write

$$\lim_{x \rightarrow a^+} f(x) = +\infty. \quad (2.10)$$

- ii. If the values of $f(x)$ decrease without bound as the values of x (where $x > a$) approach the number a , then we say that the limit as x approaches a from the right is negative infinity and we write

$$\lim_{x \rightarrow a^+} f(x) = -\infty. \quad (2.11)$$

Two-sided infinite limit: Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a .

- i. If the values of $f(x)$ increase without bound as the values of x (where $x \neq a$) approach the number a , then we say that the limit as x approaches a is positive infinity and we write

$$\lim_{x \rightarrow a} f(x) = +\infty. \quad (2.12)$$

- ii. If the values of $f(x)$ decrease without bound as the values of x (where $x \neq a$) approach the number a , then we say that the limit as x approaches a is negative infinity and we write

$$\lim_{x \rightarrow a} f(x) = -\infty. \quad (2.13)$$

It is important to understand that when we write statements such as $\lim_{x \rightarrow a} f(x) = +\infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$ we are describing the behavior of the function, as we have just defined it. We are not asserting that a limit exists. For the limit of a function $f(x)$ to exist at a , it must approach a real number L as x approaches a . That said, if, for example, $\lim_{x \rightarrow a} f(x) = +\infty$, we always write $\lim_{x \rightarrow a} f(x) = +\infty$ rather than $\lim_{x \rightarrow a} f(x)$ DNE.

EXAMPLE 2.9

Recognizing an Infinite Limit

Evaluate each of the following limits, if possible. Use a table of functional values and graph $f(x) = 1/x$ to confirm your conclusion.

- a. $\lim_{x \rightarrow 0^-} \frac{1}{x}$
- b. $\lim_{x \rightarrow 0^+} \frac{1}{x}$
- c. $\lim_{x \rightarrow 0} \frac{1}{x}$

$f(x)$ does not exist at $x = 2$. [Figure 2.18](#) shows a graph of $f(x)$ and reinforces our conclusion about these limits.

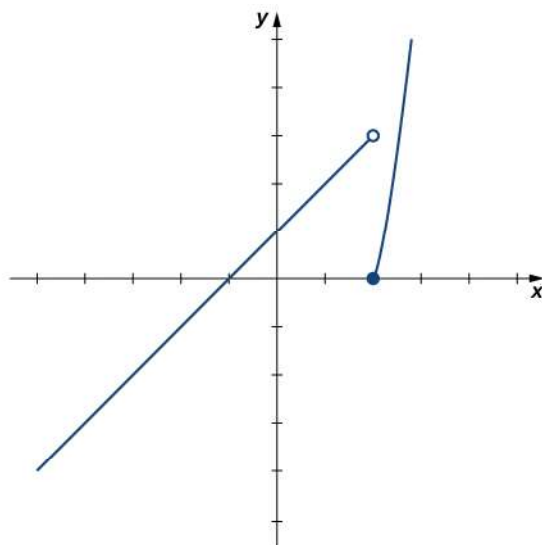


Figure 2.18 The graph of $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$ has a break at $x = 2$.

✓ 2.7 Use a table of functional values to estimate the following limits, if possible.

a. $\lim_{x \rightarrow 2^-} \frac{|x^2 - 4|}{x - 2}$

b. $\lim_{x \rightarrow 2^+} \frac{|x^2 - 4|}{x - 2}$

Let us now consider the relationship between the limit of a function at a point and the limits from the right and left at that point. It seems clear that if the limit from the right and the limit from the left have a common value, then that common value is the limit of the function at that point. Similarly, if the limit from the left and the limit from the right take on different values, the limit of the function does not exist. These conclusions are summarized in [Relating One-Sided and Two-Sided Limits](#).

Theorem 2.2

Relating One-Sided and Two-Sided Limits

Let $f(x)$ be a function defined at all values in an open interval containing a , with the possible exception of a itself, and let L be a real number. Then,

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L.$$

Infinite Limits

Evaluating the limit of a function at a point or evaluating the limit of a function from the right and left at a point helps us to characterize the behavior of a function around a given value. As we shall see, we can also describe the behavior of functions that do not have finite limits.

We now turn our attention to $h(x) = 1/(x - 2)^2$, the third and final function introduced at the beginning of this section (see [Figure 2.12\(c\)](#)). From its graph we see that as the values of x approach 2, the values of $h(x) = 1/(x - 2)^2$ become larger and larger and, in fact, become infinite. Mathematically, we say that the limit of $h(x)$ as x approaches 2 is positive infinity. Symbolically, we express this idea as

Similarly, as x approaches 2 from the right (or *from the positive side*), $g(x)$ approaches 1. Symbolically, we express this idea as

$$\lim_{x \rightarrow 2^+} g(x) = 1.$$

We can now present an informal definition of one-sided limits.

Definition

We define two types of **one-sided limits**.

Limit from the left: Let $f(x)$ be a function defined at all values in an open interval of the form (c, a) , and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x < a$) approach the number a , then we say that L is the limit of $f(x)$ as x approaches a from the left. Symbolically, we express this idea as

$$\lim_{x \rightarrow a^-} f(x) = L. \quad (2.6)$$

Limit from the right: Let $f(x)$ be a function defined at all values in an open interval of the form (a, c) , and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x > a$) approach the number a , then we say that L is the limit of $f(x)$ as x approaches a from the right. Symbolically, we express this idea as

$$\lim_{x \rightarrow a^+} f(x) = L. \quad (2.7)$$

EXAMPLE 2.8

Evaluating One-Sided Limits

For the function $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits.

- $\lim_{x \rightarrow 2^-} f(x)$
- $\lim_{x \rightarrow 2^+} f(x)$

Solution

We can use tables of functional values again [Table 2.6](#). Observe that for values of x less than 2, we use $f(x) = x + 1$ and for values of x greater than 2, we use $f(x) = x^2 - 4$.

x	$f(x) = x + 1$	x	$f(x) = x^2 - 4$
1.9	2.9	2.1	0.41
1.99	2.99	2.01	0.0401
1.999	2.999	2.001	0.004001
1.9999	2.9999	2.0001	0.00040001
1.99999	2.99999	2.00001	0.0000400001

Table 2.6 Table of Functional Values for

$$f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$$

Based on this table, we can conclude that a. $\lim_{x \rightarrow 2^-} f(x) = 3$ and b. $\lim_{x \rightarrow 2^+} f(x) = 0$. Therefore, the (two-sided) limit of

x	$\sin\left(\frac{1}{x}\right)$	x	$\sin\left(\frac{1}{x}\right)$
-0.00001	-0.035748797987	0.00001	0.035748797987
-0.000001	0.349993504187	0.000001	-0.349993504187

Table 2.5 Table of Functional Values for $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$

After examining the table of functional values, we can see that the y -values do not seem to approach any one single value. It appears the limit does not exist. Before drawing this conclusion, let's take a more systematic approach. Take the following sequence of x -values approaching 0:

$$\frac{2}{\pi}, \frac{2}{3\pi}, \frac{2}{5\pi}, \frac{2}{7\pi}, \frac{2}{9\pi}, \frac{2}{11\pi}, \dots$$

The corresponding y -values are

$$1, -1, 1, -1, 1, -1, \dots$$

At this point we can indeed conclude that $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist. (Mathematicians frequently abbreviate “does not exist” as DNE. Thus, we would write $\lim_{x \rightarrow 0} \sin(1/x)$ DNE.) The graph of $f(x) = \sin(1/x)$ is shown in [Figure 2.17](#) and it gives a clearer picture of the behavior of $\sin(1/x)$ as x approaches 0. You can see that $\sin(1/x)$ oscillates ever more wildly between -1 and 1 as x approaches 0.

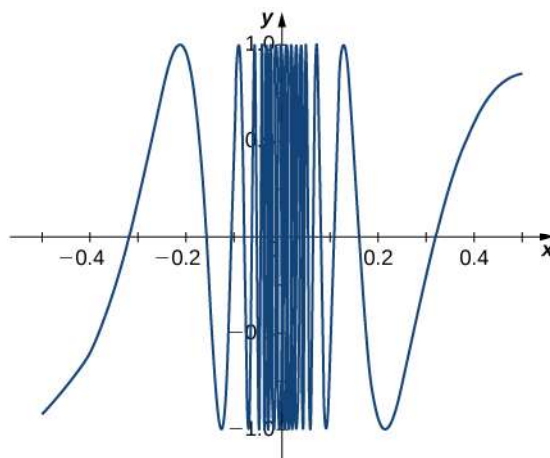


Figure 2.17 The graph of $f(x) = \sin(1/x)$ oscillates rapidly between -1 and 1 as x approaches 0.

- ✓ 2.6 Use a table of functional values to evaluate $\lim_{x \rightarrow 2} \frac{|x^2 - 4|}{x - 2}$, if possible.

One-Sided Limits

Sometimes indicating that the limit of a function fails to exist at a point does not provide us with enough information about the behavior of the function at that particular point. To see this, we now revisit the function $g(x) = |x - 2| / (x - 2)$ introduced at the beginning of the section (see [Figure 2.12\(b\)](#)). As we pick values of x close to 2, $g(x)$ does not approach a single value, so the limit as x approaches 2 does not exist—that is, $\lim_{x \rightarrow 2} g(x)$ DNE. However, this statement alone does not give us a complete picture of the behavior of the function around the x -value 2. To provide a more accurate description, we introduce the idea of a **one-sided limit**. For all values to the left of 2 (or *the negative side of 2*), $g(x) = -1$. Thus, as x approaches 2 from the left, $g(x)$ approaches -1 . Mathematically, we say that the limit as x approaches 2 from the left is -1 . Symbolically, we express this idea as

$$\lim_{x \rightarrow 2^-} g(x) = -1.$$

Theorem 2.1**Two Important Limits**

Let a be a real number and c be a constant.

i. $\lim_{x \rightarrow a} x = a$ (2.4)

ii. $\lim_{x \rightarrow a} c = c$ (2.5)

We can make the following observations about these two limits.

- For the first limit, observe that as x approaches a , so does $f(x)$, because $f(x) = x$. Consequently, $\lim_{x \rightarrow a} x = a$.
- For the second limit, consider [Table 2.4](#).

x	$f(x) = c$	x	$f(x) = c$
$a - 0.1$	c	$a + 0.1$	c
$a - 0.01$	c	$a + 0.01$	c
$a - 0.001$	c	$a + 0.001$	c
$a - 0.0001$	c	$a + 0.0001$	c

Table 2.4 Table of Functional Values for $\lim_{x \rightarrow a} c = c$

Observe that for all values of x (regardless of whether they are approaching a), the values $f(x)$ remain constant at c . We have no choice but to conclude $\lim_{x \rightarrow a} c = c$.

The Existence of a Limit

As we consider the limit in the next example, keep in mind that for the limit of a function to exist at a point, the functional values must approach a single real-number value at that point. If the functional values do not approach a single value, then the limit does not exist.

EXAMPLE 2.7**Evaluating a Limit That Fails to Exist**

Evaluate $\lim_{x \rightarrow 0} \sin(1/x)$ using a table of values.

✓ Solution

[Table 2.5](#) lists values for the function $\sin(1/x)$ for the given values of x .

x	$\sin\left(\frac{1}{x}\right)$	x	$\sin\left(\frac{1}{x}\right)$
-0.1	0.544021110889	0.1	-0.544021110889
-0.01	0.50636564111	0.01	-0.50636564111
-0.001	-0.8268795405312	0.001	0.826879540532
-0.0001	0.305614388888	0.0001	-0.305614388888

Table 2.5 Table of Functional Values for $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$

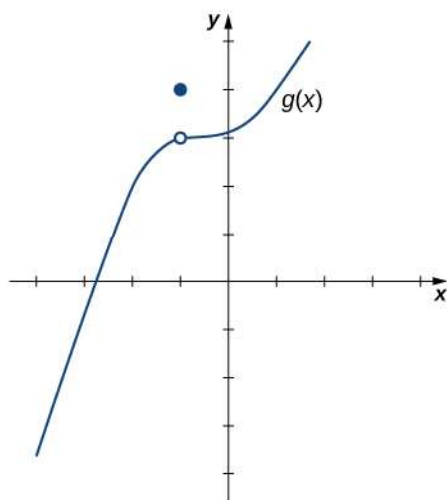


Figure 2.15 The graph of $g(x)$ includes one value not on a smooth curve.

✓ **Solution**

Despite the fact that $g(-1) = 4$, as the x -values approach -1 from either side, the $g(x)$ values approach 3. Therefore, $\lim_{x \rightarrow -1} g(x) = 3$. Note that we can determine this limit without even knowing the algebraic expression of the function.

Based on [Example 2.6](#), we make the following observation: It is possible for the limit of a function to exist at a point, and for the function to be defined at this point, but the limit of the function and the value of the function at the point may be different.

- ✓ 2.5 Use the graph of $h(x)$ in [Figure 2.16](#) to evaluate $\lim_{x \rightarrow 2} h(x)$, if possible.

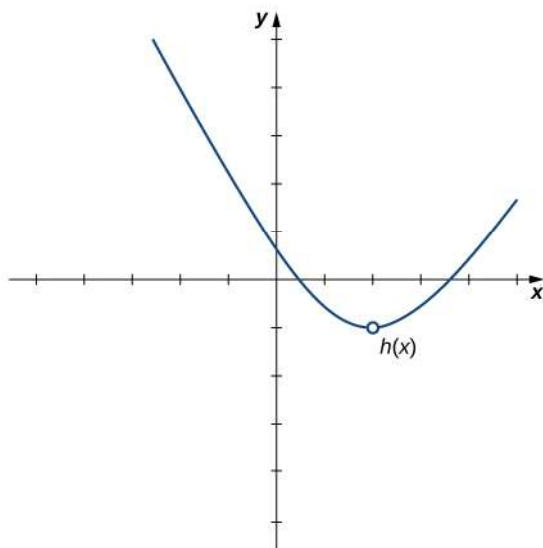


Figure 2.16

Looking at a table of functional values or looking at the graph of a function provides us with useful insight into the value of the limit of a function at a given point. However, these techniques rely too much on guesswork. We eventually need to develop alternative methods of evaluating limits. These new methods are more algebraic in nature and we explore them in the next section; however, at this point we introduce two special limits that are foundational to the techniques to come.

x	$\frac{\sqrt{x}-2}{x-4}$	x	$\frac{\sqrt{x}-2}{x-4}$
3.9999	0.250001563	4.0001	0.249998438
3.99999	0.250000016	4.00001	0.249999984

Table 2.3 Table of Functional Values for $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$

After inspecting this table, we see that the functional values less than 4 appear to be decreasing toward 0.25 whereas the functional values greater than 4 appear to be increasing toward 0.25. We conclude that $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4} = 0.25$. We confirm this estimate using the graph of $f(x) = \frac{\sqrt{x}-2}{x-4}$ shown in [Figure 2.14](#).

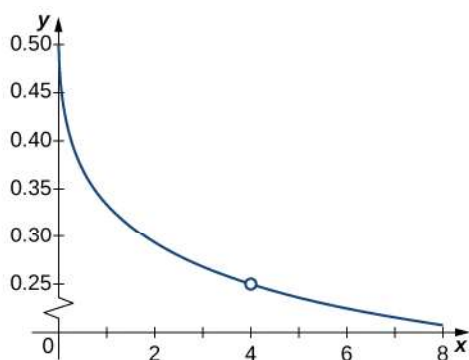


Figure 2.14 The graph of $f(x) = \frac{\sqrt{x}-2}{x-4}$ confirms the estimate from [Table 2.3](#).

- ✓ 2.4 Estimate $\lim_{x \rightarrow 1} \frac{\frac{1}{x}-1}{x-1}$ using a table of functional values. Use a graph to confirm your estimate.

At this point, we see from [Example 2.4](#) and [Example 2.5](#) that it may be just as easy, if not easier, to estimate a limit of a function by inspecting its graph as it is to estimate the limit by using a table of functional values. In [Example 2.6](#), we evaluate a limit exclusively by looking at a graph rather than by using a table of functional values.

EXAMPLE 2.6

Evaluating a Limit Using a Graph

For $g(x)$ shown in [Figure 2.15](#), evaluate $\lim_{x \rightarrow -1} g(x)$.

x	$\frac{\sin x}{x}$	x	$\frac{\sin x}{x}$
-0.1	0.998334166468	0.1	0.998334166468
-0.01	0.999983333417	0.01	0.999983333417
-0.001	0.999999833333	0.001	0.999999833333
-0.0001	0.999999998333	0.0001	0.999999998333

Table 2.2 Table of Functional Values for $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

Note: The values in this table were obtained using a calculator and using all the places given in the calculator output.

As we read down each $\frac{(\sin x)}{x}$ column, we see that the values in each column appear to be approaching one. Thus, it is fairly reasonable to conclude that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. A calculator or computer-generated graph of $f(x) = \frac{(\sin x)}{x}$ would be similar to that shown in [Figure 2.13](#), and it confirms our estimate.

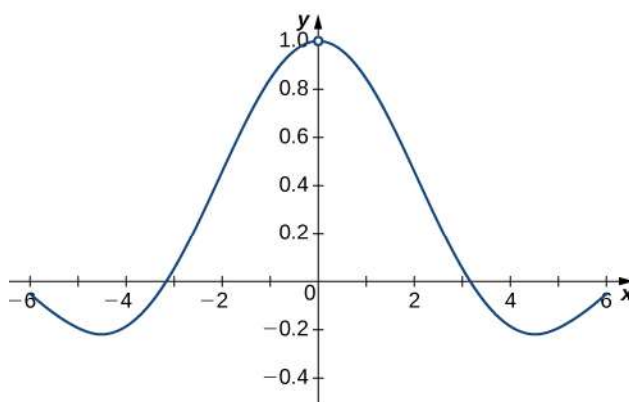


Figure 2.13 The graph of $f(x) = (\sin x)/x$ confirms the estimate from [Table 2.2](#).

EXAMPLE 2.5

Evaluating a Limit Using a Table of Functional Values 2

Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$ using a table of functional values.

Solution

As before, we use a table—in this case, [Table 2.3](#)—to list the values of the function for the given values of x .

x	$\frac{\sqrt{x}-2}{x-4}$	x	$\frac{\sqrt{x}-2}{x-4}$
3.9	0.251582341869	4.1	0.248456731317
3.99	0.25015644562	4.01	0.24984394501
3.999	0.250015627	4.001	0.249984377

Table 2.3 Table of Functional Values for $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$

Problem-Solving Strategy

Evaluating a Limit Using a Table of Functional Values

1. To evaluate $\lim_{x \rightarrow a} f(x)$, we begin by completing a table of functional values. We should choose two sets of x -values—one set of values approaching a and less than a , and another set of values approaching a and greater than a . [Table 2.1](#) demonstrates what your tables might look like.

x	$f(x)$	x	$f(x)$
$a - 0.1$	$f(a - 0.1)$	$a + 0.1$	$f(a + 0.1)$
$a - 0.01$	$f(a - 0.01)$	$a + 0.01$	$f(a + 0.01)$
$a - 0.001$	$f(a - 0.001)$	$a + 0.001$	$f(a + 0.001)$
$a - 0.0001$	$f(a - 0.0001)$	$a + 0.0001$	$f(a + 0.0001)$
Use additional values as necessary.		Use additional values as necessary.	

Table 2.1 Table of Functional Values for $\lim_{x \rightarrow a} f(x)$

2. Next, let's look at the values in each of the $f(x)$ columns and determine whether the values seem to be approaching a single value as we move down each column. In our columns, we look at the sequence $f(a - 0.1)$, $f(a - 0.01)$, $f(a - 0.001)$, $f(a - 0.0001)$, and so on, and $f(a + 0.1)$, $f(a + 0.01)$, $f(a + 0.001)$, $f(a + 0.0001)$, and so on. (Note: Although we have chosen the x -values $a \pm 0.1$, $a \pm 0.01$, $a \pm 0.001$, $a \pm 0.0001$, and so forth, and these values will probably work nearly every time, on very rare occasions we may need to modify our choices.)
3. If both columns approach a common y -value L , we state $\lim_{x \rightarrow a} f(x) = L$. We can use the following strategy to confirm the result obtained from the table or as an alternative method for estimating a limit.
4. Using a graphing calculator or computer software that allows us to graph functions, we can plot the function $f(x)$, making sure the functional values of $f(x)$ for x -values near a are in our window. We can use the trace feature to move along the graph of the function and watch the y -value readout as the x -values approach a . If the y -values approach L as our x -values approach a from both directions, then $\lim_{x \rightarrow a} f(x) = L$. We may need to zoom in on our graph and repeat this process several times.

We apply this Problem-Solving Strategy to compute a limit in [Example 2.4](#).

EXAMPLE 2.4

Evaluating a Limit Using a Table of Functional Values 1

Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using a table of functional values.

Solution

We have calculated the values of $f(x) = (\sin x)/x$ for the values of x listed in [Table 2.2](#).

circles. Yet, the formal definition of a limit—as we know and understand it today—did not appear until the late 19th century. We therefore begin our quest to understand limits, as our mathematical ancestors did, by using an intuitive approach. At the end of this chapter, armed with a conceptual understanding of limits, we examine the formal definition of a limit.

We begin our exploration of limits by taking a look at the graphs of the functions

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad g(x) = \frac{|x - 2|}{x - 2}, \quad \text{and} \quad h(x) = \frac{1}{(x - 2)^2},$$

which are shown in [Figure 2.12](#). In particular, let's focus our attention on the behavior of each graph at and around $x = 2$.

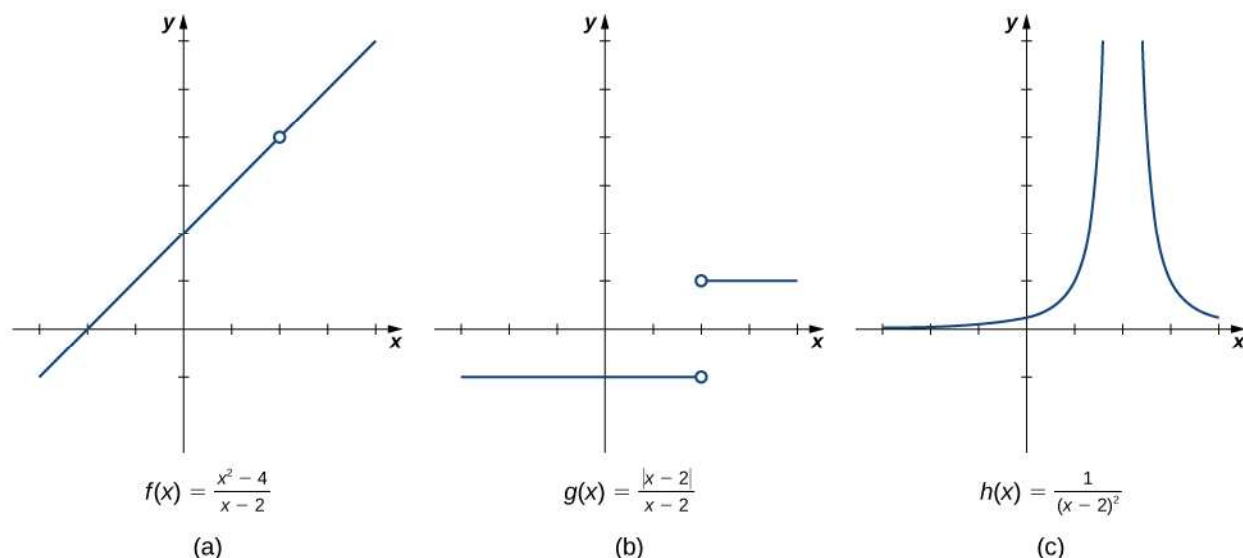


Figure 2.12 These graphs show the behavior of three different functions around $x = 2$.

Each of the three functions is undefined at $x = 2$, but if we make this statement and no other, we give a very incomplete picture of how each function behaves in the vicinity of $x = 2$. To express the behavior of each graph in the vicinity of 2 more completely, we need to introduce the concept of a limit.

Intuitive Definition of a Limit

Let's first take a closer look at how the function $f(x) = (x^2 - 4)/(x - 2)$ behaves around $x = 2$ in [Figure 2.12](#). As the values of x approach 2 from either side of 2, the values of $y = f(x)$ approach 4. Mathematically, we say that the limit of $f(x)$ as x approaches 2 is 4. Symbolically, we express this limit as

$$\lim_{x \rightarrow 2} f(x) = 4.$$

From this very brief informal look at one limit, let's start to develop an **intuitive definition of the limit**. We can think of the limit of a function at a number a as being the one real number L that the functional values approach as the x -values approach a , provided such a real number L exists. Stated more carefully, we have the following definition:

Definition

Let $f(x)$ be a function defined at all values in an open interval containing a , with the possible exception of a itself, and let L be a real number. If *all* values of the function $f(x)$ approach the real number L as the values of x ($x \neq a$) approach the number a , then we say that the limit of $f(x)$ as x approaches a is L . (More succinct, as x gets closer to a , $f(x)$ gets closer and stays close to L .) Symbolically, we express this idea as

$$\lim_{x \rightarrow a} f(x) = L. \quad (2.3)$$

We can estimate limits by constructing tables of functional values and by looking at their graphs. This process is described in the following Problem-Solving Strategy.

For the following exercises, consider an athlete running a 40-m dash. The position of the athlete is given by $d(t) = \frac{t^3}{6} + 4t$, where d is the position in meters and t is the time elapsed, measured in seconds.

22. [T] Compute the average velocity of the runner over the given time intervals.
- $[1.95, 2.05]$
 - $[1.995, 2.005]$
 - $[1.9995, 2.0005]$
 - $[2, 2.00001]$
23. Use the preceding exercise to guess the instantaneous velocity of the runner at $t = 2$ sec.

For the following exercises, consider the function $f(x) = |x|$.

24. Sketch the graph of f over the interval $[-1, 2]$ and shade the region above the x -axis.
25. Use the preceding exercise to find the approximate value of the area between the x -axis and the graph of f over the interval $[-1, 2]$ using rectangles. For the rectangles, use the square units, and approximate both above and below the lines. Use geometry to find the exact answer.

For the following exercises, consider the function $f(x) = \sqrt{1 - x^2}$. (Hint: This is the upper half of a circle of radius 1 positioned at $(0, 0)$.)

26. Sketch the graph of f over the interval $[-1, 1]$.
27. Use the preceding exercise to find the approximate area between the x -axis and the graph of f over the interval $[-1, 1]$ using rectangles. For the rectangles, use squares 0.4 by 0.4 units, and approximate both above and below the lines. Use geometry to find the exact answer.

For the following exercises, consider the function $f(x) = -x^2 + 1$.

28. Sketch the graph of f over the interval $[-1, 1]$.
29. Approximate the area of the region between the x -axis and the graph of f over the interval $[-1, 1]$.

2.2 The Limit of a Function

Learning Objectives

- 2.2.1 Using correct notation, describe the limit of a function.
- 2.2.2 Use a table of values to estimate the limit of a function or to identify when the limit does not exist.
- 2.2.3 Use a graph to estimate the limit of a function or to identify when the limit does not exist.
- 2.2.4 Define one-sided limits and provide examples.
- 2.2.5 Explain the relationship between one-sided and two-sided limits.
- 2.2.6 Using correct notation, describe an infinite limit.
- 2.2.7 Define a vertical asymptote.

The concept of a limit or limiting process, essential to the understanding of calculus, has been around for thousands of years. In fact, early mathematicians used a limiting process to obtain better and better approximations of areas of

For the following exercises, points $P(-1, -1)$ and $Q(x, y)$ are on the graph of the function $f(x) = \frac{1}{x}$.

13. [T] Complete the following table with the appropriate values: y -coordinate of Q , the point $Q(x, y)$, and the slope of the secant line passing through points P and Q . Round your answer to eight significant digits.
14. Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the line tangent to f at $x = -1$.
15. Use the value in the preceding exercise to find an equation of the tangent line at point P .

x	y	$Q(x, y)$	m_{sec}
-1.05	a.	e.	i.
-1.01	b.	f.	j.
-1.005	c.	g.	k.
-1.001	d.	h.	l.

For the following exercises, the position function of a ball dropped from the top of a 200-meter tall building is given by $s(t) = 200 - 4.9t^2$, where position s is measured in meters and time t is measured in seconds. Round your answer to eight significant digits.

16. [T] Compute the average velocity of the ball over the given time intervals.
- $[4.99, 5]$
 - $[5, 5.01]$
 - $[4.999, 5]$
 - $[5, 5.001]$
17. Use the preceding exercise to guess the instantaneous velocity of the ball at $t = 5$ sec.

For the following exercises, consider a stone tossed into the air from ground level with an initial velocity of 15 m/sec. Its height in meters at time t seconds is $h(t) = 15t - 4.9t^2$.

18. [T] Compute the average velocity of the stone over the given time intervals.
- $[1, 1.05]$
 - $[1, 1.01]$
 - $[1, 1.005]$
 - $[1, 1.001]$
19. Use the preceding exercise to guess the instantaneous velocity of the stone at $t = 1$ sec.

For the following exercises, consider a rocket shot into the air that then returns to Earth. The height of the rocket in meters is given by $h(t) = 600 + 78.4t - 4.9t^2$, where t is measured in seconds.

20. [T] Compute the average velocity of the rocket over the given time intervals.
- $[9, 9.01]$
 - $[8.99, 9]$
 - $[9, 9.001]$
 - $[8.999, 9]$
21. Use the preceding exercise to guess the instantaneous velocity of the rocket at $t = 9$ sec.

For the following exercises, points $P(4, 2)$ and $Q(x, y)$ are on the graph of the function $f(x) = \sqrt{x}$.

7. [T] Complete the following table with the appropriate values: y -coordinate of Q , the point $Q(x, y)$, and the slope of the secant line passing through points P and Q . Round your answer to eight significant digits.
8. Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the tangent line to f at $x = 4$.
9. Use the value in the preceding exercise to find an equation of the tangent line at point P .

x	y	$Q(x, y)$	m_{sec}
4.1	a.	e.	i.
4.01	b.	f.	j.
4.001	c.	g.	k.
4.0001	d.	h.	l.

For the following exercises, points $P(1.5, 0)$ and $Q(\phi, y)$ are on the graph of the function $f(\phi) = \cos(\pi\phi)$.

10. [T] Complete the following table with the appropriate values: y -coordinate of Q , the point $Q(\phi, y)$, and the slope of the secant line passing through points P and Q . Round your answer to eight significant digits.
11. Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the tangent line to f at $\phi = 1.5$.
12. Use the value in the preceding exercise to find an equation of the tangent line at point P .

ϕ	y	$Q(\phi, y)$	m_{sec}
1.4	a.	e.	i.
1.49	b.	f.	j.
1.499	c.	g.	k.
1.4999	d.	h.	l.



SECTION 2.1 EXERCISES

For the following exercises, points $P(1, 2)$ and $Q(x, y)$ are on the graph of the function $f(x) = x^2 + 1$.

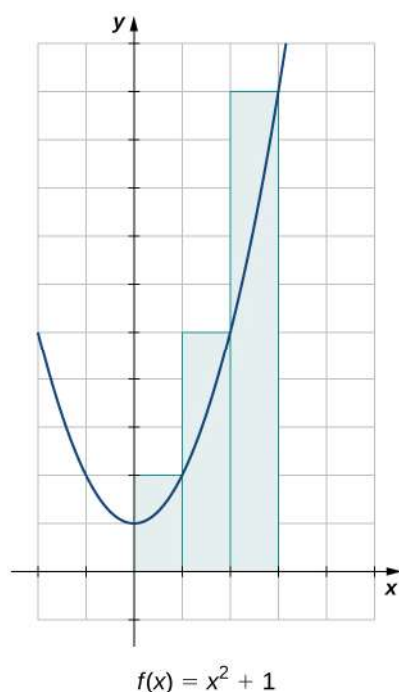
- [T]** Complete the following table with the appropriate values: y -coordinate of Q , the point $Q(x, y)$, and the slope of the secant line passing through points P and Q . Round your answer to eight significant digits.
- Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the line tangent to f at $x = 1$.
- Use the value in the preceding exercise to find an equation of the tangent line at point P . Graph $f(x)$ and the tangent line.

x	y	$Q(x, y)$	m_{sec}
1.1	a.	e.	i.
1.01	b.	f.	j.
1.001	c.	g.	k.
1.0001	d.	h.	l.

For the following exercises, points $P(1, 1)$ and $Q(x, y)$ are on the graph of the function $f(x) = x^3$.

- [T]** Complete the following table with the appropriate values: y -coordinate of Q , the point $Q(x, y)$, and the slope of the secant line passing through points P and Q . Round your answer to eight significant digits.
- Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the tangent line to f at $x = 1$.
- Use the value in the preceding exercise to find an equation of the tangent line at point P . Graph $f(x)$ and the tangent line.

x	y	$Q(x, y)$	m_{sec}
1.1	a.	e.	i.
1.01	b.	f.	j.
1.001	c.	g.	k.
1.0001	d.	h.	l.



Other Aspects of Calculus

So far, we have studied functions of one variable **only**. Such functions can be represented visually using graphs in two dimensions; however, there is no good reason to restrict our investigation to two dimensions. Suppose, for example, that instead of determining the velocity of an object moving along a coordinate axis, we want to determine the velocity of a rock fired from a catapult at a given time, or of an airplane moving in three dimensions. We might want to graph real-value functions of two variables or determine volumes of solids of the type shown in [Figure 2.11](#). These are only a few of the types of questions that can be asked and answered using **multivariable calculus**. Informally, multivariable calculus can be characterized as the study of the calculus of functions of two or more variables. However, before exploring these and other ideas, we must first lay a foundation for the study of calculus in one variable by exploring the concept of a limit.

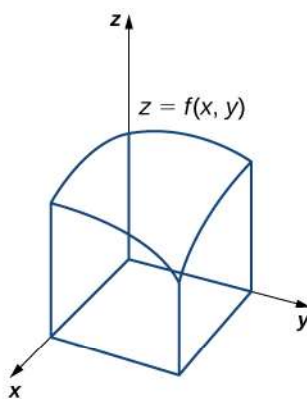


Figure 2.11 We can use multivariable calculus to find the volume between a surface defined by a function of two variables and a plane.

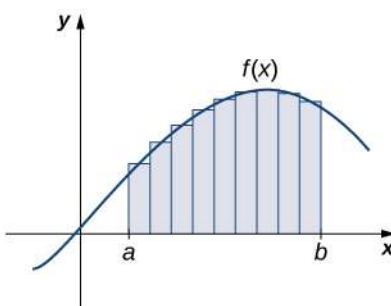


Figure 2.9 The area of the region under the curve is approximated by summing the areas of thin rectangles.

As the widths of the rectangles become smaller (approach zero), the sums of the areas of the rectangles approach the area between the graph of $f(x)$ and the x -axis over the interval $[a, b]$. Once again, we find ourselves taking a limit. Limits of this type serve as a basis for the definition of the definite integral. **Integral calculus** is the study of integrals and their applications.

EXAMPLE 2.3

Estimation Using Rectangles

Estimate the area between the x -axis and the graph of $f(x) = x^2 + 1$ over the interval $[0, 3]$ by using the three rectangles shown in [Figure 2.10](#).

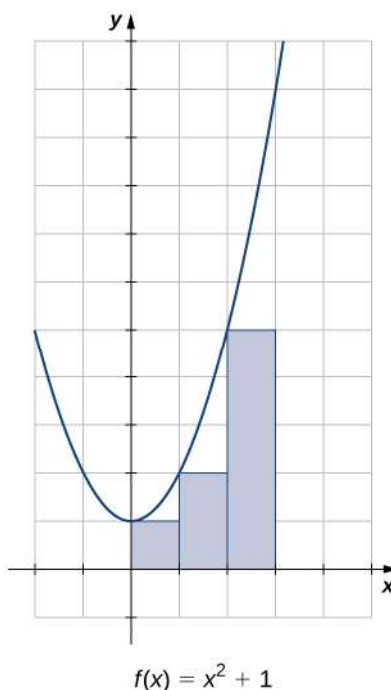


Figure 2.10 The area of the region under the curve of $f(x) = x^2 + 1$ can be estimated using rectangles.

☑ Solution

The areas of the three rectangles are 1 unit^2 , 2 unit^2 , and 5 unit^2 . Using these rectangles, our area estimate is 8 unit^2 .

-
- ☑ 2.3 Estimate the area between the x -axis and the graph of $f(x) = x^2 + 1$ over the interval $[0, 3]$ by using the three rectangles shown here:

Definition

For a position function $s(t)$, the **instantaneous velocity** at a time $t = a$ is the value that the average velocities approach on intervals of the form $[a, t]$ and $[t, a]$ as the values of t become closer to a , provided such a value exists.

[Example 2.2](#) illustrates this concept of limits and average velocity.

EXAMPLE 2.2**Finding Average Velocity**

A rock is dropped from a height of 64 ft. It is determined that its height (in feet) above ground t seconds later (for $0 \leq t \leq 2$) is given by $s(t) = -16t^2 + 64$. Find the average velocity of the rock over each of the given time intervals. Use this information to guess the instantaneous velocity of the rock at time $t = 0.5$.

- a. $[0.49, 0.5]$
- b. $[0.5, 0.51]$

✓ Solution

Substitute the data into the formula for the definition of average velocity.

$$\begin{aligned} \text{a. } v_{\text{ave}} &= \frac{s(0.5) - s(0.49)}{0.5 - 0.49} = -15.84 \\ \text{b. } v_{\text{ave}} &= \frac{s(0.51) - s(0.5)}{0.51 - 0.5} = -16.16 \end{aligned}$$

The instantaneous velocity is somewhere between -15.84 and -16.16 ft/sec. A good guess might be -16 ft/sec.

- ✓ 2.2 An object moves along a coordinate axis so that its position at time t is given by $s(t) = t^3$. Estimate its instantaneous velocity at time $t = 2$ by computing its average velocity over the time interval $[2, 2.001]$.

The Area Problem and Integral Calculus

We now turn our attention to a classic question from calculus. Many quantities in physics—for example, quantities of work—may be interpreted as the area under a curve. This leads us to ask the question: How can we find the area between the graph of a function and the x -axis over an interval ([Figure 2.8](#))?

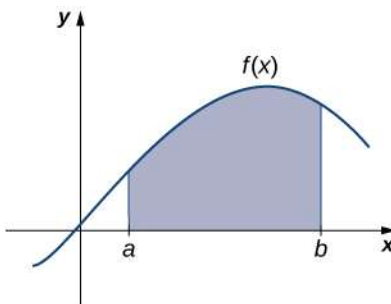


Figure 2.8 The Area Problem: How do we find the area of the shaded region?

As in the answer to our previous questions on velocity, we first try to approximate the solution. We approximate the area by dividing up the interval $[a, b]$ into smaller intervals in the shape of rectangles. The approximation of the area comes from adding up the areas of these rectangles ([Figure 2.9](#)).

$$\text{a. } m_{\text{sec}} = \frac{4-1}{2-1} = 3$$

$$\text{b. } m_{\text{sec}} = \frac{\frac{9}{4}-1}{\frac{3}{2}-1} = \frac{5}{2} = 2.5$$

The point in part b. is closer to the point $(1, 1)$, so the slope of 2.5 is closer to the slope of the tangent line. A good estimate for the slope of the tangent would be in the range of 2 to 2.5 (Figure 2.7).

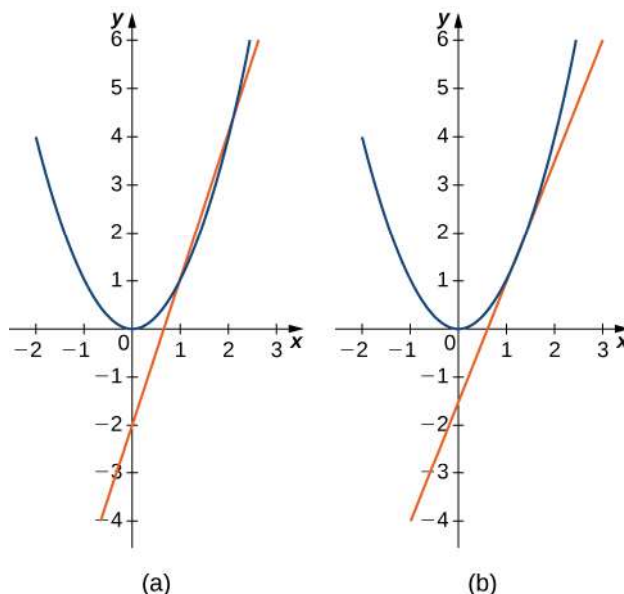


Figure 2.7 The secant lines to $f(x) = x^2$ at $(1, 1)$ through (a) $(2, 4)$ and (b) $(\frac{3}{2}, \frac{9}{4})$ provide successively closer approximations to the tangent line to $f(x) = x^2$ at $(1, 1)$.

- ✓ 2.1 Estimate the slope of the tangent line (rate of change) to $f(x) = x^2$ at $x = 1$ by finding the slope of the secant line through $(1, 1)$ and the point $(\frac{5}{4}, \frac{25}{16})$ on the graph of $f(x) = x^2$.

We continue our investigation by exploring a related question. Keeping in mind that velocity may be thought of as the rate of change of position, suppose that we have a function, $s(t)$, that gives the position of an object along a coordinate axis at any given time t . Can we use these same ideas to create a reasonable definition of the instantaneous velocity at a given time $t = a$? We start by approximating the instantaneous velocity with an average velocity. First, recall that the speed of an object traveling at a constant rate is the ratio of the distance traveled to the length of time it has traveled. We define the **average velocity** of an object over a time period to be the change in its position divided by the length of the time period.

Definition

Let $s(t)$ be the position of an object moving along a coordinate axis at time t . The **average velocity** of the object over a time interval $[a, t]$ where $a < t$ (or $[t, a]$ if $t < a$) is

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}. \quad (2.2)$$

As t is chosen closer to a , the average velocity becomes closer to the instantaneous velocity. Note that finding the average velocity of a position function over a time interval is essentially the same as finding the slope of a secant line to a function. Furthermore, to find the slope of a tangent line at a point a , we let the x -values approach a in the slope of the secant line. Similarly, to find the instantaneous velocity at time a , we let the t -values approach a in the average velocity. This process of letting x or t approach a in an expression is called taking a **limit**. Thus, we may define the **instantaneous velocity** as follows.

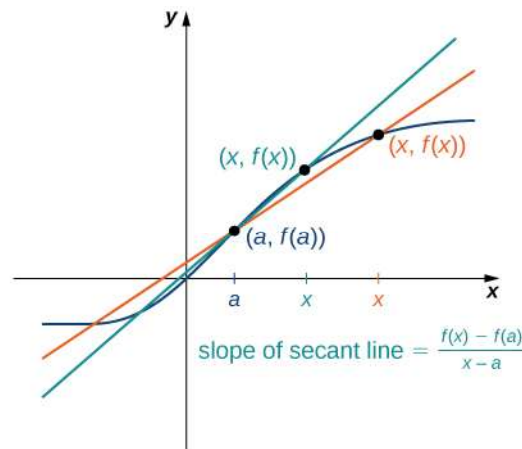


Figure 2.5 As x gets closer to a , the slope of the secant line becomes a better approximation to the rate of change of the function $f(x)$ at a .

The secant lines themselves approach a line that is called the **tangent** to the function $f(x)$ at a (Figure 2.6). The slope of the tangent line to the graph at a measures the rate of change of the function at a . This value also represents the derivative of the function $f(x)$ at a , or the rate of change of the function at a . This derivative is denoted by $f'(a)$.

Differential calculus is the field of calculus concerned with the study of derivatives and their applications.

▶ MEDIA

For an interactive demonstration of the slope of a secant line that you can manipulate yourself, visit this applet (Note: this site requires a Java browser plugin): [Math Insight \(http://www.openstax.org/l/20_mathinsight\)](http://www.openstax.org/l/20_mathinsight).

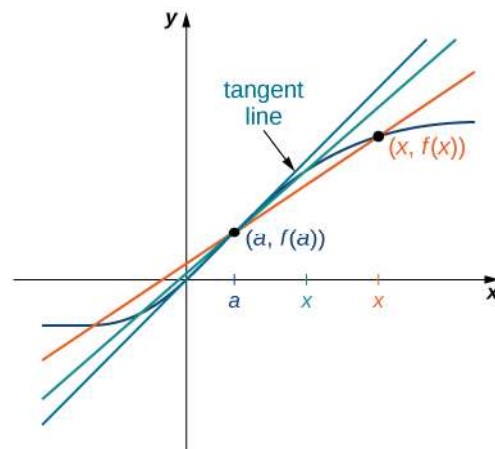


Figure 2.6 Solving the Tangent Problem: As x approaches a , the secant lines approach the tangent line.

[Example 2.1](#) illustrates how to find slopes of secant lines. These slopes estimate the slope of the tangent line or, equivalently, the rate of change of the function at the point at which the slopes are calculated.

EXAMPLE 2.1

Finding Slopes of Secant Lines

Estimate the slope of the tangent line (rate of change) to $f(x) = x^2$ at $x = 1$ by finding slopes of secant lines through $(1, 1)$ and each of the following points on the graph of $f(x) = x^2$.

- $(2, 4)$
- $(\frac{3}{2}, \frac{9}{4})$

✓ Solution

Use the formula for the slope of a secant line from the definition.

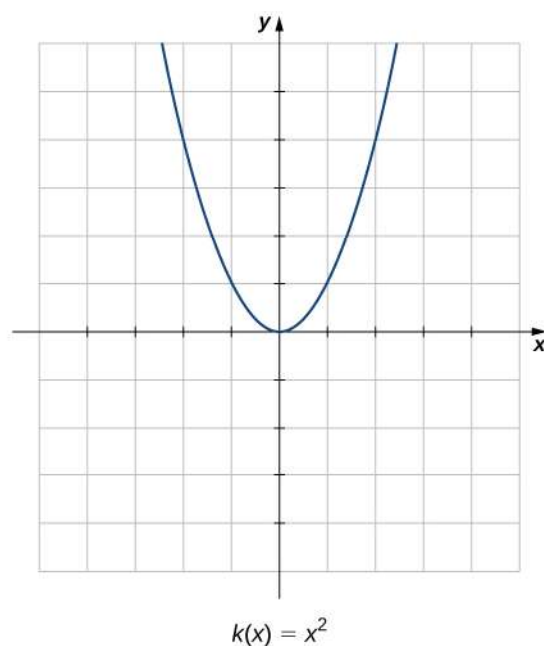


Figure 2.3 The function $k(x) = x^2$ does not have a constant rate of change.

We can approximate the rate of change of a function $f(x)$ at a point $(a, f(a))$ on its graph by taking another point $(x, f(x))$ on the graph of $f(x)$, drawing a line through the two points, and calculating the slope of the resulting line. Such a line is called a **secant** line. [Figure 2.4](#) shows a secant line to a function $f(x)$ at a point $(a, f(a))$.

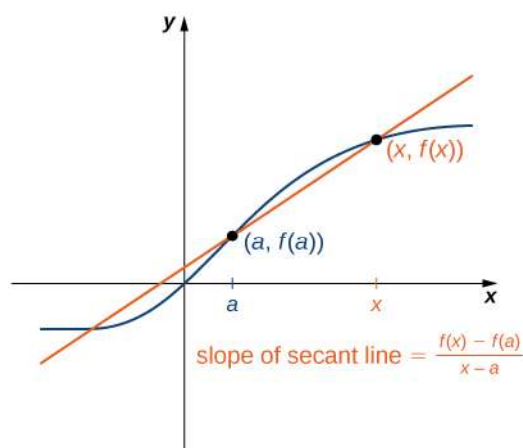


Figure 2.4 The slope of a secant line through a point $(a, f(a))$ estimates the rate of change of the function at the point $(a, f(a))$.

We formally define a secant line as follows:

Definition

The **secant** to the function $f(x)$ through the points $(a, f(a))$ and $(x, f(x))$ is the line passing through these points. Its slope is given by

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a}. \quad (2.1)$$

The accuracy of approximating the rate of change of the function with a secant line depends on how close x is to a . As we see in [Figure 2.5](#), if x is closer to a , the slope of the secant line is a better measure of the rate of change of $f(x)$ at a .

go on to describe how to find the limit of a function at a given point. Not all functions have limits at all points, and we discuss what this means and how we can tell if a function does or does not have a limit at a particular value. This chapter has been created in an informal, intuitive fashion, but this is not always enough if we need to prove a mathematical statement involving limits. The last section of this chapter presents the more precise definition of a limit and shows how to prove whether a function has a limit.

2.1 A Preview of Calculus

Learning Objectives

- 2.1.1** Describe the tangent problem and how it led to the idea of a derivative.
- 2.1.2** Explain how the idea of a limit is involved in solving the tangent problem.
- 2.1.3** Recognize a tangent to a curve at a point as the limit of secant lines.
- 2.1.4** Identify instantaneous velocity as the limit of average velocity over a small time interval.
- 2.1.5** Describe the area problem and how it was solved by the integral.
- 2.1.6** Explain how the idea of a limit is involved in solving the area problem.
- 2.1.7** Recognize how the ideas of limit, derivative, and integral led to the studies of infinite series and multivariable calculus.

As we embark on our study of calculus, we shall see how its development arose from common solutions to practical problems in areas such as engineering physics—like the space travel problem posed in the chapter opener. Two key problems led to the initial formulation of calculus: (1) the tangent problem, or how to determine the slope of a line tangent to a curve at a point; and (2) the area problem, or how to determine the area under a curve.

The Tangent Problem and Differential Calculus

Rate of change is one of the most critical concepts in calculus. We begin our investigation of rates of change by looking at the graphs of the three lines $f(x) = -2x - 3$, $g(x) = \frac{1}{2}x + 1$, and $h(x) = 2$, shown in [Figure 2.2](#).

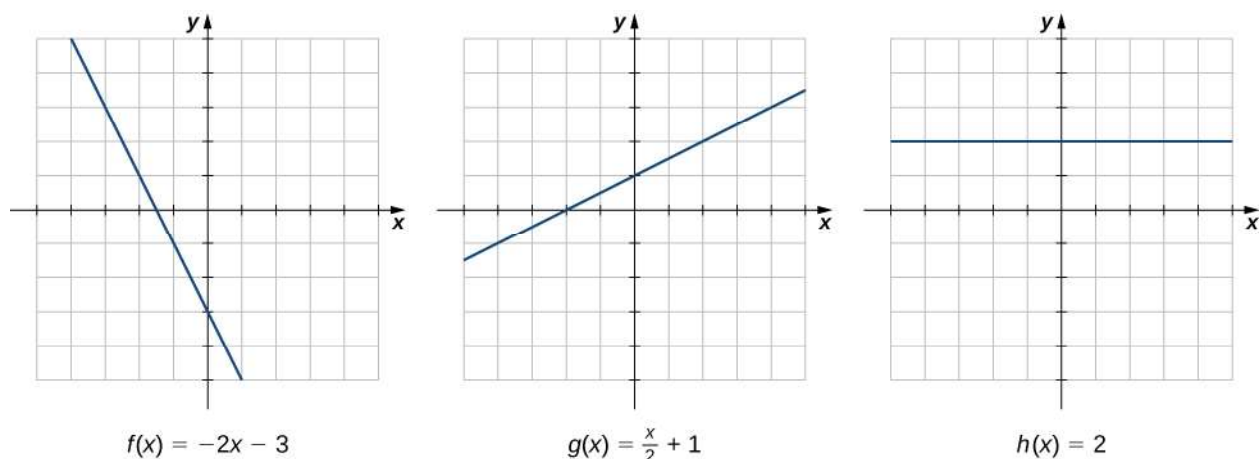


Figure 2.2 The rate of change of a linear function is constant in each of these three graphs, with the constant determined by the slope.

As we move from left to right along the graph of $f(x) = -2x - 3$, we see that the graph decreases at a constant rate. For every 1 unit we move to the right along the x -axis, the y -coordinate decreases by 2 units. This rate of change is determined by the slope (-2) of the line. Similarly, the slope of $1/2$ in the function $g(x)$ tells us that for every change in x of 1 unit there is a corresponding change in y of $1/2$ unit. The function $h(x) = 2$ has a slope of zero, indicating that the values of the function remain constant. We see that the slope of each linear function indicates the rate of change of the function.

Compare the graphs of these three functions with the graph of $k(x) = x^2$ ([Figure 2.3](#)). The graph of $k(x) = x^2$ starts from the left by decreasing rapidly, then begins to decrease more slowly and level off, and then finally begins to increase—slowly at first, followed by an increasing rate of increase as it moves toward the right. Unlike a linear function, no single number represents the rate of change for this function. We quite naturally ask: How do we measure the rate of change of a nonlinear function?

2

LIMITS



Figure 2.1 The vision of human exploration by the National Aeronautics and Space Administration (NASA) to distant parts of the universe illustrates the idea of space travel at high speeds. But, is there a limit to how fast a spacecraft can go? (credit: NASA)

Chapter Outline

- 2.1 A Preview of Calculus
- 2.2 The Limit of a Function
- 2.3 The Limit Laws
- 2.4 Continuity
- 2.5 The Precise Definition of a Limit



Introduction

Science fiction writers often imagine spaceships that can travel to far-off planets in distant galaxies. However, back in 1905, Albert Einstein showed that a limit exists to how fast any object can travel. The problem is that the faster an object moves, the more mass it attains (in the form of energy), according to the equation

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}},$$

where m_0 is the object's mass at rest, v is its speed, and c is the speed of light. What is this speed limit? (We explore this problem further in [Example 2.12](#).)

The idea of a limit is central to all of calculus. We begin this chapter by examining why limits are so important. Then, we

For the following problems, consider radioactive dating. A human skeleton is found in an archeological dig. Carbon dating is implemented to determine how old the skeleton is by using the equation $y = e^{rt}$, where y is the ratio of radiocarbon still present in the material, t is the number of years passed, and $r = -0.0001210$ is the decay rate of radiocarbon.

- 335.** If the skeleton is expected to be 2000 years old, what percentage of radiocarbon should be present?
- 336.** Find the inverse of the carbon-dating equation. What does it mean? If there is 25% radiocarbon, how old is the skeleton?

Solve the following trigonometric equations on the interval $x = [-2\pi, 2\pi]$ exactly.

322. $6\cos^2 x - 3 = 0$

323. $\sec^2 x - 2\sec x + 1 = 0$

Solve the following logarithmic equations.

324. $5^x = 16$

325. $\log_2(x + 4) = 3$

Are the following functions one-to-one over their domain of existence? Does the function have an inverse? If so, find the inverse $f^{-1}(x)$ of the function. Justify your answer.

326. $f(x) = x^2 + 2x + 1$

327. $f(x) = \frac{1}{x}$

For the following problems, determine the largest domain on which the function is one-to-one and find the inverse on that domain.

328. $f(x) = \sqrt{9 - x}$

329. $f(x) = x^2 + 3x + 4$

330. A car is racing along a circular track with diameter of 1 mi. A trainer standing in the center of the circle marks his progress every 5 sec. After 5 sec, the trainer has to turn 55° to keep up with the car. How fast is the car traveling?

For the following problems, consider a restaurant owner who wants to sell T-shirts advertising his brand. He recalls that there is a fixed cost and variable cost, although he does not remember the values. He does know that the T-shirt printing company charges \$440 for 20 shirts and \$1000 for 100 shirts.

331. a. Find the equation $C = f(x)$ that describes the total cost as a function of number of shirts and b. determine how many shirts he must sell to break even if he sells the shirts for \$10 each.

332. a. Find the inverse function $x = f^{-1}(C)$ and describe the meaning of this function. b. Determine how many shirts the owner can buy if he has \$8000 to spend.

For the following problems, consider the population of Ocean City, New Jersey, which is cyclical by season.

333. The population can be modeled by $P(t) = 82.5 - 67.5\cos[(\pi/6)t]$, where t is time in months ($t = 0$ represents January 1) and P is population (in thousands). During a year, in what intervals is the population less than 20,000? During what intervals is the population more than 140,000?

334. In reality, the overall population is most likely increasing or decreasing throughout each year. Let's reformulate the model as $P(t) = 82.5 - 67.5\cos[(\pi/6)t] + t$, where t is time in months ($t = 0$ represents January 1) and P is population (in thousands). When is the first time the population reaches 200,000?

measure 1. An angle with a degree measure of 180° has a radian measure of π rad.

- For acute angles θ , the values of the trigonometric functions are defined as ratios of two sides of a right triangle in which one of the acute angles is θ .
- For a general angle θ , let (x, y) be a point on a circle of radius r corresponding to this angle θ . The trigonometric functions can be written as ratios involving x , y , and r .
- The trigonometric functions are periodic. The sine, cosine, secant, and cosecant functions have period 2π . The tangent and cotangent functions have period π .

1.4 Inverse Functions

- For a function to have an inverse, the function must be one-to-one. Given the graph of a function, we can determine whether the function is one-to-one by using the horizontal line test.
- If a function is not one-to-one, we can restrict the domain to a smaller domain where the function is one-to-one and then define the inverse of the function on the smaller domain.
- For a function f and its inverse f^{-1} , $f(f^{-1}(x)) = x$ for all x in the domain of f^{-1} and $f^{-1}(f(x)) = x$ for all x in the domain of f .
- Since the trigonometric functions are periodic, we need to restrict their domains to define the inverse trigonometric functions.
- The graph of a function f and its inverse f^{-1} are symmetric about the line $y = x$.

1.5 Exponential and Logarithmic Functions

- The exponential function $y = b^x$ is increasing if $b > 1$ and decreasing if $0 < b < 1$. Its domain is $(-\infty, \infty)$ and its range is $(0, \infty)$.
- The logarithmic function $y = \log_b(x)$ is the inverse of $y = b^x$. Its domain is $(0, \infty)$ and its range is $(-\infty, \infty)$.
- The natural exponential function is $y = e^x$ and the natural logarithmic function is $y = \ln x = \log_e x$.
- Given an exponential function or logarithmic function in base a , we can make a change of base to convert this function to any base $b > 0, b \neq 1$. We typically convert to base e .
- The hyperbolic functions involve combinations of the exponential functions e^x and e^{-x} . As a result, the inverse hyperbolic functions involve the natural logarithm.

Review Exercises

True or False? Justify your answer with a proof or a counterexample.

- 310.** A function is always one-to-one. **311.** $f \circ g = g \circ f$, assuming f and g are functions. **312.** A relation that passes the horizontal and vertical line tests is a one-to-one function.
- 313.** A relation passing the horizontal line test is a function.

For the following problems, state the domain and range of the given functions:

$$f = x^2 + 2x - 3, \quad g = \ln(x - 5), \quad h = \frac{1}{x+4}$$

- 314.** h **315.** g **316.** $h \circ f$
- 317.** $g \circ f$

Find the degree, y-intercept, and zeros for the following polynomial functions.

- 318.** $f(x) = 2x^2 + 9x - 5$ **319.** $f(x) = x^3 + 2x^2 - 2x$

Simplify the following trigonometric expressions.

- 320.** $\frac{\tan^2 x}{\sec^2 x} + \cos^2 x$ **321.** $\cos^2 x - \sin^2 x$

table of values a table containing a list of inputs and their corresponding outputs

transcendental function a function that cannot be expressed by a combination of basic arithmetic operations

transformation of a function a shift, scaling, or reflection of a function

trigonometric functions functions of an angle defined as ratios of the lengths of the sides of a right triangle

trigonometric identity an equation involving trigonometric functions that is true for all angles θ for which the functions in the equation are defined

vertical line test given the graph of a function, every vertical line intersects the graph, at most, once

zeros of a function when a real number x is a zero of a function f , $f(x) = 0$

Key Equations

Composition of two functions	$(g \circ f)(x) = g(f(x))$
Absolute value function	$f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$
Point-slope equation of a line	$y - y_1 = m(x - x_1)$
Slope-intercept form of a line	$y = mx + b$
Standard form of a line	$ax + by = c$
Polynomial function	$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$
Generalized sine function	$f(x) = A \sin(B(x - \alpha)) + C$
Inverse functions	$f^{-1}(f(x)) = x$ for all x in D , and $f(f^{-1}(y)) = y$ for all y in R .

Key Concepts

1.1 Review of Functions

- A function is a mapping from a set of inputs to a set of outputs with exactly one output for each input.
- If no domain is stated for a function $y = f(x)$, the domain is considered to be the set of all real numbers x for which the function is defined.
- When sketching the graph of a function f , each vertical line may intersect the graph, at most, once.
- A function may have any number of zeros, but it has, at most, one y -intercept.
- To define the composition $g \circ f$, the range of f must be contained in the domain of g .
- Even functions are symmetric about the y -axis whereas odd functions are symmetric about the origin.

1.2 Basic Classes of Functions

- The power function $f(x) = x^n$ is an even function if n is even and $n \neq 0$, and it is an odd function if n is odd.
- The root function $f(x) = x^{1/n}$ has the domain $[0, \infty)$ if n is even and the domain $(-\infty, \infty)$ if n is odd. If n is odd, then $f(x) = x^{1/n}$ is an odd function.
- The domain of the rational function $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomial functions, is the set of x such that $q(x) \neq 0$.
- Functions that involve the basic operations of addition, subtraction, multiplication, division, and powers are algebraic functions. All other functions are transcendental. Trigonometric, exponential, and logarithmic functions are examples of transcendental functions.
- A polynomial function f with degree $n \geq 1$ satisfies $f(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$. The sign of the output as $x \rightarrow \infty$ depends on the sign of the leading coefficient only and on whether n is even or odd.
- Vertical and horizontal shifts, vertical and horizontal scalings, and reflections about the x - and y -axes are examples of transformations of functions.

1.3 Trigonometric Functions

- Radian measure is defined such that the angle associated with the arc of length 1 on the unit circle has radian

Chapter Review

Key Terms

absolute value function $f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

algebraic function a function involving any combination of only the basic operations of addition, subtraction, multiplication, division, powers, and roots applied to an input variable x

base the number b in the exponential function $f(x) = b^x$ and the logarithmic function $f(x) = \log_b x$

composite function given two functions f and g , a new function, denoted $g \circ f$, such that $(g \circ f)(x) = g(f(x))$

cubic function a polynomial of degree 3; that is, a function of the form $f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$

decreasing on the interval I a function decreasing on the interval I if, for all $x_1, x_2 \in I$, $f(x_1) \geq f(x_2)$ if $x_1 < x_2$

degree for a polynomial function, the value of the largest exponent of any term

dependent variable the output variable for a function

domain the set of inputs for a function

even function a function is even if $f(-x) = f(x)$ for all x in the domain of f

exponent the value x in the expression b^x

function a set of inputs, a set of outputs, and a rule for mapping each input to exactly one output

graph of a function the set of points (x, y) such that x is in the domain of f and $y = f(x)$

horizontal line test a function f is one-to-one if and only if every horizontal line intersects the graph of f , at most, once

hyperbolic functions the functions denoted $\sinh$, $\cosh$, $\tanh$, csch , sech , and coth , which involve certain combinations of e^x and e^{-x}

increasing on the interval I a function increasing on the interval I if for all $x_1, x_2 \in I$, $f(x_1) \leq f(x_2)$ if $x_1 < x_2$

independent variable the input variable for a function

inverse function for a function f , the inverse function f^{-1} satisfies $f^{-1}(y) = x$ if $f(x) = y$

inverse hyperbolic functions the inverses of the hyperbolic functions where $\cosh$ and sech are restricted to the domain $[0, \infty)$; each of these functions can be expressed in terms of a composition of the natural logarithm function and an algebraic function

inverse trigonometric functions the inverses of the trigonometric functions are defined on restricted domains where they are one-to-one functions

linear function a function that can be written in the form $f(x) = mx + b$

logarithmic function a function of the form $f(x) = \log_b(x)$ for some base $b > 0$, $b \neq 1$ such that $y = \log_b(x)$ if and only if $b^y = x$

mathematical model A method of simulating real-life situations with mathematical equations

natural exponential function the function $f(x) = e^x$

natural logarithm the function $\ln x = \log_e x$

number e as m gets larger, the quantity $(1 + (1/m))^m$ gets closer to some real number; we define that real number to be e ; the value of e is approximately 2.718282

odd function a function is odd if $f(-x) = -f(x)$ for all x in the domain of f

one-to-one function a function f is one-to-one if $f(x_1) \neq f(x_2)$ if $x_1 \neq x_2$

periodic function a function is periodic if it has a repeating pattern as the values of x move from left to right

piecewise-defined function a function that is defined differently on different parts of its domain

point-slope equation equation of a linear function indicating its slope and a point on the graph of the function

polynomial function a function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

power function a function of the form $f(x) = x^n$ for any positive integer $n \geq 1$

quadratic function a polynomial of degree 2; that is, a function of the form $f(x) = ax^2 + bx + c$ where $a \neq 0$

radians for a circular arc of length s on a circle of radius 1, the radian measure of the associated angle θ is s

range the set of outputs for a function

rational function a function of the form $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomials

restricted domain a subset of the domain of a function f

root function a function of the form $f(x) = x^{1/n}$ for any integer $n \geq 2$

slope the change in y for each unit change in x

slope-intercept form equation of a linear function indicating its slope and y -intercept

symmetry about the origin the graph of a function f is symmetric about the origin if $(-x, -y)$ is on the graph of f whenever (x, y) is on the graph

symmetry about the y -axis the graph of a function f is symmetric about the y -axis if $(-x, y)$ is on the graph of f whenever (x, y) is on the graph

- 307. [T]** A bacterial colony grown in a lab is known to double in number in 12 hours. Suppose, initially, there are 1000 bacteria present.
- Use the exponential function $Q = Q_0 e^{kt}$ to determine the value k , which is the growth rate of the bacteria. Round to four decimal places.
 - Determine approximately how long it takes for 200,000 bacteria to grow.
- 308. [T]** The rabbit population on a game reserve doubles every 6 months. Suppose there were 120 rabbits initially.
- Use the exponential function $P = P_0 a^t$ to determine the growth rate constant a . Round to four decimal places.
 - Use the function in part a. to determine approximately how long it takes for the rabbit population to reach 3500.
- 309. [T]** The 1906 earthquake in San Francisco had a magnitude of 8.3 on the Richter scale. At the same time, in Japan, an earthquake with magnitude 4.9 caused only minor damage. Approximately how much more intense was the San Francisco earthquake than the Japanese earthquake? See the definition of Richter Scale in [\[link\]](#) in this section.

- 301. [T]** The amount A of a \$100,000 investment paying continuously and compounded for t years is given by
 $A(t) = 100,000 \cdot e^{0.055t}$.
 Find the amount A accumulated in 5 years.

- 302. [T]** An investment is compounded monthly, quarterly, or yearly and is given by the function
 $A = P\left(1 + \frac{j}{n}\right)^{nt}$, where
 A is the value of the investment at time t , P is the initial principle that was invested, j is the annual interest rate, and n is the number of time the interest is compounded per year. Given a yearly interest rate of 3.5% and an initial principle of \$100,000, find the amount A accumulated in 5 years for interest that is compounded a. daily, b., monthly, c. quarterly, and d. yearly.

- 303. [T]** The concentration of hydrogen ions in a substance is denoted by $[H^+]$, measured in moles per liter. The pH of a substance is defined by the logarithmic function $pH = -\log [H^+]$. This function is used to measure the acidity of a substance. The pH of water is 7. A substance with a pH less than 7 is an acid, whereas one that has a pH of more than 7 is a base.
- Find the pH of the following substances. Round answers to one digit.
 - Determine whether the substance is an acid or a base.
 - Eggs:
 $[H^+] = 1.6 \times 10^{-8}$
 mol/L
 - Beer:
 $[H^+] = 3.16 \times 10^{-3}$
 mol/L
 - Tomato Juice:
 $[H^+] = 7.94 \times 10^{-5}$
 mol/L

- 304. [T]** Iodine-131 is a radioactive substance that decays according to the function
 $Q(t) = Q_0 \cdot e^{-0.08664t}$,
 where Q_0 is the initial quantity of a sample of the substance and t is in days. Determine how long it takes (to the nearest day) for 95% of a quantity to decay.

- 305. [T]** According to the World Bank, at the end of 2013 ($t = 0$) the U.S. population was 316 million and was increasing according to the following model:
 $P(t) = 316e^{0.0074t}$, where
 P is measured in millions of people and t is measured in years after 2013.
- Based on this model, what will be the population of the United States in 2020?
 - Determine when the U.S. population will be twice what it is in 2013.

- 306. [T]** The amount A accumulated after 1000 dollars is invested for t years at an interest rate of 4% is modeled by the function
 $A(t) = 1000(1.04)^t$.
- Find the amount accumulated after 5 years and 10 years.
 - Determine how long it takes for the original investment to triple.

267. $f(x) = 1 - \ln x$

268. $f(x) = \log x - 1$

269. $f(x) = \ln(x + 1)$

For the following exercises, use properties of logarithms to write the expressions as a sum, difference, and/or product of logarithms.

270. $\log_3 x^4 y$

271. $\log_3 \frac{9a^3}{b}$

272. $\ln a \sqrt[3]{b}$

273. $\log_5 \sqrt{125xy^3}$

274. $\log_4 \frac{\sqrt[3]{xy}}{64}$

275. $\ln \left(\frac{6}{\sqrt{e^3}} \right)$

For the following exercises, solve the exponential equation exactly.

276. $5^x = 125$

277. $e^{3x} - 15 = 0$

278. $8^x = 4$

279. $4^{x+1} - 32 = 0$

280. $3^{x/14} = \frac{1}{10}$

281. $10^x = 7.21$

282. $4 \cdot 2^{3x} - 20 = 0$

283. $7^{3x-2} = 11$

For the following exercises, solve the logarithmic equation exactly, if possible.

284. $\log_3 x = 0$

285. $\log_5 x = -2$

286. $\log_4 (x + 5) = 0$

287. $\log (2x - 7) = 0$

288. $\ln \sqrt{x + 3} = 2$

289. $\log_6 (x + 9) + \log_6 x = 2$

290. $\log_4 (x + 2) - \log_4 (x - 1) = 0$

291. $\ln x + \ln (x - 2) = \ln 4$

For the following exercises, use the change-of-base formula and either base 10 or base e to evaluate the given expressions. Answer in exact form and in approximate form, rounding to four decimal places.

292. $\log_5 47$

293. $\log_7 82$

294. $\log_6 103$

295. $\log_{0.5} 211$

296. $\log_2 \pi$

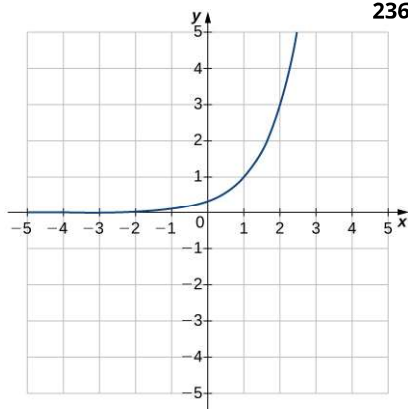
297. $\log_{0.2} 0.452$

298. Rewrite the following expressions in terms of exponentials and simplify.
 a. $2 \cosh (\ln x)$ b. $\cosh 4x + \sinh 4x$ c. $\cosh 2x - \sinh 2x$ d. $\ln (\cosh x + \sinh x) + \ln (\cosh x - \sinh x)$

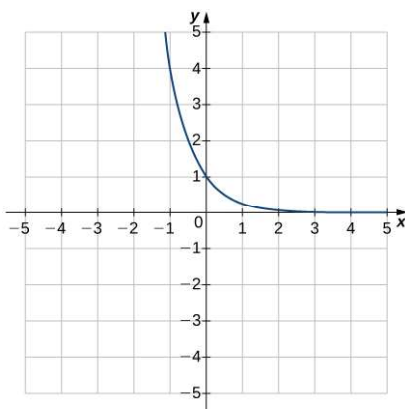
299. [T] The number of bacteria N in a culture after t days can be modeled by the function $N(t) = 1300 \cdot (2)^{t/4}$. Find the number of bacteria present after 15 days.

300. [T] The demand D (in millions of barrels) for oil in an oil-rich country is given by the function $D(p) = 150 \cdot (2.7)^{-0.25p}$, where p is the price (in dollars) of a barrel of oil. Find the amount of oil demanded (to the nearest million barrels) when the price is between \$15 and \$20.

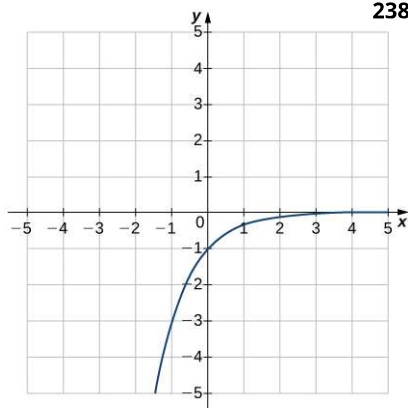
235.



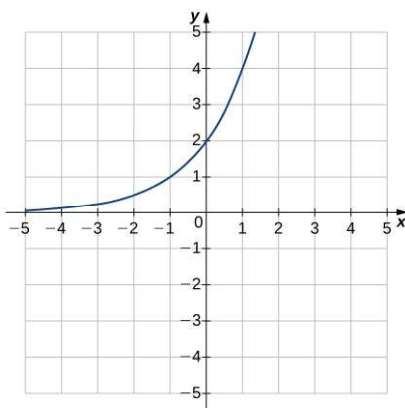
236.



237.



238.



For the following exercises, sketch the graph of the exponential function. Determine the domain, range, and horizontal asymptote.

239. $f(x) = e^x + 2$

240. $f(x) = -2^x$

241. $f(x) = 3^{x+1}$

242. $f(x) = 4^x - 1$

243. $f(x) = 1 - 2^{-x}$

244. $f(x) = 5^{x+1} + 2$

245. $f(x) = e^{-x} - 1$

For the following exercises, write the equation in equivalent exponential form.

246. $\log_3 81 = 4$

247. $\log_8 2 = \frac{1}{3}$

248. $\log_5 1 = 0$

249. $\log_5 25 = 2$

250. $\log 0.1 = -1$

251. $\ln\left(\frac{1}{e^3}\right) = -3$

252. $\log_9 3 = 0.5$

253. $\ln 1 = 0$

For the following exercises, write the equation in equivalent logarithmic form.

254. $2^3 = 8$

255. $4^{-2} = \frac{1}{16}$

256. $10^2 = 100$

257. $9^0 = 1$

258. $\left(\frac{1}{3}\right)^3 = \frac{1}{27}$

259. $\sqrt[3]{64} = 4$

260. $e^x = y$

261. $9^y = 150$

262. $b^3 = 45$

263. $4^{-3/2} = 0.125$

For the following exercises, sketch the graph of the logarithmic function. Determine the domain, range, and vertical asymptote.

264. $f(x) = 3 + \ln x$

265. $f(x) = \ln(x - 1)$

266. $f(x) = \ln(-x)$

EXAMPLE 1.41**Evaluating Inverse Hyperbolic Functions**

Evaluate each of the following expressions.

$$\sinh^{-1}(2)$$

$$\tanh^{-1}(1/4)$$

✓ Solution

$$\sinh^{-1}(2) = \ln\left(2 + \sqrt{2^2 + 1}\right) = \ln\left(2 + \sqrt{5}\right) \approx 1.4436$$

$$\tanh^{-1}(1/4) = \frac{1}{2} \ln\left(\frac{1+1/4}{1-1/4}\right) = \frac{1}{2} \ln\left(\frac{5/4}{3/4}\right) = \frac{1}{2} \ln\left(\frac{5}{3}\right) \approx 0.2554$$

☒ 1.35 Evaluate $\tanh^{-1}(1/2)$.**SECTION 1.5 EXERCISES**

For the following exercises, evaluate the given exponential functions as indicated, accurate to two significant digits after the decimal.

229. $f(x) = 5^x$ a. $x = 3$ b.
 $x = \frac{1}{2}$ c. $x = \sqrt{2}$

230. $f(x) = (0.3)^x$ a. $x = -1$
b. $x = 4$ c. $x = -1.5$

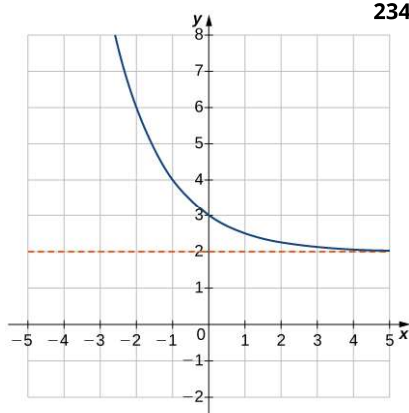
231. $f(x) = 10^x$ a. $x = -2$ b.
 $x = 4$ c. $x = \frac{5}{3}$

232. $f(x) = e^x$ a. $x = 2$ b.
 $x = -3.2$ c. $x = \pi$

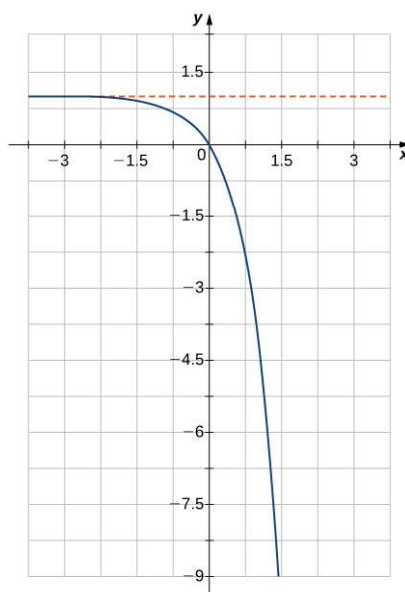
For the following exercises, match the exponential equation to the correct graph.

- a. $y = 4^{-x}$
- b. $y = 3^{x-1}$
- c. $y = 2^{x+1}$
- d. $y = \left(\frac{1}{2}\right)^x + 2$
- e. $y = -3^{-x}$
- f. $y = 1 - 5^x$

233.



234.



EXAMPLE 1.40**Evaluating Hyperbolic Functions**

- a. Simplify $\sinh(5 \ln x)$.
 b. If $\sinh x = 3/4$, find the values of the remaining five hyperbolic functions.

✓ Solution

- a. Using the definition of the $\sinh$ function, we write

$$\sinh(5 \ln x) = \frac{e^{5 \ln x} - e^{-5 \ln x}}{2} = \frac{e^{\ln(x^5)} - e^{\ln(x^{-5})}}{2} = \frac{x^5 - x^{-5}}{2}.$$

- b. Using the identity $\cosh^2 x - \sinh^2 x = 1$, we see that

$$\cosh^2 x = 1 + \left(\frac{3}{4}\right)^2 = \frac{25}{16}.$$

Since $\cosh x \geq 1$ for all x , we must have $\cosh x = 5/4$. Then, using the definitions for the other hyperbolic functions, we conclude that $\tanh x = 3/5$, $\operatorname{csch} x = 4/3$, $\operatorname{sech} x = 4/5$, and $\operatorname{coth} x = 5/3$.

✓ 1.34 Simplify $\cosh(2 \ln x)$.

Inverse Hyperbolic Functions

From the graphs of the hyperbolic functions, we see that all of them are one-to-one except $\cosh x$ and $\operatorname{sech} x$. If we restrict the domains of these two functions to the interval $[0, \infty)$, then all the hyperbolic functions are one-to-one, and we can define the **inverse hyperbolic functions**. Since the hyperbolic functions themselves involve exponential functions, the inverse hyperbolic functions involve logarithmic functions.

Definition**Inverse Hyperbolic Functions**

$$\begin{aligned} \sinh^{-1} x &= \operatorname{arcsinh} x = \ln \left(x + \sqrt{x^2 + 1} \right) & \cosh^{-1} x &= \operatorname{arccosh} x = \ln \left(x + \sqrt{x^2 - 1} \right) \\ \tanh^{-1} x &= \operatorname{arctanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) & \coth^{-1} x &= \operatorname{arccoth} x = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right) \\ \operatorname{sech}^{-1} x &= \operatorname{arcsech} x = \ln \left(\frac{1+\sqrt{1-x^2}}{x} \right) & \operatorname{csch}^{-1} x &= \operatorname{arccsch} x = \ln \left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|} \right) \end{aligned}$$

Let's look at how to derive the first equation. The others follow similarly. Suppose $y = \sinh^{-1} x$. Then, $x = \sinh y$ and, by the definition of the hyperbolic sine function, $x = \frac{e^y - e^{-y}}{2}$. Therefore,

$$e^y - 2x - e^{-y} = 0.$$

Multiplying this equation by e^y , we obtain

$$e^{2y} - 2xe^y - 1 = 0.$$

This can be solved like a quadratic equation, with the solution

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}.$$

Since $e^y > 0$, the only solution is the one with the positive sign. Applying the natural logarithm to both sides of the equation, we conclude that

$$y = \ln \left(x + \sqrt{x^2 + 1} \right).$$

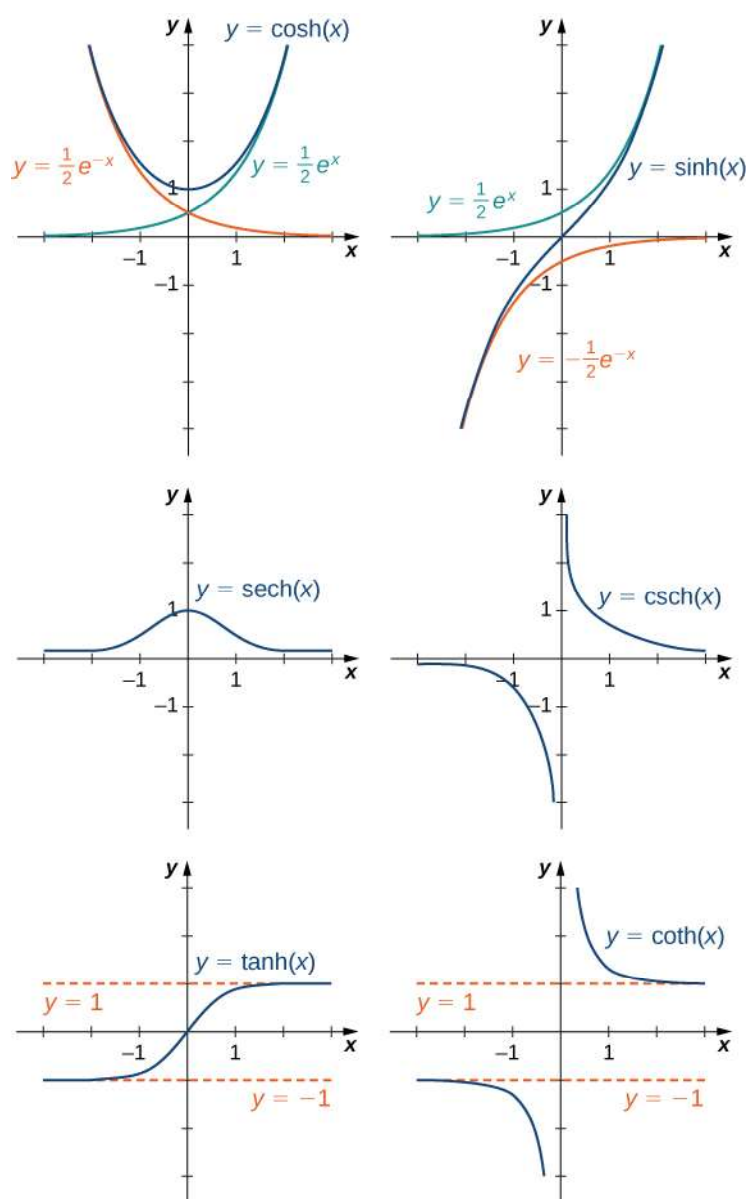


Figure 1.51 The hyperbolic functions involve combinations of e^x and e^{-x} .

Identities Involving Hyperbolic Functions

The identity $\cosh^2 t - \sinh^2 t$, shown in [Figure 1.50](#), is one of several identities involving the hyperbolic functions, some of which are listed next. The first four properties follow easily from the definitions of hyperbolic sine and hyperbolic cosine. Except for some differences in signs, most of these properties are analogous to identities for trigonometric functions.

Rule: Identities Involving Hyperbolic Functions

1. $\cosh(-x) = \cosh x$
2. $\sinh(-x) = -\sinh x$
3. $\cosh x + \sinh x = e^x$
4. $\cosh x - \sinh x = e^{-x}$
5. $\cosh^2 x - \sinh^2 x = 1$
6. $1 - \tanh^2 x = \operatorname{sech}^2 x$
7. $\coth^2 x - 1 = \operatorname{csch}^2 x$
8. $\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$
9. $\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

This identity is the analog of the trigonometric identity $\cos^2 t + \sin^2 t = 1$. Here, given a value t , the point $(x, y) = (\cosh t, \sinh t)$ lies on the unit hyperbola $x^2 - y^2 = 1$ (Figure 1.50).

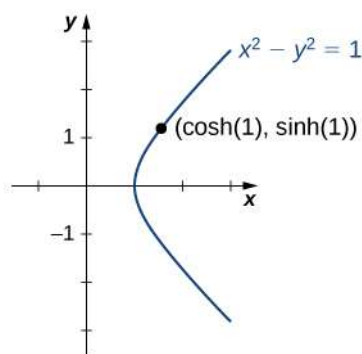


Figure 1.50 The unit hyperbola $\cosh^2 t - \sinh^2 t = 1$.

Graphs of Hyperbolic Functions

To graph $\cosh x$ and $\sinh x$, we make use of the fact that both functions approach $(1/2)e^x$ as $x \rightarrow \infty$, since $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$. As $x \rightarrow -\infty$, $\cosh x$ approaches $1/2e^{-x}$, whereas $\sinh x$ approaches $-1/2e^{-x}$. Therefore, using the graphs of $1/2e^x$, $1/2e^{-x}$, and $-1/2e^{-x}$ as guides, we graph $\cosh x$ and $\sinh x$. To graph $\tanh x$, we use the fact that $\tanh(0) = 0$, $-1 < \tanh(x) < 1$ for all x , $\tanh x \rightarrow 1$ as $x \rightarrow \infty$, and $\tanh x \rightarrow -1$ as $x \rightarrow -\infty$. The graphs of the other three hyperbolic functions can be sketched using the graphs of $\cosh x$, $\sinh x$, and $\tanh x$ (Figure 1.51).



Figure 1.49 The shape of a strand of silk in a spider's web can be described in terms of a hyperbolic function. The same shape applies to a chain or cable hanging from two supports with only its own weight. (credit: "Mtpaley", Wikimedia Commons)

Definition

Hyperbolic cosine

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Hyperbolic sine

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

Hyperbolic tangent

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Hyperbolic cosecant

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

Hyperbolic secant

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

Hyperbolic cotangent

$$\operatorname{coth} x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

The name *cosh* rhymes with “gosh,” whereas the name *sinh* is pronounced “cinch.” *Tanh*, *sech*, *csch*, and *coth* are pronounced “tanch,” “seech,” “coseech,” and “cotanch,” respectively.

Using the definition of $\cosh(x)$ and principles of physics, it can be shown that the height of a hanging chain, such as the one in [Figure 1.49](#), can be described by the function $h(x) = a \cosh(x/a) + c$ for certain constants a and c .

But why are these functions called *hyperbolic functions*? To answer this question, consider the quantity $\cosh^2 t - \sinh^2 t$. Using the definition of $\cosh$ and $\sinh$, we see that

$$\cosh^2 t - \sinh^2 t = \frac{e^{2t} + 2 + e^{-2t}}{4} - \frac{e^{2t} - 2 + e^{-2t}}{4} = 1.$$

$$R_1 - R_2 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Consider an earthquake that measures 8 on the Richter scale and an earthquake that measures 7 on the Richter scale. Then,

$$8 - 7 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Therefore,

$$\log_{10} \left(\frac{A_1}{A_2} \right) = 1,$$

which implies $A_1/A_2 = 10$ or $A_1 = 10A_2$. Since A_1 is 10 times the size of A_2 , we say that the first earthquake is 10 times as intense as the second earthquake. On the other hand, if one earthquake measures 8 on the Richter scale and another measures 6, then the relative intensity of the two earthquakes satisfies the equation

$$\log_{10} \left(\frac{A_1}{A_2} \right) = 8 - 6 = 2.$$

Therefore, $A_1 = 100A_2$. That is, the first earthquake is 100 times more intense than the second earthquake.

How can we use logarithmic functions to compare the relative severity of the magnitude 9 earthquake in Japan in 2011 with the magnitude 7.3 earthquake in Haiti in 2010?

Solution

To compare the Japan and Haiti earthquakes, we can use an equation presented earlier:

$$9 - 7.3 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Therefore, $A_1/A_2 = 10^{1.7}$, and we conclude that the earthquake in Japan was approximately 50 times more intense than the earthquake in Haiti.

-  1.33 Compare the relative severity of a magnitude 8.4 earthquake with a magnitude 7.4 earthquake.

Hyperbolic Functions

The hyperbolic functions are defined in terms of certain combinations of e^x and e^{-x} . These functions arise naturally in various engineering and physics applications, including the study of water waves and vibrations of elastic membranes. Another common use for a hyperbolic function is the representation of a hanging chain or cable, also known as a catenary (Figure 1.49). If we introduce a coordinate system so that the low point of the chain lies along the y-axis, we can describe the height of the chain in terms of a hyperbolic function. First, we define the **hyperbolic functions**.

$$2. \log_a x = \frac{\log_b x}{\log_b a} \text{ for any real number } x > 0.$$

If $b = e$, this equation reduces to $\log_a x = \frac{\ln x}{\ln a}$.

Proof

For the first change-of-base formula, we begin by making use of the power property of logarithmic functions. We know that for any base $b > 0$, $b \neq 1$, $\log_b(a^x) = x \log_b a$. Therefore,

$$b^{\log_b(a^x)} = b^{x \log_b a}.$$

In addition, we know that b^x and $\log_b(x)$ are inverse functions. Therefore,

$$b^{\log_b(a^x)} = a^x.$$

Combining these last two equalities, we conclude that $a^x = b^{x \log_b a}$.

To prove the second property, we show that

$$(\log_b a) \cdot (\log_a x) = \log_b x.$$

Let $u = \log_b a$, $v = \log_a x$, and $w = \log_b x$. We will show that $u \cdot v = w$. By the definition of logarithmic functions, we know that $b^u = a$, $a^v = x$, and $b^w = x$. From the previous equations, we see that

$$b^{uv} = (b^u)^v = a^v = x = b^w.$$

Therefore, $b^{uv} = b^w$. Since exponential functions are one-to-one, we can conclude that $u \cdot v = w$.

□

EXAMPLE 1.38

Changing Bases

Use a calculating utility to evaluate $\log_3 7$ with the change-of-base formula presented earlier.

✓ Solution

Use the second equation with $a = 3$ and $e = 3$:

$$\log_3 7 = \frac{\ln 7}{\ln 3} \approx 1.77124.$$

✓ 1.32 Use the change-of-base formula and a calculating utility to evaluate $\log_4 6$.

EXAMPLE 1.39

Chapter Opener: The Richter Scale for Earthquakes



Figure 1.48 (credit: modification of work by Robb Hannawacker, NPS)

In 1935, Charles Richter developed a scale (now known as the *Richter scale*) to measure the magnitude of an earthquake. The scale is a base-10 logarithmic scale, and it can be described as follows: Consider one earthquake with magnitude R_1 on the Richter scale and a second earthquake with magnitude R_2 on the Richter scale. Suppose $R_1 > R_2$, which means the earthquake of magnitude R_1 is stronger, but how much stronger is it than the other earthquake? A way of measuring the intensity of an earthquake is by using a seismograph to measure the amplitude of the earthquake waves. If A_1 is the amplitude measured for the first earthquake and A_2 is the amplitude measured for the second earthquake, then the amplitudes and magnitudes of the two earthquakes satisfy the following equation:

Therefore, the solutions satisfy $e^x = 3$ and $e^x = 2$. Taking the natural logarithm of both sides gives us the solutions $x = \ln 3, \ln 2$.

✓ 1.30 Solve $e^{2x}/(3 + e^{2x}) = 1/2$.

EXAMPLE 1.37

Solving Equations Involving Logarithmic Functions

Solve each of the following equations for x .

- $\ln\left(\frac{1}{x}\right) = 4$
- $\log_{10} \sqrt{x} + \log_{10} x = 2$
- $\ln(2x) - 3\ln(x^2) = 0$

✓ Solution

- a. By the definition of the natural logarithm function,

$$\ln\left(\frac{1}{x}\right) = 4 \text{ if and only if } e^4 = \frac{1}{x}.$$

Therefore, the solution is $x = 1/e^4$.

- b. Using the product and power properties of logarithmic functions, rewrite the left-hand side of the equation as

$$\log_{10} \sqrt{x} + \log_{10} x = \log_{10} x \sqrt{x} = \log_{10} x^{3/2} = \frac{3}{2} \log_{10} x.$$

Therefore, the equation can be rewritten as

$$\frac{3}{2} \log_{10} x = 2 \text{ or } \log_{10} x = \frac{4}{3}.$$

The solution is $x = 10^{4/3} = 10\sqrt[3]{10}$.

- c. Using the power property of logarithmic functions, we can rewrite the equation as $\ln(2x) - \ln(x^6) = 0$.
Using the quotient property, this becomes

$$\ln\left(\frac{2}{x^5}\right) = 0.$$

Therefore, $2/x^5 = 1$, which implies $x = \sqrt[5]{2}$. We should then check for any extraneous solutions.

✓ 1.31 Solve $\ln(x^3) - 4\ln(x) = 1$.

When evaluating a logarithmic function with a calculator, you may have noticed that the only options are $\log_{10}$ or $\log$, called the *common logarithm*, or $\ln$, which is the natural logarithm. However, exponential functions and logarithm functions can be expressed in terms of any desired base b . If you need to use a calculator to evaluate an expression with a different base, you can apply the change-of-base formulas first. Using this change of base, we typically write a given exponential or logarithmic function in terms of the natural exponential and natural logarithmic functions.

Rule: Change-of-Base Formulas

Let $a > 0$, $b > 0$, and $a \neq 1, b \neq 1$.

- $a^x = b^{x \log_b a}$ for any real number x .
If $b = e$, this equation reduces to $a^x = e^{x \log_e a} = e^{x \ln a}$.

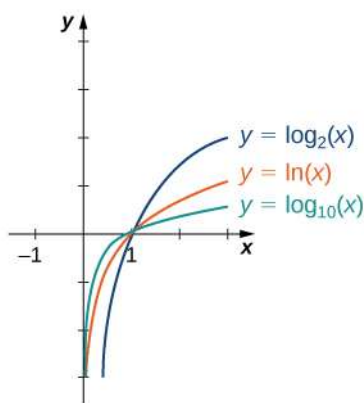


Figure 1.47 Graphs of $y = \log_b(x)$ are depicted for $b = 2, e, 10$.

Before solving some equations involving exponential and logarithmic functions, let's review the basic properties of logarithms.

Rule: Properties of Logarithms

If $a, b, c > 0$, $b \neq 1$, and r is any real number, then

1. $\log_b(ac) = \log_b(a) + \log_b(c)$ (Product property)
2. $\log_b\left(\frac{a}{c}\right) = \log_b(a) - \log_b(c)$ (Quotient property)
3. $\log_b(a^r) = r\log_b(a)$ (Power property)

EXAMPLE 1.36

Solving Equations Involving Exponential Functions

Solve each of the following equations for x .

- a. $5^x = 2$
- b. $e^x + 6e^{-x} = 5$

✓ Solution

- a. Applying the natural logarithm function to both sides of the equation, we have

$$\ln 5^x = \ln 2.$$

Using the power property of logarithms,

$$x \ln 5 = \ln 2.$$

Therefore, $x = \ln 2 / \ln 5$.

- b. Multiplying both sides of the equation by e^x , we arrive at the equation

$$e^{2x} + 6 = 5e^x.$$

Rewriting this equation as

$$e^{2x} - 5e^x + 6 = 0,$$

we can then rewrite it as a quadratic equation in e^x :

$$(e^x)^2 - 5(e^x) + 6 = 0.$$

Now we can solve the quadratic equation. Factoring this equation, we obtain

$$(e^x - 3)(e^x - 2) = 0.$$

The exponential function $f(x) = b^x$ is one-to-one, with domain $(-\infty, \infty)$ and range $(0, \infty)$. Therefore, it has an inverse function, called the *logarithmic function with base b* . For any $b > 0$, $b \neq 1$, the logarithmic function with base b , denoted $\log_b$, has domain $(0, \infty)$ and range $(-\infty, \infty)$, and satisfies

$$\log_b(x) = y \text{ if and only if } b^y = x.$$

For example,

$$\begin{array}{lll} \log_2(8) = 3 & \text{since} & 2^3 = 8, \\ \log_{10}\left(\frac{1}{100}\right) = -2 & \text{since} & 10^{-2} = \frac{1}{10^2} = \frac{1}{100}, \\ \log_b(1) = 0 & \text{since} & b^0 = 1 \text{ for any base } b > 0. \end{array}$$

Furthermore, since $y = \log_b(x)$ and $y = b^x$ are inverse functions,

$$\log_b(b^x) = x \text{ and } b^{\log_b(x)} = x.$$

The most commonly used logarithmic function is the function $\log_e$. Since this function uses natural e as its base, it is called the **natural logarithm**. Here we use the notation $\ln(x)$ or $\ln x$ to mean $\log_e(x)$. For example,

$$\ln(e) = \log_e(e) = 1, \ln(e^3) = \log_e(e^3) = 3, \ln(1) = \log_e(1) = 0.$$

Since the functions $f(x) = e^x$ and $g(x) = \ln(x)$ are inverses of each other,

$$\ln(e^x) = x \text{ and } e^{\ln x} = x,$$

and their graphs are symmetric about the line $y = x$ (Figure 1.46).

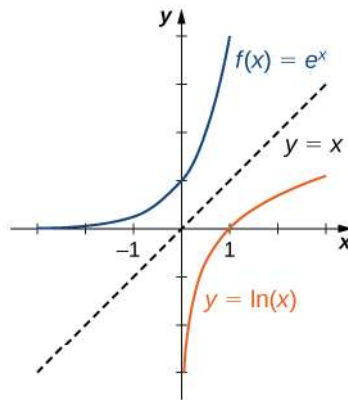


Figure 1.46 The functions $y = e^x$ and $y = \ln(x)$ are inverses of each other, so their graphs are symmetric about the line $y = x$.



At this [site \(http://www.openstax.org/l/20_logscale\)](http://www.openstax.org/l/20_logscale) you can see an example of a base-10 logarithmic scale.

In general, for any base $b > 0$, $b \neq 1$, the function $g(x) = \log_b(x)$ is symmetric about the line $y = x$ with the function $f(x) = b^x$. Using this fact and the graphs of the exponential functions, we graph functions $\log_b$ for several values of $b > 1$ (Figure 1.47).

$(1 + 1/m)^m$ does approach some number as $m \rightarrow \infty$. We call this **number** e . To six decimal places of accuracy,

$$e \approx 2.718282.$$

The letter e was first used to represent this number by the Swiss mathematician Leonhard Euler during the 1720s. Although Euler did not discover the number, he showed many important connections between e and logarithmic functions. We still use the notation e today to honor Euler's work because it appears in many areas of mathematics and because we can use it in many practical applications.

Returning to our savings account example, we can conclude that if a person puts P dollars in an account at an annual interest rate r , compounded continuously, then $A(t) = Pe^{rt}$. This function may be familiar. Since functions involving base e arise often in applications, we call the function $f(x) = e^x$ the **natural exponential function**. Not only is this function interesting because of the definition of the number e , but also, as discussed next, its graph has an important property.

Since $e > 1$, we know e^x is increasing on $(-\infty, \infty)$. In [Figure 1.45](#), we show a graph of $f(x) = e^x$ along with a *tangent line* to the graph of at $x = 0$. We give a precise definition of tangent line in the next chapter; but, informally, we say a tangent line to a graph of f at $x = a$ is a line that passes through the point $(a, f(a))$ and has the same “slope” as f at that point. The function $f(x) = e^x$ is the only exponential function b^x with tangent line at $x = 0$ that has a slope of 1. As we see later in the text, having this property makes the natural exponential function the most simple exponential function to use in many instances.

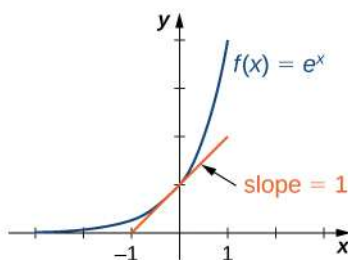


Figure 1.45 The graph of $f(x) = e^x$ has a tangent line with slope 1 at $x = 0$.

EXAMPLE 1.35

Compounding Interest

Suppose \$500 is invested in an account at an annual interest rate of $r = 5.5\%$, compounded continuously.

- Let t denote the number of years after the initial investment and $A(t)$ denote the amount of money in the account at time t . Find a formula for $A(t)$.
- Find the amount of money in the account after 10 years and after 20 years.

Solution

- If P dollars are invested in an account at an annual interest rate r , compounded continuously, then $A(t) = Pe^{rt}$. Here $P = \$500$ and $r = 0.055$. Therefore, $A(t) = 500e^{0.055t}$.
- After 10 years, the amount of money in the account is

$$A(10) = 500e^{0.055 \cdot 10} = 500e^{0.55} \approx \$866.63.$$

After 20 years, the amount of money in the account is

$$A(20) = 500e^{0.055 \cdot 20} = 500e^{1.1} \approx \$1,502.08.$$

-  1.29 If \$750 is invested in an account at an annual interest rate of 4%, compounded continuously, find a formula for the amount of money in the account after t years. Find the amount of money after 30 years.

Logarithmic Functions

Using our understanding of exponential functions, we can discuss their inverses, which are the logarithmic functions. These come in handy when we need to consider any phenomenon that varies over a wide range of values, such as pH in chemistry or decibels in sound levels.

$$\frac{(2x^{2/3})^3}{(4x^{-1/3})^2} = \frac{2^3(x^{2/3})^3}{4^2(x^{-1/3})^2} = \frac{8x^2}{16x^{-2/3}} = \frac{x^2x^{2/3}}{2} = \frac{x^{8/3}}{2}.$$

b. We can simplify as follows:

$$\frac{(x^3y^{-1})^2}{(xy^2)^{-2}} = \frac{(x^3)^2(y^{-1})^2}{x^{-2}(y^2)^{-2}} = \frac{x^6y^{-2}}{x^{-2}y^{-4}} = x^6x^2y^{-2}y^4 = x^8y^2.$$

✓ 1.28 Use the laws of exponents to simplify $(6x^{-3}y^2)/(12x^{-4}y^5)$.

The Number e

A special type of exponential function appears frequently in real-world applications. To describe it, consider the following example of exponential growth, which arises from compounding interest in a savings account. Suppose a person invests P dollars in a savings account with an annual interest rate r , compounded annually. The amount of money after 1 year is

$$A(1) = P + rP = P(1 + r).$$

The amount of money after 2 years is

$$A(2) = A(1) + rA(1) = P(1 + r) + rP(1 + r) = P(1 + r)^2.$$

More generally, the amount after t years is

$$A(t) = P(1 + r)^t.$$

If the money is compounded 2 times per year, the amount of money after half a year is

$$A\left(\frac{1}{2}\right) = P + \left(\frac{r}{2}\right)P = P\left(1 + \left(\frac{r}{2}\right)\right).$$

The amount of money after 1 year is

$$A(1) = A\left(\frac{1}{2}\right) + \left(\frac{r}{2}\right)A\left(\frac{1}{2}\right) = P\left(1 + \frac{r}{2}\right) + \frac{r}{2}\left(P\left(1 + \frac{r}{2}\right)\right) = P\left(1 + \frac{r}{2}\right)^2.$$

After t years, the amount of money in the account is

$$A(t) = P\left(1 + \frac{r}{2}\right)^{2t}.$$

More generally, if the money is compounded n times per year, the amount of money in the account after t years is given by the function

$$A(t) = P\left(1 + \frac{r}{n}\right)^{nt}.$$

What happens as $n \rightarrow \infty$? To answer this question, we let $m = n/r$ and write

$$\left(1 + \frac{r}{n}\right)^{nt} = \left(1 + \frac{1}{m}\right)^{mrt},$$

and examine the behavior of $(1 + 1/m)^m$ as $m \rightarrow \infty$, using a table of values (Table 1.12).

m	10	100	1000	10,000	100,000	1,000,000
$\left(1 + \frac{1}{m}\right)^m$	2.5937	2.7048	2.71692	2.71815	2.718268	2.718280

Table 1.12 Values of $\left(1 + \frac{1}{m}\right)^m$ as $m \rightarrow \infty$

Looking at this table, it appears that $(1 + 1/m)^m$ is approaching a number between 2.7 and 2.8 as $m \rightarrow \infty$. In fact,

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Go to [Population Balance \(http://www.openstax.org/l/20_exponengrow\)](http://www.openstax.org/l/20_exponengrow) for another example of exponential population growth.

Graphing Exponential Functions

For any base $b > 0$, $b \neq 1$, the exponential function $f(x) = b^x$ is defined for all real numbers x and $b^x > 0$. Therefore, the domain of $f(x) = b^x$ is $(-\infty, \infty)$ and the range is $(0, \infty)$. To graph b^x , we note that for $b > 1$, b^x is increasing on $(-\infty, \infty)$ and $b^x \rightarrow \infty$ as $x \rightarrow \infty$, whereas $b^x \rightarrow 0$ as $x \rightarrow -\infty$. On the other hand, if $0 < b < 1$, $f(x) = b^x$ is decreasing on $(-\infty, \infty)$ and $b^x \rightarrow 0$ as $x \rightarrow \infty$ whereas $b^x \rightarrow \infty$ as $x \rightarrow -\infty$ (Figure 1.44).

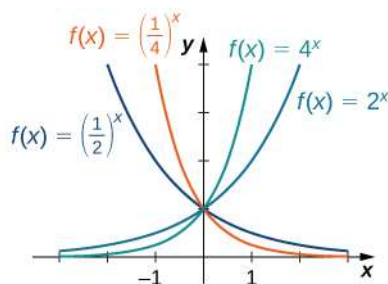


Figure 1.44 If $b > 1$, then b^x is increasing on $(-\infty, \infty)$. If $0 < b < 1$, then b^x is decreasing on $(-\infty, \infty)$.

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Visit this [site \(http://www.openstax.org/l/20_inverse\)](http://www.openstax.org/l/20_inverse) for more exploration of the graphs of exponential functions.

Note that exponential functions satisfy the general laws of exponents. To remind you of these laws, we state them as rules.

Rule: Laws of Exponents

For any constants $a > 0$, $b > 0$, and for all x and y ,

1. $b^x \cdot b^y = b^{x+y}$
2. $\frac{b^x}{b^y} = b^{x-y}$
3. $(b^x)^y = b^{xy}$
4. $(ab)^x = a^x b^x$
5. $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$

EXAMPLE 1.34

Using the Laws of Exponents

Use the laws of exponents to simplify each of the following expressions.

- a. $\frac{(2x^{2/3})^3}{(4x^{-1/3})^2}$
- b. $\frac{(x^3y^{-1})^2}{(xy^2)^{-2}}$

 Solution

- a. We can simplify as follows:

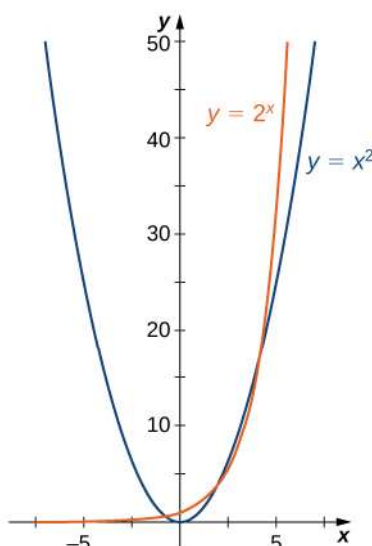


Figure 1.43 Both 2^x and x^2 approach infinity as $x \rightarrow \infty$, but 2^x grows more rapidly than x^2 . As $x \rightarrow -\infty$, $x^2 \rightarrow \infty$, whereas $2^x \rightarrow 0$.

Evaluating Exponential Functions

Recall the properties of exponents: If x is a positive integer, then we define $b^x = b \cdot b \cdots b$ (with x factors of b). If x is a negative integer, then $x = -y$ for some positive integer y , and we define $b^x = b^{-y} = 1/b^y$. Also, b^0 is defined to be 1. If x is a rational number, then $x = p/q$, where p and q are integers and $b^x = b^{p/q} = \sqrt[q]{b^p}$. For example, $9^{3/2} = \sqrt{9^3} = 27$.

However, how is b^x defined if x is an irrational number? For example, what do we mean by $2^{\sqrt{2}}$? This is too complex a question for us to answer fully right now; however, we can make an approximation. In [Table 1.11](#), we list some rational numbers approaching $\sqrt{2}$, and the values of 2^x for each rational number x are presented as well. We claim that if we choose rational numbers x getting closer and closer to $\sqrt{2}$, the values of 2^x get closer and closer to some number L . We define that number L to be $2^{\sqrt{2}}$.

x	1.4	1.41	1.414	1.4142	1.41421	1.414213
2^x	2.639	2.65737	2.66475	2.665119	2.665138	2.665143

Table 1.11 Values of 2^x for a List of Rational Numbers Approximating $\sqrt{2}$

EXAMPLE 1.33

Bacterial Growth

Suppose a particular population of bacteria is known to double in size every 4 hours. If a culture starts with 1000 bacteria, the number of bacteria after 4 hours is $n(4) = 1000 \cdot 2$. The number of bacteria after 8 hours is $n(8) = n(4) \cdot 2 = 1000 \cdot 2^2$. In general, the number of bacteria after $4m$ hours is $n(4m) = 1000 \cdot 2^m$. Letting $t = 4m$, we see that the number of bacteria after t hours is $n(t) = 1000 \cdot 2^{t/4}$. Find the number of bacteria after 6 hours, 10 hours, and 24 hours.

Solution

The number of bacteria after 6 hours is given by $n(6) = 1000 \cdot 2^{6/4} \approx 2828$ bacteria. The number of bacteria after 10 hours is given by $n(10) = 1000 \cdot 2^{10/4} \approx 5657$ bacteria. The number of bacteria after 24 hours is given by $n(24) = 1000 \cdot 2^6 = 64,000$ bacteria.

- 1.27 Given the exponential function $f(x) = 100 \cdot 3^{x/2}$, evaluate $f(4)$ and $f(10)$.

1.5 Exponential and Logarithmic Functions

Learning Objectives

- 1.5.1** Identify the form of an exponential function.
- 1.5.2** Explain the difference between the graphs of x^b and b^x .
- 1.5.3** Recognize the significance of the number e .
- 1.5.4** Identify the form of a logarithmic function.
- 1.5.5** Explain the relationship between exponential and logarithmic functions.
- 1.5.6** Describe how to calculate a logarithm to a different base.
- 1.5.7** Identify the hyperbolic functions, their graphs, and basic identities.

In this section we examine exponential and logarithmic functions. We use the properties of these functions to solve equations involving exponential or logarithmic terms, and we study the meaning and importance of the number e . We also define hyperbolic and inverse hyperbolic functions, which involve combinations of exponential and logarithmic functions. (Note that we present alternative definitions of exponential and logarithmic functions in the chapter [Applications of Integrations](#), and prove that the functions have the same properties with either definition.)

Exponential Functions

Exponential functions arise in many applications. One common example is population growth.

For example, if a population starts with P_0 individuals and then grows at an annual rate of 2%, its population after 1 year is

$$P(1) = P_0 + 0.02P_0 = P_0(1 + 0.02) = P_0(1.02).$$

Its population after 2 years is

$$P(2) = P(1) + 0.02P(1) = P(1)(1.02) = P_0(1.02)^2.$$

In general, its population after t years is

$$P(t) = P_0(1.02)^t,$$

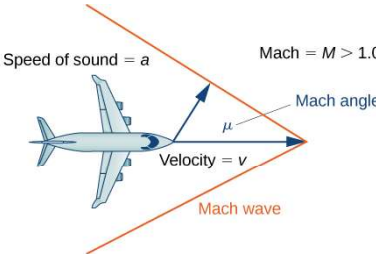
which is an exponential function. More generally, any function of the form $f(x) = b^x$, where $b > 0$, $b \neq 1$, is an exponential function with **base** b and **exponent** x . Exponential functions have constant bases and variable exponents. Note that a function of the form $f(x) = x^b$ for some constant b is not an exponential function but a power function.

To see the difference between an exponential function and a power function, we compare the functions $y = x^2$ and $y = 2^x$. In [Table 1.10](#), we see that both 2^x and x^2 approach infinity as $x \rightarrow \infty$. Eventually, however, 2^x becomes larger than x^2 and grows more rapidly as $x \rightarrow \infty$. In the opposite direction, as $x \rightarrow -\infty$, $x^2 \rightarrow \infty$, whereas $2^x \rightarrow 0$. The line $y = 0$ is a horizontal asymptote for $y = 2^x$.

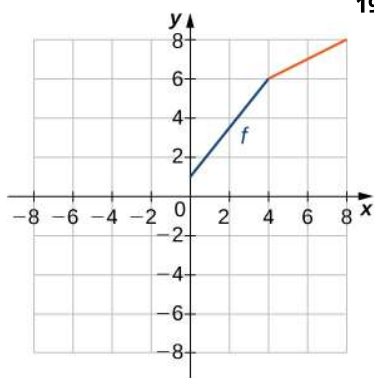
x	-3	-2	-1	0	1	2	3	4	5	6
x^2	9	4	1	0	1	4	9	16	25	36
2^x	1/8	1/4	1/2	1	2	4	8	16	32	64

Table 1.10 Values of x^2 and 2^x

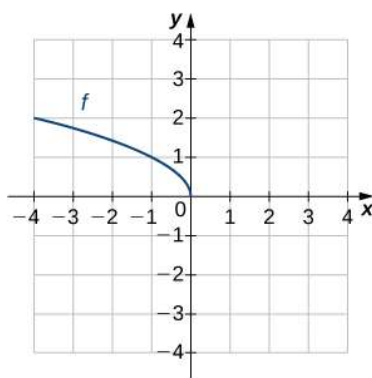
In [Figure 1.43](#), we graph both $y = x^2$ and $y = 2^x$ to show how the graphs differ.

- 219. [T]** The cost to remove a toxin from a lake is modeled by the function $C(p) = 75p/(85 - p)$, where C is the cost (in thousands of dollars) and p is the amount of toxin in a small lake (measured in parts per billion [ppb]). This model is valid only when the amount of toxin is less than 85 ppb.
- Find the cost to remove 25 ppb, 40 ppb, and 50 ppb of the toxin from the lake.
 - Find the inverse function.
 - Use part b. to determine how much of the toxin is removed for \$50,000.
- 220. [T]** A race car is accelerating at a velocity given by $v(t) = \frac{25}{4}t + 54$, where v is the velocity (in feet per second) at time t .
- Find the velocity of the car at 10 sec.
 - Find the inverse function.
 - Use part b. to determine how long it takes for the car to reach a speed of 150 ft/sec.
- 221. [T]** An airplane's Mach number M is the ratio of its speed to the speed of sound. When a plane is flying at a constant altitude, then its Mach angle is given by $\mu = 2\sin^{-1}\left(\frac{1}{M}\right)$. Find the Mach angle (to the nearest degree) for the following Mach numbers.
- 
- $M = 1.4$
 - $M = 2.8$
 - $M = 4.3$
- 222. [T]** Using $\mu = 2\sin^{-1}\left(\frac{1}{M}\right)$, find the Mach number M for the following angles.
- $\mu = \frac{\pi}{6}$
 - $\mu = \frac{2\pi}{7}$
 - $\mu = \frac{3\pi}{8}$
- 223. [T]** The average temperature (in degrees Celsius) of a city in the northern United States can be modeled by the function $T(x) = 5 + 18\sin\left[\frac{\pi}{6}(x - 4.6)\right]$, where x is time in months and $x = 1.00$ corresponds to January 1. Determine the day(s) (month and day) when the average temperature is 21°C . Use the integer portion of your answer(s) as the month and calculate the day of the month from the decimal portion.
- 224. [T]** The depth (in feet) of water at a dock changes with the rise and fall of tides. It is modeled by the function $D(t) = 5\sin\left(\frac{\pi}{6}t - \frac{7\pi}{6}\right) + 8$, where t is the number of hours after midnight. Determine the first time after midnight when the depth is 11.75 ft.
- 225. [T]** An object moving in simple harmonic motion is modeled by the function $s(t) = -6\cos\left(\frac{\pi t}{2}\right)$, where s is measured in inches and t is measured in seconds. Determine the first time when the distance moved is 4.5 in.
- 226. [T]** A local art gallery has a portrait 3 ft in height that is hung 2.5 ft above the eye level of an average person. The viewing angle θ can be modeled by the function $\theta = \tan^{-1}\frac{5.5}{x} - \tan^{-1}\frac{2.5}{x}$, where x is the distance (in feet) from the portrait. Find the viewing angle when a person is 4 ft from the portrait.
- 227. [T]** Use a calculator to evaluate $\tan^{-1}(\tan(2.1))$ and $\cos^{-1}(\cos(2.1))$. Explain the results of each.
- 228. [T]** Use a calculator to evaluate $\sin(\sin^{-1}(-2))$ and $\tan(\tan^{-1}(-2))$. Explain the results of each.

197.



198.



For the following exercises, use composition to determine which pairs of functions are inverses.

199. $f(x) = 8x$, $g(x) = \frac{x}{8}$ 200. $f(x) = 8x + 3$, $g(x) = \frac{x-3}{8}$ 201. $f(x) = 5x - 7$, $g(x) = \frac{x+5}{7}$

202. $f(x) = \frac{2}{3}x + 2$, $g(x) = \frac{3}{2}x + 3$ 203. $f(x) = \frac{1}{x-1}$, $x \neq 1$, $g(x) = \frac{1}{x} + 1$, $x \neq 0$

204. $f(x) = x^3 + 1$, $g(x) = (x-1)^{1/3}$ 205. $f(x) = x^2 + 2x + 1$, $x \geq -1$, $g(x) = -1 + \sqrt{x}$, $x \geq 0$

206. $f(x) = \sqrt{4-x^2}$, $0 \leq x \leq 2$, $g(x) = \sqrt{4-x^2}$, $0 \leq x \leq 2$

For the following exercises, evaluate the functions. Give the exact value.

207. $\tan^{-1}\left(\frac{\sqrt{3}}{3}\right)$

208. $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$

209. $\cot^{-1}(1)$

210. $\sin^{-1}(-1)$

211. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

212. $\cos\left(\tan^{-1}(\sqrt{3})\right)$

213. $\sin\left(\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)\right)$

214. $\sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$

215. $\tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right)$

216. The function $C = T(F) = (5/9)(F - 32)$ converts degrees Fahrenheit to degrees Celsius.

- Find the inverse function $F = T^{-1}(C)$
- What is the inverse function used for?

217. [T] The velocity V (in centimeters per second) of blood in an artery at a distance x cm from the center of the artery can be modeled by the function $V = f(x) = 500(0.04 - x^2)$ for $0 \leq x \leq 0.2$.

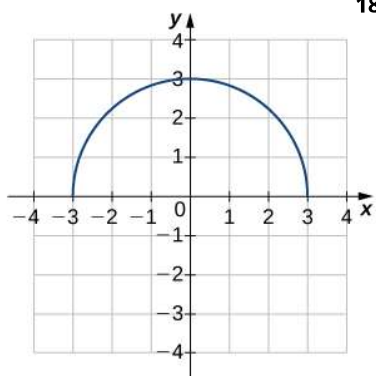
- Find $x = f^{-1}(V)$.
- Interpret what the inverse function is used for.
- Find the distance from the center of an artery with a velocity of 15 cm/sec, 10 cm/sec, and 5 cm/sec.

218. A function that converts dress sizes in the United States to those in Europe is given by

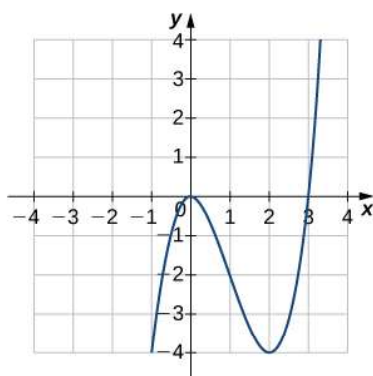
$$D(x) = 2x + 24.$$

- Find the European dress sizes that correspond to sizes 6, 8, 10, and 12 in the United States.
- Find the function that converts European dress sizes to U.S. dress sizes.
- Use part b. to find the dress sizes in the United States that correspond to 46, 52, 62, and 70.

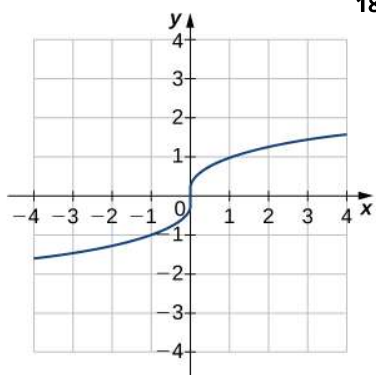
185.



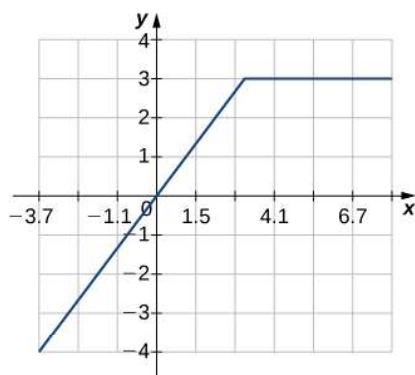
186.



187.



188.



For the following exercises, a. find the inverse function, and b. find the domain and range of the inverse function.

189. $f(x) = x^2 - 4, x \geq 0$

190. $f(x) = \sqrt[3]{x-4}$

191. $f(x) = x^3 + 1$

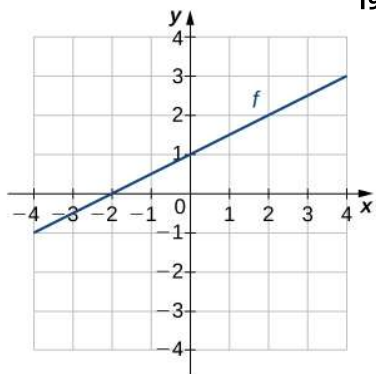
192. $f(x) = (x-1)^2, x \leq 1$

193. $f(x) = \sqrt{x-1}$

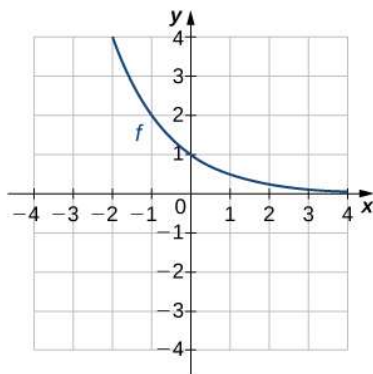
194. $f(x) = \frac{1}{x+2}$

For the following exercises, use the graph of f to sketch the graph of its inverse function.

195.



196.



A	B	x	y	A	B	x	y
0	1			$\sqrt{3}$	1		
1	0			1	$\sqrt{3}$		
1	1			12	5		
1	2			5	12		
2	1						
2	2						
3	4						
4	3						

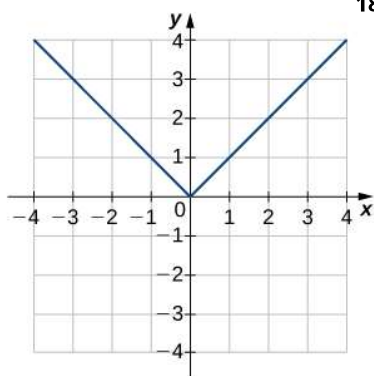
- Try to figure out the formula for the y-values.
- The formula for the x-values is a little harder. The most helpful points from the table are $(1, 1)$, $(1, \sqrt{3})$, $(\sqrt{3}, 1)$. (Hint: Consider inverse trigonometric functions.)
- If you found formulas for parts (5) and (6), show that they work together. That is, substitute the x-value formula you found into $y = A \sin x + B \cos x$ and simplify it to arrive at the y-value formula you found.



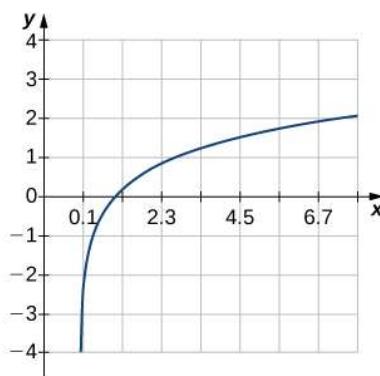
SECTION 1.4 EXERCISES

For the following exercises, use the horizontal line test to determine whether each of the given graphs is one-to-one.

183.



184.



✓ **Solution**

- Evaluating $\sin^{-1}(-\sqrt{3}/2)$ is equivalent to finding the angle θ such that $\sin \theta = -\sqrt{3}/2$ and $-\pi/2 \leq \theta \leq \pi/2$. The angle $\theta = -\pi/3$ satisfies these two conditions. Therefore, $\sin^{-1}(-\sqrt{3}/2) = -\pi/3$.
- First we use the fact that $\tan^{-1}(-1/\sqrt{3}) = -\pi/6$. Then $\tan(-\pi/6) = -1/\sqrt{3}$. Therefore, $\tan(\tan^{-1}(-1/\sqrt{3})) = -1/\sqrt{3}$.
- To evaluate $\cos^{-1}(\cos(5\pi/4))$, first use the fact that $\cos(5\pi/4) = -\sqrt{2}/2$. Then we need to find the angle θ such that $\cos(\theta) = -\sqrt{2}/2$ and $0 \leq \theta \leq \pi$. Since $3\pi/4$ satisfies both these conditions, we have $\cos^{-1}(\cos(5\pi/4)) = \cos^{-1}(-\sqrt{2}/2) = 3\pi/4$.
- Since $\cos(2\pi/3) = -1/2$, we need to evaluate $\sin^{-1}(-1/2)$. That is, we need to find the angle θ such that $\sin(\theta) = -1/2$ and $-\pi/2 \leq \theta \leq \pi/2$. Since $-\pi/6$ satisfies both these conditions, we can conclude that $\sin^{-1}(\cos(2\pi/3)) = \sin^{-1}(-1/2) = -\pi/6$.

STUDENT PROJECT

The Maximum Value of a Function

In many areas of science, engineering, and mathematics, it is useful to know the maximum value a function can obtain, even if we don't know its exact value at a given instant. For instance, if we have a function describing the strength of a roof beam, we would want to know the maximum weight the beam can support without breaking. If we have a function that describes the speed of a train, we would want to know its maximum speed before it jumps off the rails. Safe design often depends on knowing maximum values.

This project describes a simple example of a function with a maximum value that depends on two equation coefficients. We will see that maximum values can depend on several factors other than the independent variable x .

- Consider the graph in [Figure 1.42](#) of the function $y = \sin x + \cos x$. Describe its overall shape. Is it periodic? How do you know?

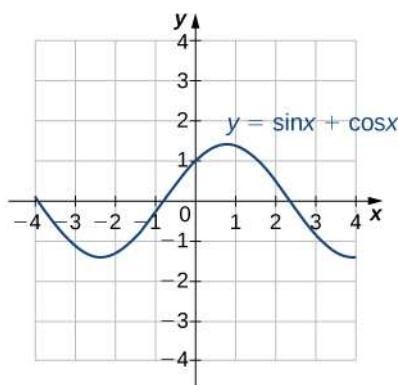


Figure 1.42 The graph of $y = \sin x + \cos x$.

Using a graphing calculator or other graphing device, estimate the x - and y -values of the maximum point for the graph (the first such point where $x > 0$). It may be helpful to express the x -value as a multiple of π .

- Now consider other graphs of the form $y = A \sin x + B \cos x$ for various values of A and B . Sketch the graph when $A = 2$ and $B = 1$, and find the x - and y -values for the maximum point. (Remember to express the x -value as a multiple of π , if possible.) Has it moved?
- Repeat for $A = 1$, $B = 2$. Is there any relationship to what you found in part (2)?
- Complete the following table, adding a few choices of your own for A and B :

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Go to the [following site \(http://www.openstax.org/l/20_inversefun\)](http://www.openstax.org/l/20_inversefun) for more comparisons of functions and their inverses.

When evaluating an inverse trigonometric function, the output is an angle. For example, to evaluate $\cos^{-1}\left(\frac{1}{2}\right)$, we need to find an angle θ such that $\cos \theta = \frac{1}{2}$. Clearly, many angles have this property. However, given the definition of $\cos^{-1}$, we need the angle θ that not only solves this equation, but also lies in the interval $[0, \pi]$. We conclude that $\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$.

We now consider a composition of a trigonometric function and its inverse. For example, consider the two expressions $\sin\left(\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)\right)$ and $\sin^{-1}(\sin(\pi))$. For the first one, we simplify as follows:

$$\sin\left(\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}.$$

For the second one, we have

$$\sin^{-1}(\sin(\pi)) = \sin^{-1}(0) = 0.$$

The inverse function is supposed to “undo” the original function, so why isn't $\sin^{-1}(\sin(\pi)) = \pi$? Recalling our definition of inverse functions, a function f and its inverse f^{-1} satisfy the conditions $f(f^{-1}(y)) = y$ for all y in the domain of f^{-1} and $f^{-1}(f(x)) = x$ for all x in the domain of f , so what happened here? The issue is that the inverse sine function, $\sin^{-1}$, is the inverse of the *restricted* sine function defined on the domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Therefore, for x in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, it is true that $\sin^{-1}(\sin x) = x$. However, for values of x outside this interval, the equation does not hold, even though $\sin^{-1}(\sin x)$ is defined for all real numbers x .

What about $\sin(\sin^{-1} y)$? Does that have a similar issue? The answer is *no*. Since the domain of $\sin^{-1}$ is the interval $[-1, 1]$, we conclude that $\sin(\sin^{-1} y) = y$ if $-1 \leq y \leq 1$ and the expression is not defined for other values of y . To summarize,

$$\sin(\sin^{-1} y) = y \text{ if } -1 \leq y \leq 1$$

and

$$\sin^{-1}(\sin x) = x \text{ if } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}.$$

Similarly, for the cosine function,

$$\cos(\cos^{-1} y) = y \text{ if } -1 \leq y \leq 1$$

and

$$\cos^{-1}(\cos x) = x \text{ if } 0 \leq x \leq \pi.$$

Similar properties hold for the other trigonometric functions and their inverses.

EXAMPLE 1.32

Evaluating Expressions Involving Inverse Trigonometric Functions

Evaluate each of the following expressions.

- $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$
- $\tan\left(\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)$
- $\cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$
- $\sin^{-1}\left(\cos\left(\frac{2\pi}{3}\right)\right)$

Definition

The inverse sine function, denoted $\sin^{-1}$ or arcsin, and the inverse cosine function, denoted $\cos^{-1}$ or arccos, are defined on the domain $D = \{x \mid -1 \leq x \leq 1\}$ as follows:

$$\begin{aligned}\sin^{-1}(x) &= y \text{ if and only if } \sin(y) = x \text{ and } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}; \\ \cos^{-1}(x) &= y \text{ if and only if } \cos(y) = x \text{ and } 0 \leq y \leq \pi.\end{aligned}\tag{1.12}$$

The inverse tangent function, denoted $\tan^{-1}$ or arctan, and inverse cotangent function, denoted $\cot^{-1}$ or arccot, are defined on the domain $D = \{x \mid -\infty < x < \infty\}$ as follows:

$$\begin{aligned}\tan^{-1}(x) &= y \text{ if and only if } \tan(y) = x \text{ and } -\frac{\pi}{2} < y < \frac{\pi}{2}; \\ \cot^{-1}(x) &= y \text{ if and only if } \cot(y) = x \text{ and } 0 < y < \pi.\end{aligned}\tag{1.13}$$

The inverse cosecant function, denoted $\csc^{-1}$ or arcsc, and inverse secant function, denoted $\sec^{-1}$ or arcsec, are defined on the domain $D = \{x \mid |x| \geq 1\}$ as follows:

$$\begin{aligned}\csc^{-1}(x) &= y \text{ if and only if } \csc(y) = x \text{ and } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0; \\ \sec^{-1}(x) &= y \text{ if and only if } \sec(y) = x \text{ and } 0 \leq y \leq \pi, y \neq \pi/2.\end{aligned}\tag{1.14}$$

To graph the inverse trigonometric functions, we use the graphs of the trigonometric functions restricted to the domains defined earlier and reflect the graphs about the line $y = x$ (Figure 1.41).

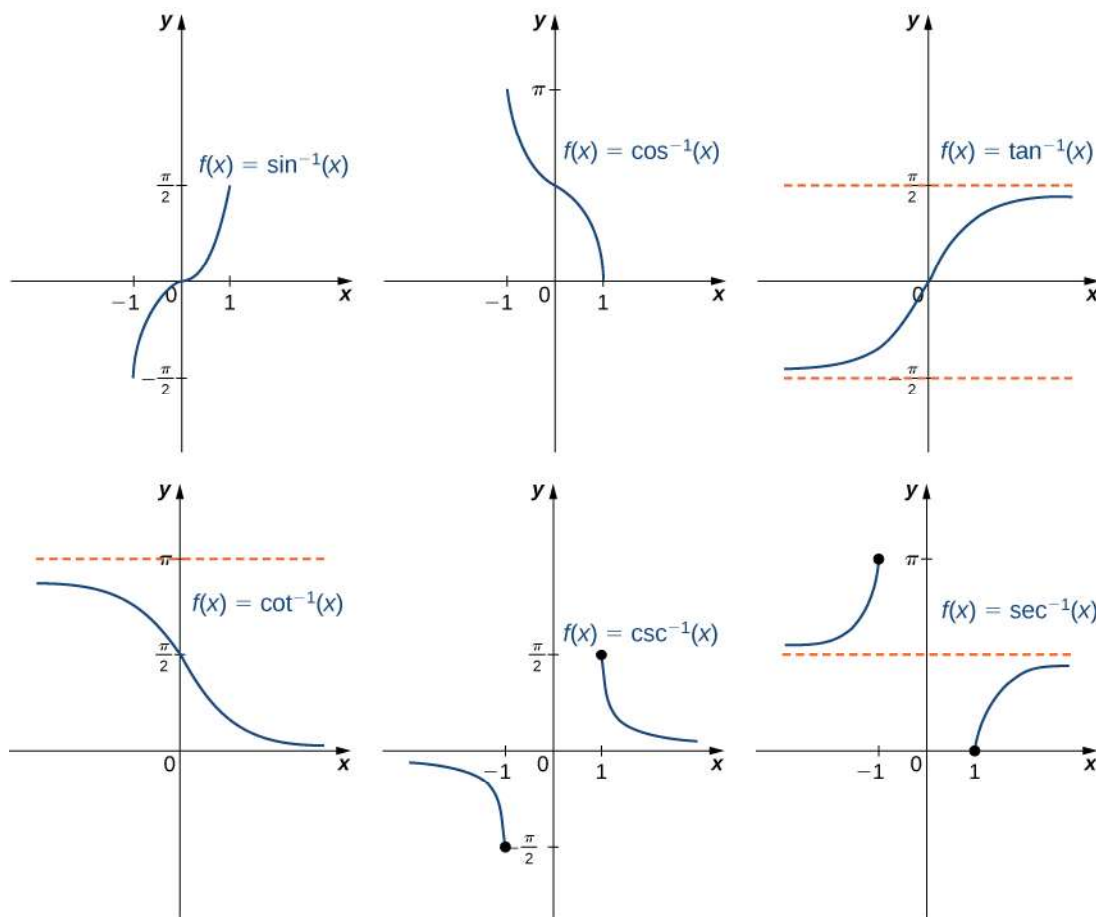
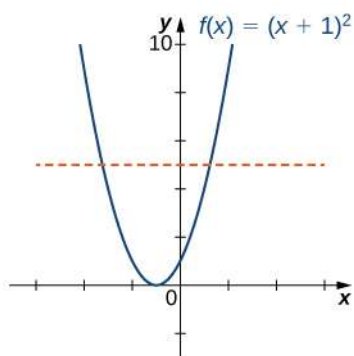
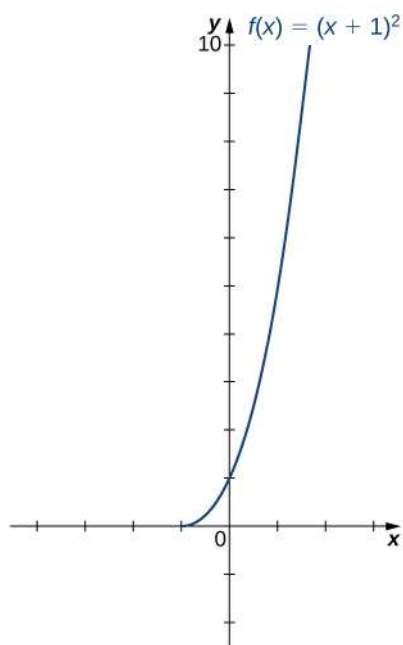


Figure 1.41 The graph of each of the inverse trigonometric functions is a reflection about the line $y = x$ of the corresponding restricted trigonometric function.



- b. On the interval $[-1, \infty)$, f is one-to-one.



The domain and range of f^{-1} are given by the range and domain of f , respectively. Therefore, the domain of f^{-1} is $[0, \infty)$ and the range of f^{-1} is $[-1, \infty)$. To find a formula for f^{-1} , solve the equation $y = (x + 1)^2$ for x . If $y = (x + 1)^2$, then $x = -1 \pm \sqrt{y}$. Since we are restricting the domain to the interval where $x \geq -1$, we need $\pm\sqrt{y} \geq 0$. Therefore, $x = -1 + \sqrt{y}$. Interchanging x and y , we write $y = -1 + \sqrt{x}$ and conclude that $f^{-1}(x) = -1 + \sqrt{x}$.

- ✓ 1.26 Consider $f(x) = 1/x^2$ restricted to the domain $(-\infty, 0)$. Verify that f is one-to-one on this domain. Determine the domain and range of the inverse of f and find a formula for f^{-1} .

Inverse Trigonometric Functions

The six basic trigonometric functions are periodic, and therefore they are not one-to-one. However, if we restrict the domain of a trigonometric function to an interval where it is one-to-one, we can define its inverse. Consider the sine function (Figure 1.34). The sine function is one-to-one on an infinite number of intervals, but the standard convention is to restrict the domain to the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. By doing so, we define the inverse sine function on the domain $[-1, 1]$ such that for any x in the interval $[-1, 1]$, the inverse sine function tells us which angle θ in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ satisfies $\sin \theta = x$. Similarly, we can restrict the domains of the other trigonometric functions to define **inverse trigonometric functions**, which are functions that tell us which angle in a certain interval has a specified trigonometric value.

- ✓ 1.25 Sketch the graph of $f(x) = 2x + 3$ and the graph of its inverse using the symmetry property of inverse functions.

Restricting Domains

As we have seen, $f(x) = x^2$ does not have an inverse function because it is not one-to-one. However, we can choose a subset of the domain of f such that the function is one-to-one. This subset is called a **restricted domain**. By restricting the domain of f , we can define a new function g such that the domain of g is the restricted domain of f and $g(x) = f(x)$ for all x in the domain of g . Then we can define an inverse function for g on that domain. For example, since $f(x) = x^2$ is one-to-one on the interval $[0, \infty)$, we can define a new function g such that the domain of g is $[0, \infty)$ and $g(x) = x^2$ for all x in its domain. Since g is a one-to-one function, it has an inverse function, given by the formula $g^{-1}(x) = \sqrt{x}$. On the other hand, the function $f(x) = x^2$ is also one-to-one on the domain $(-\infty, 0]$. Therefore, we could also define a new function h such that the domain of h is $(-\infty, 0]$ and $h(x) = x^2$ for all x in the domain of h . Then h is a one-to-one function and must also have an inverse. Its inverse is given by the formula $h^{-1}(x) = -\sqrt{x}$ (Figure 1.40).

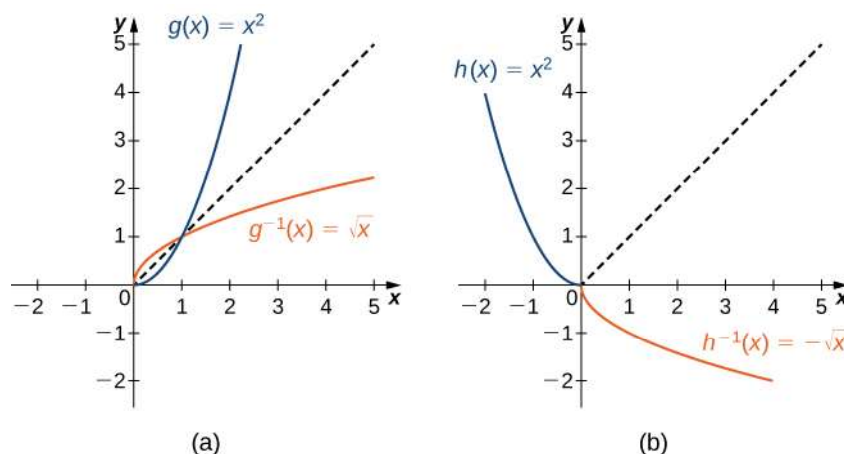


Figure 1.40 (a) For $g(x) = x^2$ restricted to $[0, \infty)$, $g^{-1}(x) = \sqrt{x}$. (b) For $h(x) = x^2$ restricted to $(-\infty, 0]$, $h^{-1}(x) = -\sqrt{x}$.

EXAMPLE 1.31

Restricting the Domain

Consider the function $f(x) = (x + 1)^2$.

- Sketch the graph of f and use the horizontal line test to show that f is not one-to-one.
- Show that f is one-to-one on the restricted domain $[-1, \infty)$. Determine the domain and range for the inverse of f on this restricted domain and find a formula for f^{-1} .

✓ Solution

- The graph of f is the graph of $y = x^2$ shifted left 1 unit. Since there exists a horizontal line intersecting the graph more than once, f is not one-to-one.

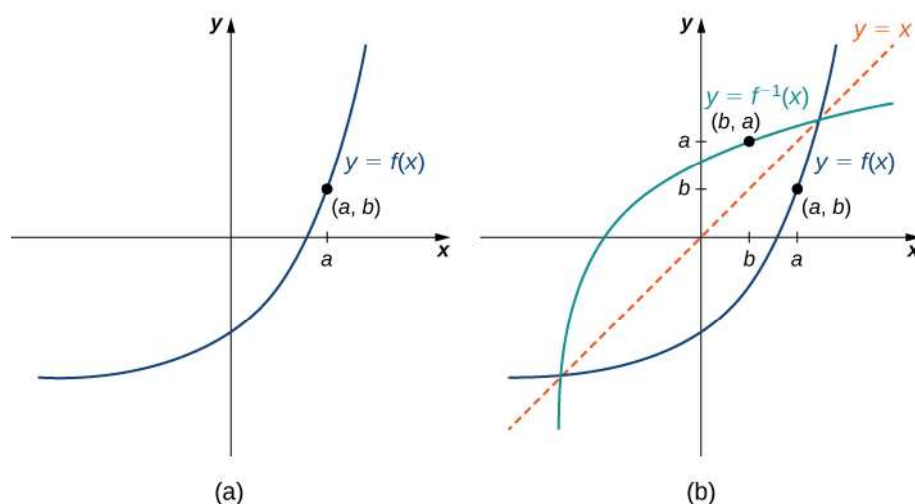
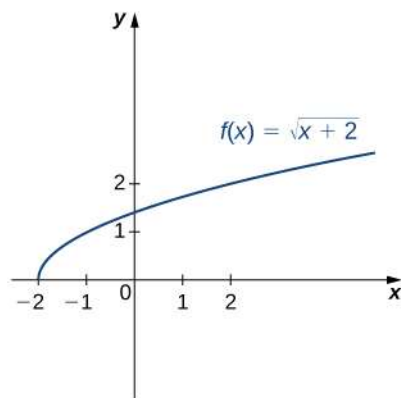


Figure 1.39 (a) The graph of this function f shows point (a, b) on the graph of f . (b) Since (a, b) is on the graph of f , the point (b, a) is on the graph of f^{-1} . The graph of f^{-1} is a reflection of the graph of f about the line $y = x$.

EXAMPLE 1.30

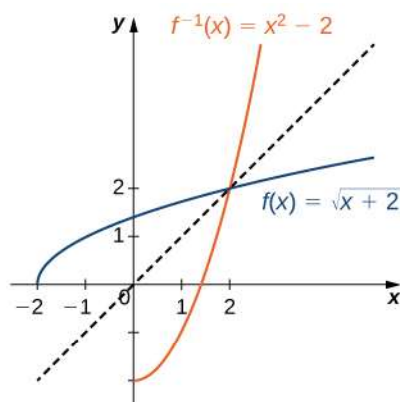
Sketching Graphs of Inverse Functions

For the graph of f in the following image, sketch a graph of f^{-1} by sketching the line $y = x$ and using symmetry. Identify the domain and range of f^{-1} .



Solution

Reflect the graph about the line $y = x$. The domain of f^{-1} is $[0, \infty)$. The range of f^{-1} is $[-2, \infty)$. By using the preceding strategy for finding inverse functions, we can verify that the inverse function is $f^{-1}(x) = x^2 - 2$, as shown in the graph.



Finding a Function's Inverse

We can now consider one-to-one functions and show how to find their inverses. Recall that a function maps elements in the domain of f to elements in the range of f . The inverse function maps each element from the range of f back to its corresponding element from the domain of f . Therefore, to find the inverse function of a one-to-one function f , given any y in the range of f , we need to determine which x in the domain of f satisfies $f(x) = y$. Since f is one-to-one, there is exactly one such value x . We can find that value x by solving the equation $f(x) = y$ for x . Doing so, we are able to write x as a function of y where the domain of this function is the range of f and the range of this new function is the domain of f . Consequently, this function is the inverse of f , and we write $x = f^{-1}(y)$. Since we typically use the variable x to denote the independent variable and y to denote the dependent variable, we often interchange the roles of x and y , and write $y = f^{-1}(x)$. Representing the inverse function in this way is also helpful later when we graph a function f and its inverse f^{-1} on the same axes.

Problem-Solving Strategy

Finding an Inverse Function

1. Solve the equation $y = f(x)$ for x .
2. Interchange the variables x and y and write $y = f^{-1}(x)$.

EXAMPLE 1.29

Finding an Inverse Function

Find the inverse for the function $f(x) = 3x - 4$. State the domain and range of the inverse function. Verify that $f^{-1}(f(x)) = x$.

Solution

Follow the steps outlined in the strategy.

Step 1. If $y = 3x - 4$, then $3x = y + 4$ and $x = \frac{1}{3}y + \frac{4}{3}$.

Step 2. Rewrite as $y = \frac{1}{3}x + \frac{4}{3}$ and let $y = f^{-1}(x)$.

Therefore, $f^{-1}(x) = \frac{1}{3}x + \frac{4}{3}$.

Since the domain of f is $(-\infty, \infty)$, the range of f^{-1} is $(-\infty, \infty)$. Since the range of f is $(-\infty, \infty)$, the domain of f^{-1} is $(-\infty, \infty)$.

You can verify that $f^{-1}(f(x)) = x$ by writing

$$f^{-1}(f(x)) = f^{-1}(3x - 4) = \frac{1}{3}(3x - 4) + \frac{4}{3} = x - \frac{4}{3} + \frac{4}{3} = x.$$

Note that for $f^{-1}(x)$ to be the inverse of $f(x)$, both $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$ for all x in the domain of the inside function.

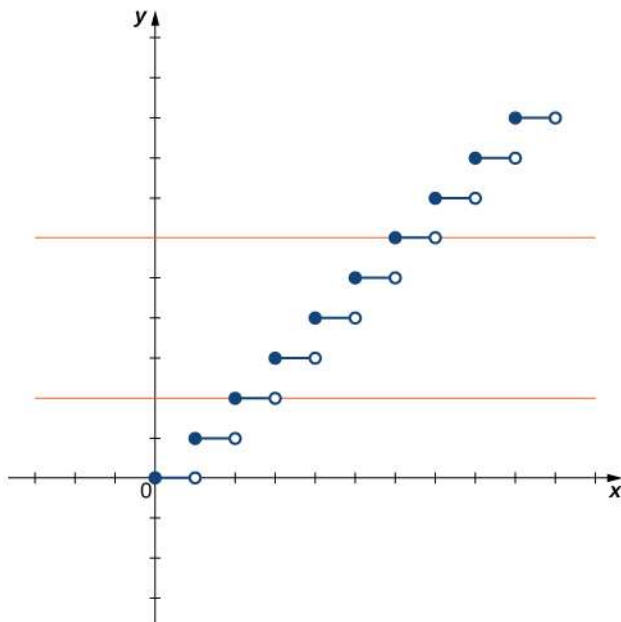
-  1.24 Find the inverse of the function $f(x) = 3x/(x - 2)$. State the domain and range of the inverse function.

Graphing Inverse Functions

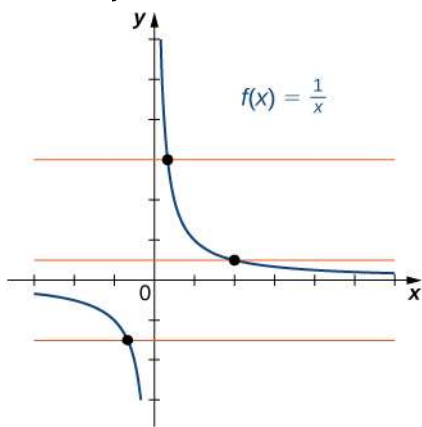
Let's consider the relationship between the graph of a function f and the graph of its inverse. Consider the graph of f shown in [Figure 1.39](#) and a point (a, b) on the graph. Since $b = f(a)$, then $f^{-1}(b) = a$. Therefore, when we graph f^{-1} , the point (b, a) is on the graph. As a result, the graph of f^{-1} is a reflection of the graph of f about the line $y = x$.

✓ **Solution**

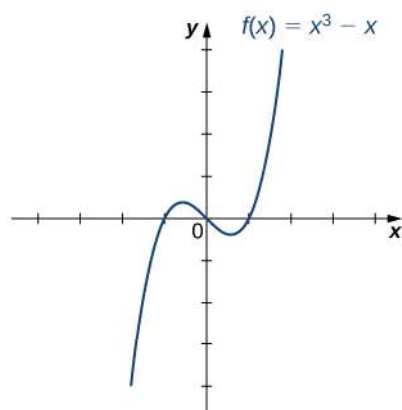
- a. Since the horizontal line $y = n$ for any integer $n \geq 0$ intersects the graph more than once, this function is not one-to-one.



- b. Since every horizontal line intersects the graph once (at most), this function is one-to-one.



- ✓ 1.23 Is the function f graphed in the following image one-to-one?



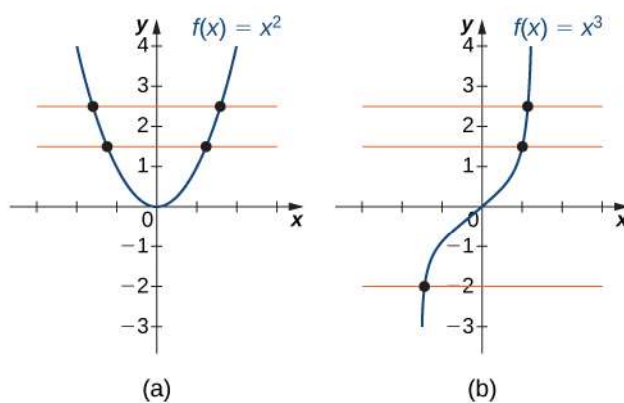


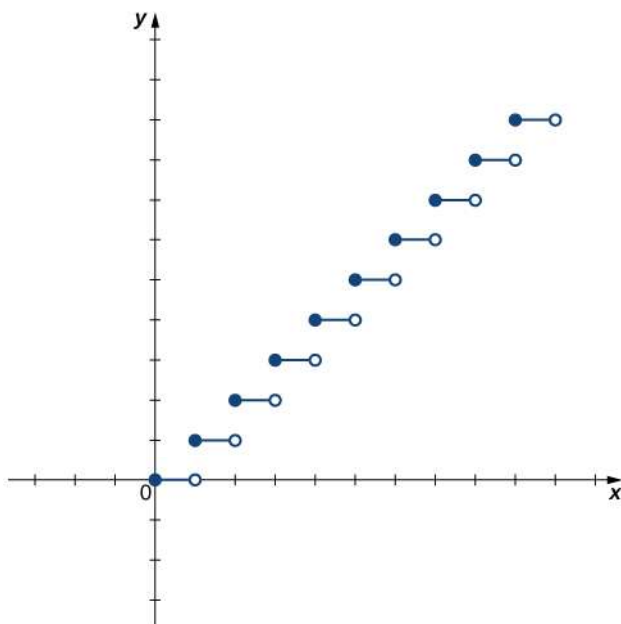
Figure 1.38 (a) The function $f(x) = x^2$ is not one-to-one because it fails the horizontal line test. (b) The function $f(x) = x^3$ is one-to-one because it passes the horizontal line test.

EXAMPLE 1.28

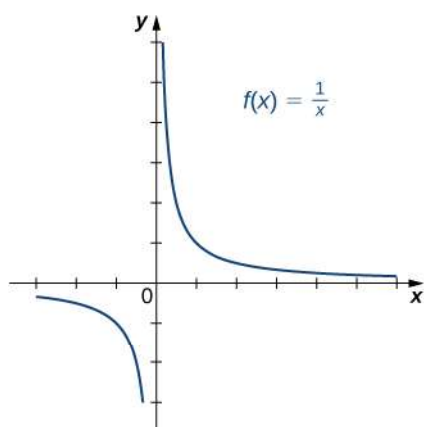
Determining Whether a Function Is One-to-One

For each of the following functions, use the horizontal line test to determine whether it is one-to-one.

a.



b.



Definition

Given a function f with domain D and range R , its **inverse function** (if it exists) is the function f^{-1} with domain R and range D such that $f^{-1}(y) = x$ if $f(x) = y$. In other words, for a function f and its inverse f^{-1} ,

$$f^{-1}(f(x)) = x \text{ for all } x \text{ in } D, \text{ and } f(f^{-1}(y)) = y \text{ for all } y \text{ in } R. \quad (1.11)$$

Note that f^{-1} is read as “ f inverse.” Here, the -1 is not used as an exponent and $f^{-1}(x) \neq 1/f(x)$. [Figure 1.37](#) shows the relationship between the domain and range of f and the domain and range of f^{-1} .

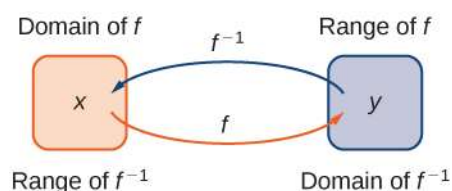


Figure 1.37 Given a function f and its inverse f^{-1} , $f^{-1}(y) = x$ if and only if $f(x) = y$. The range of f becomes the domain of f^{-1} and the domain of f becomes the range of f^{-1} .

Recall that a function has exactly one output for each input. Therefore, to define an inverse function, we need to map each input to exactly one output. For example, let's try to find the inverse function for $f(x) = x^2$. Solving the equation $y = x^2$ for x , we arrive at the equation $x = \pm\sqrt{y}$. This equation does not describe x as a function of y because there are two solutions to this equation for every $y > 0$. The problem with trying to find an inverse function for $f(x) = x^2$ is that two inputs are sent to the same output for each output $y > 0$. The function $f(x) = x^3 + 4$ discussed earlier did not have this problem. For that function, each input was sent to a different output. A function that sends each input to a *different* output is called a **one-to-one function**.

Definition

We say a f is a **one-to-one function** if $f(x_1) \neq f(x_2)$ when $x_1 \neq x_2$.

One way to determine whether a function is one-to-one is by looking at its graph. If a function is one-to-one, then no two inputs can be sent to the same output. Therefore, if we draw a horizontal line anywhere in the xy -plane, according to the **horizontal line test**, it cannot intersect the graph more than once. We note that the horizontal line test is different from the vertical line test. The vertical line test determines whether a graph is the graph of a function. The horizontal line test determines whether a function is one-to-one ([Figure 1.38](#)).

Rule: Horizontal Line Test

A function f is one-to-one if and only if every horizontal line intersects the graph of f no more than once.

- 179. [T]** An alternating current for outlets in a home has voltage given by the function $V(t) = 150\cos 368t$, where V is the voltage in volts at time t in seconds.
- Find the period of the function and interpret its meaning.
 - Determine the number of periods that occur when 1 sec has passed.
- 180. [T]** The number of hours of daylight in a northeast city is modeled by the function $N(t) = 12 + 3\sin\left[\frac{2\pi}{365}(t - 79)\right]$, where t is the number of days after January 1.
- Find the amplitude and period.
 - Determine the number of hours of daylight on the longest day of the year.
 - Determine the number of hours of daylight on the shortest day of the year.
 - Determine the number of hours of daylight 90 days after January 1.
 - Sketch the graph of the function for one period starting on January 1.
- 181. [T]** Suppose that $T = 50 + 10\sin\left[\frac{\pi}{12}(t - 8)\right]$ is a mathematical model of the temperature (in degrees Fahrenheit) at t hours after midnight on a certain day of the week.
- Determine the amplitude and period.
 - Find the temperature 7 hours after midnight.
 - At what time does $T = 60^\circ$?
 - Sketch the graph of T over $0 \leq t \leq 24$.
- 182. [T]** The function $H(t) = 8\sin\left(\frac{\pi}{6}t\right)$ models the height H (in feet) of the tide t hours after midnight. Assume that $t = 0$ is midnight.
- Find the amplitude and period.
 - Graph the function over one period.
 - What is the height of the tide at 4:30 a.m.?

1.4 Inverse Functions

Learning Objectives

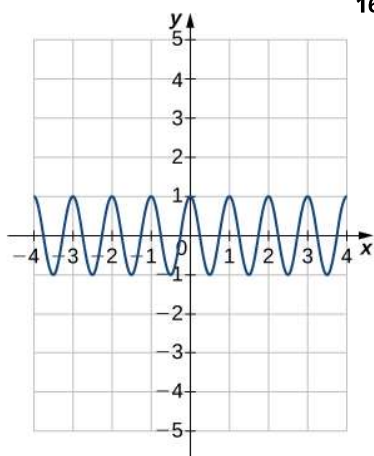
- 1.4.1** Determine the conditions for when a function has an inverse.
- 1.4.2** Use the horizontal line test to recognize when a function is one-to-one.
- 1.4.3** Find the inverse of a given function.
- 1.4.4** Draw the graph of an inverse function.
- 1.4.5** Evaluate inverse trigonometric functions.

An inverse function reverses the operation done by a particular function. In other words, whatever a function does, the inverse function undoes it. In this section, we define an inverse function formally and state the necessary conditions for an inverse function to exist. We examine how to find an inverse function and study the relationship between the graph of a function and the graph of its inverse. Then we apply these ideas to define and discuss properties of the inverse trigonometric functions.

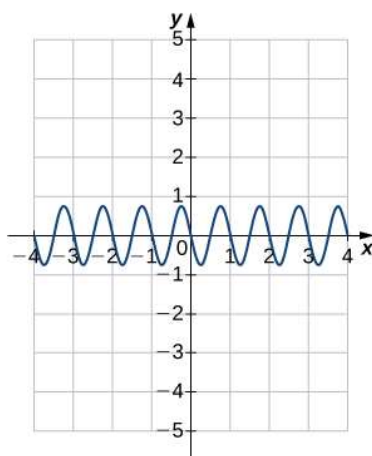
Existence of an Inverse Function

We begin with an example. Given a function f and an output $y = f(x)$, we are often interested in finding what value or values x were mapped to y by f . For example, consider the function $f(x) = x^3 + 4$. Since any output $y = x^3 + 4$, we can solve this equation for x to find that the input is $x = \sqrt[3]{y-4}$. This equation defines x as a function of y . Denoting this function as f^{-1} , and writing $x = f^{-1}(y) = \sqrt[3]{y-4}$, we see that for any x in the domain of f , $f^{-1}(f(x)) = f^{-1}(x^3 + 4) = x$. Thus, this new function, f^{-1} , “undid” what the original function f did. A function with this property is called the inverse function of the original function.

165.



166.



For the following exercises, find a. the amplitude, b. the period, and c. the phase shift with direction for each function.

167. $y = \sin\left(x - \frac{\pi}{4}\right)$

168. $y = 3 \cos(2x + 3)$

169. $y = -\frac{1}{2} \sin\left(\frac{1}{4}x\right)$

170. $y = 2 \cos\left(x - \frac{\pi}{3}\right)$

171. $y = -3 \sin(\pi x + 2)$

172. $y = 4 \cos\left(2x - \frac{\pi}{2}\right)$

173. [T] The diameter of a wheel rolling on the ground is 40 in. If the wheel rotates through an angle of 120° , how many inches does it move? Approximate to the nearest whole inch.

174. [T] Find the length of the arc intercepted by central angle θ in a circle of radius r . Round to the nearest hundredth.
 a. $r = 12.8$ cm, $\theta = \frac{5\pi}{6}$ rad
 b. $r = 4.378$ cm, $\theta = \frac{7\pi}{6}$ rad
 c. $r = 0.964$ cm, $\theta = 50^\circ$
 d. $r = 8.55$ cm, $\theta = 325^\circ$

175. [T] As a point P moves around a circle, the measure of the angle changes. The measure of how fast the angle is changing is called *angular speed*, ω , and is given by $\omega = \theta/t$, where θ is in radians and t is time. Find the angular speed for the given data. Round to the nearest thousandth.
 a. $\theta = \frac{7\pi}{4}$ rad, $t = 10$ sec
 b. $\theta = \frac{3\pi}{5}$ rad, $t = 8$ sec
 c. $\theta = \frac{2\pi}{9}$ rad, $t = 1$ min
 d. $\theta = 23.76$ rad, $t = 14$ min

176. [T] A total of $250,000 \text{ m}^2$ of land is needed to build a nuclear power plant. Suppose it is decided that the area on which the power plant is to be built should be circular.
 a. Find the radius of the circular land area.
 b. If the land area is to form a 45° sector of a circle instead of a whole circle, find the length of the curved side.

177. [T] The area of an isosceles triangle with equal sides of length x is $\frac{1}{2}x^2 \sin \theta$, where θ is the angle formed by the two sides. Find the area of an isosceles triangle with equal sides of length 8 in. and angle $\theta = 5\pi/12$ rad.

178. [T] A particle travels in a circular path at a constant angular speed ω . The angular speed is modeled by the function $\omega = 9 |\cos(\pi t - \pi/12)|$. Determine the angular speed at $t = 9$ sec.

134. $b = 28, c = 35$

For the following exercises, P is a point on the unit circle. a. Find the (exact) missing coordinate value of each point and b. find the values of the six trigonometric functions for the angle θ with a terminal side that passes through point P . Rationalize denominators.

135. $P\left(\frac{7}{25}, y\right), y > 0$

136. $P\left(\frac{-15}{17}, y\right), y < 0$

137. $P\left(x, \frac{\sqrt{7}}{3}\right), x < 0$

138. $P\left(x, \frac{-\sqrt{15}}{4}\right), x > 0$

For the following exercises, simplify each expression by writing it in terms of sines and cosines, then simplify. The final answer does not have to be in terms of sine and cosine only.

139. $\tan^2 x + \sin x \csc x$

140. $\sec x \sin x \cot x$

141. $\frac{\tan^2 x}{\sec^2 x}$

142. $\sec x - \cos x$

143. $(1 + \tan \theta)^2 - 2 \tan \theta$

144. $\sin x (\csc x - \sin x)$

145. $\frac{\cos t}{\sin t} + \frac{\sin t}{1 + \cos t}$

146. $\frac{1 + \tan^2 \alpha}{1 + \cot^2 \alpha}$

For the following exercises, verify that each equation is an identity.

147. $\frac{\tan \theta \cot \theta}{\csc \theta} = \sin \theta$

148. $\frac{\sec^2 \theta}{\tan \theta} = \sec \theta \csc \theta$

149. $\frac{\sin t}{\csc t} + \frac{\cos t}{\sec t} = 1$

150. $\frac{\sin x}{\cos x + 1} + \frac{\cos x - 1}{\sin x} = 0$

151. $\cot \gamma + \tan \gamma = \sec \gamma \csc \gamma$

152. $\sin^2 \beta + \tan^2 \beta + \cos^2 \beta = \sec^2 \beta$

153. $\frac{1}{1 - \sin \alpha} + \frac{1}{1 + \sin \alpha} = 2 \sec^2 \alpha$

154. $\frac{\tan \theta - \cot \theta}{\sin \theta \cos \theta} = \sec^2 \theta - \csc^2 \theta$

For the following exercises, solve the trigonometric equations on the interval $0 \leq \theta < 2\pi$.

155. $2 \sin \theta - 1 = 0$

156. $1 + \cos \theta = \frac{1}{2}$

157. $2 \tan^2 \theta = 2$

158. $4 \sin^2 \theta - 2 = 0$

159. $\sqrt{3} \cot \theta + 1 = 0$

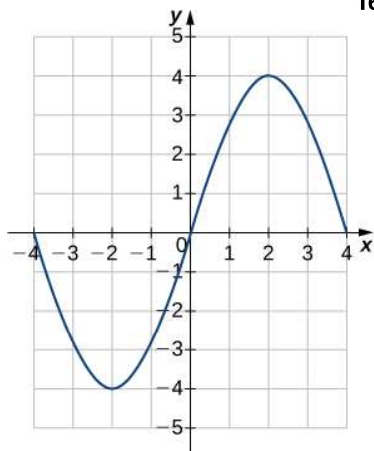
160. $3 \sec \theta - 2\sqrt{3} = 0$

161. $2 \cos \theta \sin \theta = \sin \theta$

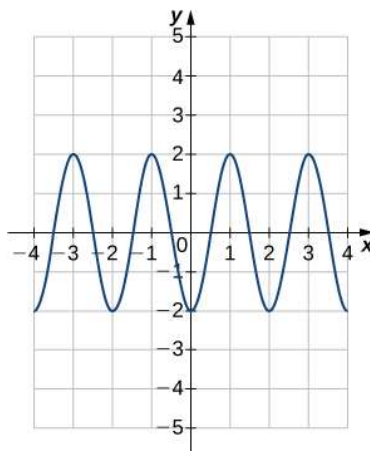
162. $\csc^2 \theta + 2 \csc \theta + 1 = 0$

For the following exercises, each graph is of the form $y = A \sin Bx$ or $y = A \cos Bx$, where $B > 0$. Write the equation of the graph.

163.

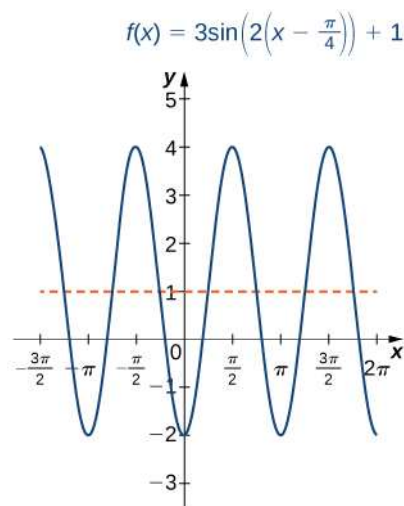


164.



 Solution

This graph is a horizontal compression by a factor of 2, a phase shift to the right by $\pi/4$ units, followed by a vertical stretch by a factor of 3, and then a vertical shift by 1 unit. The period of f is π .



-  1.22 Describe the relationship between the graph of $f(x) = 3\sin(4x) - 5$ and the graph of $y = \sin(x)$.



SECTION 1.3 EXERCISES

For the following exercises, convert each angle in degrees to radians. Write the answer as a multiple of π .

113. 240° 114. 15° 115. -60°
116. -225° 117. 330°

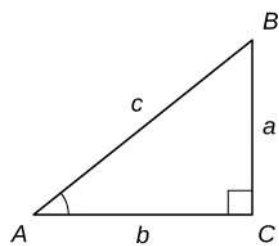
For the following exercises, convert each angle in radians to degrees.

118. $\frac{\pi}{2}$ rad 119. $\frac{7\pi}{6}$ rad 120. $\frac{11\pi}{2}$ rad
121. -3π rad 122. $\frac{5\pi}{12}$ rad

Evaluate the following functional values.

123. $\cos\left(\frac{4\pi}{3}\right)$ 124. $\tan\left(\frac{19\pi}{4}\right)$ 125. $\sin\left(-\frac{3\pi}{4}\right)$
126. $\sec\left(\frac{\pi}{6}\right)$ 127. $\sin\left(\frac{\pi}{12}\right)$ 128. $\cos\left(\frac{5\pi}{12}\right)$

For the following exercises, consider triangle ABC, a right triangle with a right angle at C. a. Find the missing side of the triangle. b. Find the six trigonometric function values for the angle at A. Where necessary, simplify to a fraction or round to three decimal places.



129. $a = 4, c = 7$ 130. $a = 21, c = 29$
131. $a = 85.3, b = 125.5$ 132. $b = 40, c = 41$ 133. $a = 84, b = 13$

$$f(x) = A \cos(B(x - \alpha)) + C. \quad (1.10)$$

In [Figure 1.35](#), the constant α causes a horizontal or phase shift. The factor B changes the period. This transformed sine function will have a period $2\pi/|B|$. The factor A results in a vertical stretch by a factor of $|A|$. We say $|A|$ is the “amplitude of f .” The constant C causes a vertical shift.

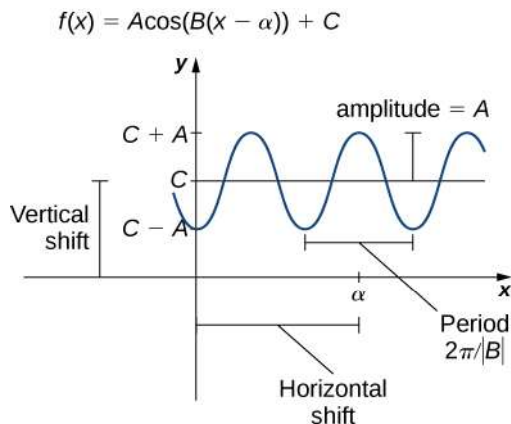


Figure 1.35 A graph of a general cosine function.

Notice in [Figure 1.34](#) that the graph of $y = \cos x$ is the graph of $y = \sin x$ shifted to the left $\pi/2$ units. Therefore, we can write $\cos x = \sin(x + \pi/2)$. Similarly, we can view the graph of $y = \sin x$ as the graph of $y = \cos x$ shifted right $\pi/2$ units, and state that $\sin x = \cos(x - \pi/2)$.

A shifted sine curve arises naturally when graphing the number of hours of daylight in a given location as a function of the day of the year. For example, suppose a city reports that June 21 is the longest day of the year with 15.7 hours and December 21 is the shortest day of the year with 8.3 hours. It can be shown that the function

$$h(t) = 3.7 \sin\left(\frac{2\pi}{365}(t - 80.5)\right) + 12$$

is a model for the number of hours of daylight h as a function of day of the year t ([Figure 1.36](#)).

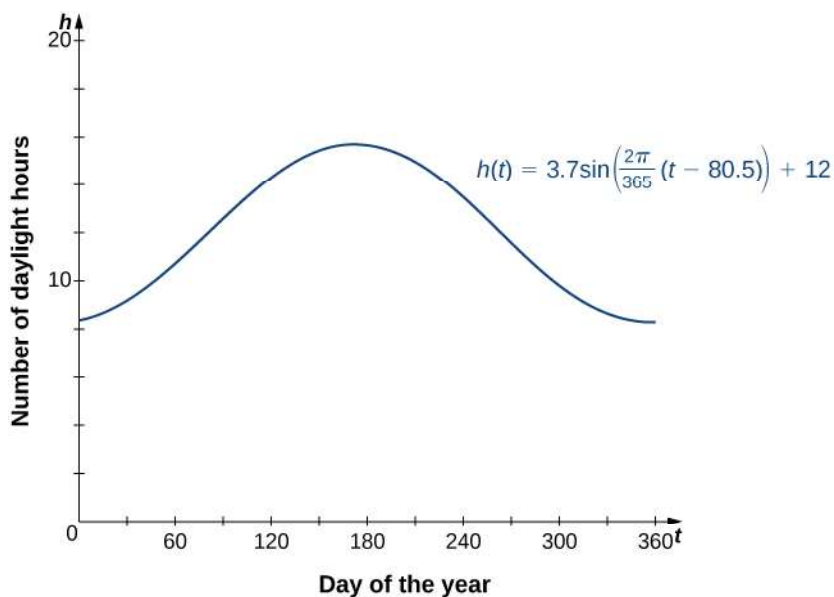


Figure 1.36 The hours of daylight as a function of day of the year can be modeled by a shifted sine curve.

EXAMPLE 1.27

Sketching the Graph of a Transformed Sine Curve

Sketch a graph of $f(x) = 3 \sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$.

 Solution

We start with the identity

$$\sin^2 \theta + \cos^2 \theta = 1.$$

Dividing both sides of this equation by $\cos^2 \theta$, we obtain

$$\frac{\sin^2 \theta}{\cos^2 \theta} + 1 = \frac{1}{\cos^2 \theta}.$$

Since $\sin \theta / \cos \theta = \tan \theta$ and $1 / \cos \theta = \sec \theta$, we conclude that

$$\tan^2 \theta + 1 = \sec^2 \theta.$$

 1.21 Prove the trigonometric identity $1 + \cot^2 \theta = \csc^2 \theta$.

Graphs and Periods of the Trigonometric Functions

We have seen that as we travel around the unit circle, the values of the trigonometric functions repeat. We can see this pattern in the graphs of the functions. Let $P = (x, y)$ be a point on the unit circle and let θ be the corresponding angle. Since the angle θ and $\theta + 2\pi$ correspond to the same point P , the values of the trigonometric functions at θ and at $\theta + 2\pi$ are the same. Consequently, the trigonometric functions are **periodic functions**. The period of a function f is defined to be the smallest positive value p such that $f(x + p) = f(x)$ for all values x in the domain of f . The sine, cosine, secant, and cosecant functions have a period of 2π . Since the tangent and cotangent functions repeat on an interval of length π , their period is π (Figure 1.34).

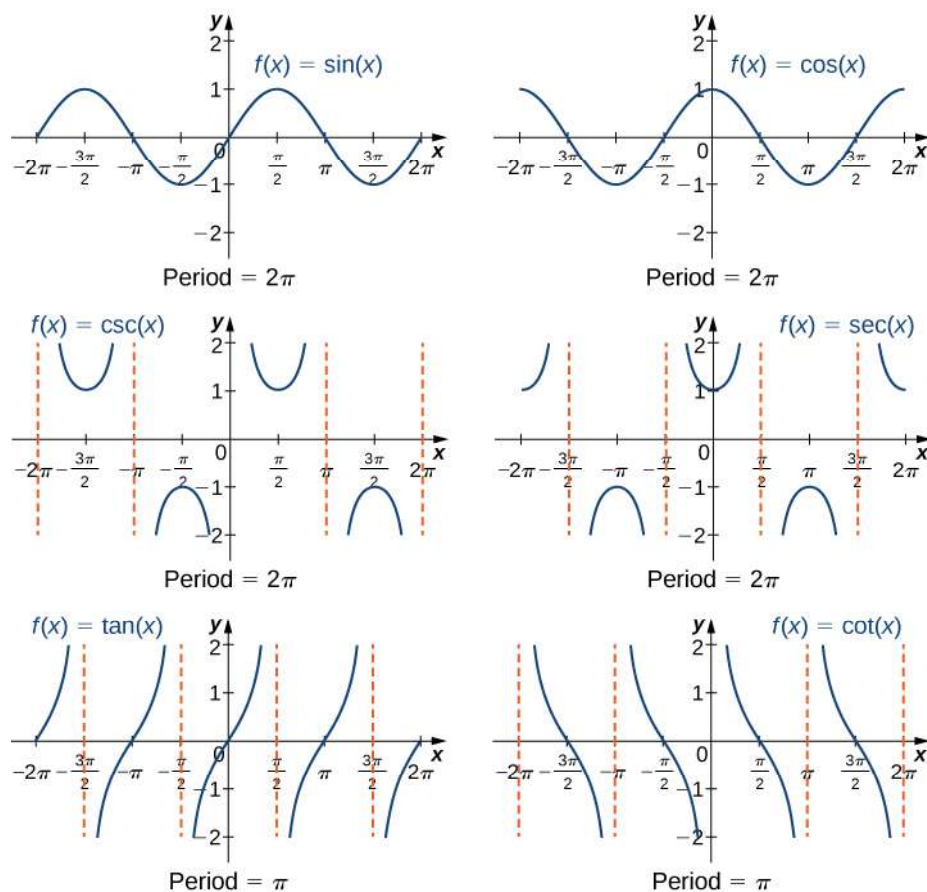


Figure 1.34 The six trigonometric functions are periodic.

Just as with algebraic functions, we can apply transformations to trigonometric functions. In particular, consider the following function: